ACCA

Paper F2

Management accounting

Essential text

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Contents

Paper Introduction

How to Use the Materials

These Kaplan Publishing learning materials have been carefully designed to make your learning experience as easy as possible and to give you the best chances of success in your examinations.

The product range contains a number of features to help you in the study process. They include:

(1) Detailed study guide and syllabus objectives

(2) Description of the examination

(3) Study skills and revision guidance

(4) Complete text or essential text

(5) Question practice

The sections on the study guide, the syllabus objectives, the examination and study skills should all be read before you commence your studies. They are designed to familiarise you with the nature and content of the examination and give you tips on how to best to approach your learning.

The **complete text or essential text** comprises the main learning materials and gives guidance as to the importance of topics and where other related resources can be found. Each chapter includes:

- The **learning objectives** contained in each chapter, which have been carefully mapped to the examining body's own syllabus learning objectives or outcomes. You should use these to check you have a clear understanding of all the topics on which you might be assessed in the examination.
- The **chapter diagram** provides a visual reference for the content in the chapter, giving an overview of the topics and how they link together.

- The **content** for each topic area commences with a brief explanation or definition to put the topic into context before covering the topic in detail. You should follow your studying of the content with a review of the illustration/s. These are worked examples which will help you to understand better how to apply the content for the topic.
- **Test your understanding** sections provide an opportunity to assess your understanding of the key topics by applying what you have learned to short questions. Answers can be found at the back of each chapter.
- **Summary diagrams** complete each chapter to show the important links between topics and the overall content of the paper. These diagrams should be used to check that you have covered and understood the core topics before moving on.
- **Question practice** is provided at the back of each text.

Icon Explanations

Definition - Key definitions that you will need to learn from the core content.

Key Point - Identifies topics that are key to success and are often examined.

Expandable Text - Expandable text provides you with additional information about a topic area and may help you gain a better understanding of the core content. Essential text users can access this additional content on-line (read it where you need further guidance or skip over when you are happy with the topic)

Illustration - Worked examples help you understand the core content better.

Test Your Understanding - Exercises for you to complete to ensure that you have understood the topics just learned.

Tricky topic - When reviewing these areas care should be taken and all illustrations and test your understanding exercises should be completed to ensure that the topic is understood.

For more details about the syllabus and the format of your exam please see your Complete Text or go online.

Online

subscribers

The Syllabus

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Further reading

You can find further reading and technical articles under the student section of ACCA's website.

chapter

1

The nature and purpose of management accounting

Chapter learning objectives

Upon completion of this chapter you will be able to:

- describe the difference between data and information
- explain, using the 'ACCURATE' acronym the attributes of good information
- describe, in overview, the managerial processes of: planning, decision making and control
- explain the characteristics of and difference between strategic, tactical and operational planning
- explain the characteristics and differences between cost, profit, investment and revenue centres
- describe the different information needs of managers of cost, profit, investment and revenue centres
- describe the purpose and role of cost and management accounting within an organisation's management information system
- compare and contrast, for a business, financial accounting with cost and management accounting.

1 The nature of good information

Data and information

'Data' means facts. Data consists of numbers, letters, symbols, raw facts, events and transactions which have been recorded but not yet processed into a form suitable for use.

Information is data which has been processed in such a way that it is meaningful to the person who receives it (for making decisions).

- The terms data and information are often used interchangeably in everyday language. Make sure that you can distinguish between the two.
- As data is converted into information, some of the detail of the data is eliminated and replaced by summaries which are easier to understand.

Expandable text

Attributes of good information

Information is provided to management to assist them with planning, controlling operations and making decisions. Management decisions are likely to be better when they are provided with better quality information.

The attributes of good information can be identified by the 'ACCURATE' acronym as shown below:

A. Accurate

C. Complete

C. Cost-effective

U. Understandable

R. Relevant

A. Accessible

T. Timely

E. Easy to use!

Expandable text

2 The managerial processes of decision making and control

The main functions that management are involved with are:

- planning
- decision making
- control.

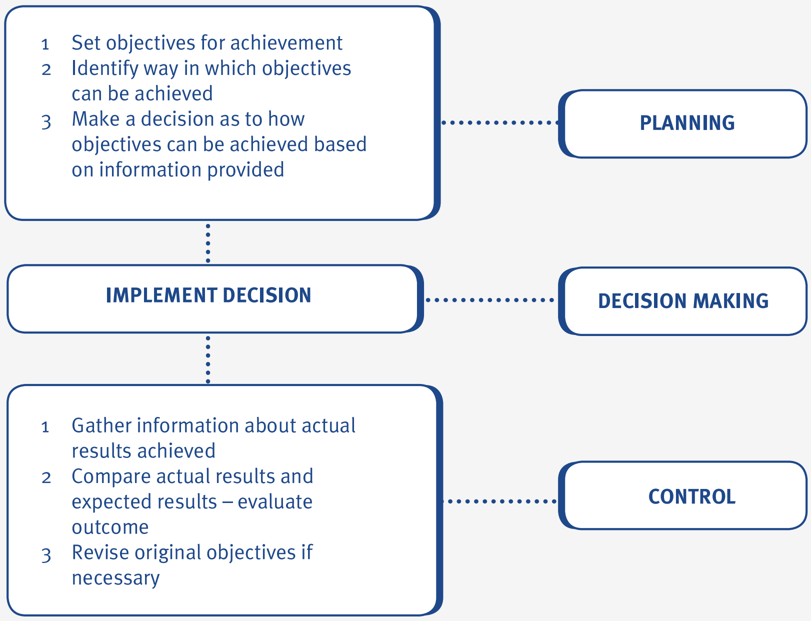
Planning

- Planning involves establishing the objectives of an organisation and formulating relevant strategies that can be used to achieve those objectives.
- Planning can be either short-term (tactical planning) or long-term (strategic planning).
- Planning is looked at in more detail in the next section of this chapter.

Decision making

Decision making involves considering information that has been provided and making an informed decision.

- In most situations, decision making involves making a choice between two or more alternatives.
- The first part of the decision-making process is planning.
- The second part of the decision-making process is control.

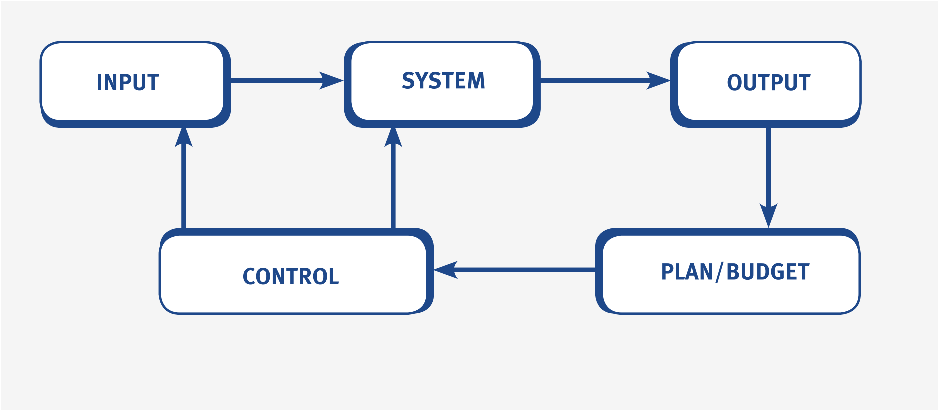


Control

Control is the second part of the decision-making process. Information relating to the actual results of an organisation is reported to managers.

- Managers use the information relating to actual results to take control measures and to re-assess and amend their original budgets or plans.
- Internally-sourced information, produced largely for control purposes, is called feedback.
- The 'feedback loop' is demonstrated in the following illustration.

Illustration 1 - The managerial processes of planning, decision



Here, management prepare a plan, which is put into action by the managers with control over the input resources (labour, money, materials, equipment and so on). Output from operations is measured and reported ('fed back') to management, and actual results are compared against the plan in control reports. Managers take corrective action where appropriate, especially in the case of exceptionally bad or good performance. Feedback can also be used to revise plans or prepare the plan for the next period.

Test your understanding 1

	Planning	Control	Decision making
Preparation of the annual budget for a cost centre			
Revise budgets for next period for a cost centre			
Implement decisions based on information provided			
Set organisation's objectives for next period			
Compare actual and expected results for a period			

Required:

Complete the table shown above, identifying each function as either planning, decision making or control.

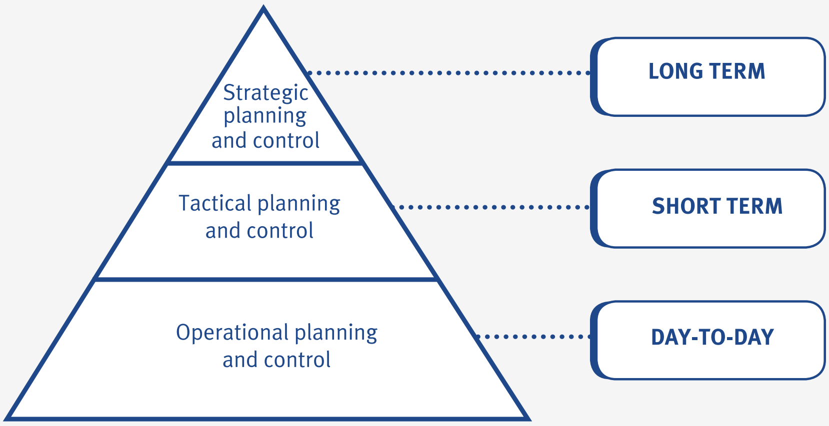
Expandable text

3 Strategic, tactical and operational planning

Levels of planning

There are three different levels of planning (known as 'planning horizons').These three levels differ according to their time span and the seniority of the manager responsible for the tasks involved.

- Strategic planning – senior managers formulate long-term objectives (goals) and plans (strategies) for an organisation.
- Tactical planning – senior managers make short-term plans for the next year.
- Operational planning – all managers (including junior managers) are involved in making day-to-day decisions about what to do next and how to deal with problems as they arise.
- A simple hierarchy of management tasks can be presented as follows.



Expandable text

4 Cost centres, profit centres, investment centres and revenue centres

Responsibility accounting

Responsibility accounting is based on identifying individual parts of a business which are the responsibility of a single manager.

A **responsibility centre** is an individual part of a business whose manager has personal responsibility for its performance. The main responsibility centres are:

- cost centre
- profit centre
- investment centre
- revenue centre.

Cost centres

A **cost centre** is a production or service location, function, activity or item of equipment whose costs are identified and recorded.

- For a paint manufacturer cost centres might be: mixing department; packaging department; administration; or selling and marketing departments.
- For an accountancy firm, the cost centres might be: audit; taxation; accountancy; word processing; administration; canteen. Alternatively, they might be the various geographical locations, e.g. the London office, the Cardiff office, the Plymouth office.
- Cost centre managers need to have information about costs that are incurred and charged to their cost centres.
- The performance of a cost centre manager is judged on the extent to which cost targets have been achieved.

Profit centres

A **profit centre** is a part of the business for which both the costs incurred and the revenues earned are identified.

- Profit centres are often found in large organisations with a divisionalised structure, and each division is treated as a profit centre.
- Within each profit centre, there could be several costs centres and revenue centres.
- The performance of a profit centre manager is measured in terms of the profit made by the centre.
- The manager must therefore be responsible for both costs and revenues and in a position to plan and control both.
- Data and information relating to both costs and revenues must be collected and allocated to the relevant profit centres.

Expandable text

Investment centres

Managers of investment centres are responsible for investment decisions as well as decisions affecting costs and revenues.

- Investment centre managers are therefore accountable for the performance of capital employed as well as profits (costs and revenues).
- The performance of investment centres is measured in terms of the profit earned relative to the capital invested (employed). This is known as the return on capital employed (ROCE).
- ROCE = Profit/Capital employed.

Revenue centres

A **revenue centre** is a part of the organisation that earns sales revenue. It is similar to a cost centre, but only accountable for revenues, and not costs.

- Revenue centres are generally associated with selling activities, for example, a regional sales managers may have responsibility for the regional sales revenues generated.
- Each regional manager would probably have sales targets to reach and would be held responsible for reaching these targets.
- Sales revenues earned must be able to be traced back to individual (regional) revenue centres so that the performance of individual revenue centre managers can be assessed.

5 Management accounting and management information

Financial accounting

Financial accounting involves recording the financial transactions of an organisation and summarising them in periodic financial statements for external users who wish to analyse and interpret the financial position of the organisation.

- The main duties of the financial accountant include: maintaining the bookkeeping system of the nominal ledger, payables control account, receivables control account and so on and to prepare financial statements as required by law and accounting standards.
- Information produced by the financial accounting system is usually insufficient for the needs of management. Managers usually want to know about: the costs of individual products and services and the profits made by individual products and services
- In order to obtain this information, details are needed for each cost, profit, investment and revenue centre. Such information is provided by cost accounting and management accounting systems.

Expandable text

Cost accounting

Cost accounting is a system for recording data and producing information about costs for the products produced by an organisation and/or the services it provides. It is also used to establish costs for particular activities or responsibility centres.

- Cost accounting involves a careful evaluation of the resources used within the enterprise.
- The techniques employed in cost accounting are designed to provide financial information about the performance of the enterprise and possibly the direction that future operations should take.
- The terms 'cost accounting' and 'management accounting' are often used to mean the same thing.

Management accounting

Management accounting has cost accounting at its essential foundation.

The main differences between management accounting and cost accounting are as follows:

- Cost accounting is mainly concerned with establishing the historical cost of a product/service.
- Management accounting is concerned with historical information but it is also forward-looking. It is concerned with both historical and future costs of products/services. (For example, budgets and forecasts).
- Management accounting is also concerned with providing non-financial information to managers.
- Management accounting is essentially concerned with offering advice to management based upon information collected (management information).
- Management accounting may include involvement in planning, decision making and control.

Expandable text

Differences between management accounting and financial accounting

The following illustration compares management accounting with financial accounting.

e.g

Illustration 2– Management accounting and management

	Management accounting	**Financial accounting**
Information mainly produced for	Internal use: e.g. managers and employees	External use: e.g. shareholders, creditors, lenders, banks, government.
Purpose of information	To aid planning, controlling and decision making	To record the financial performance in a period and the financial position at the end of that period.
Legal requirements	None	Limited companies must produce financial accounts.
Formats	Management decide on the information they require and the most useful way of presenting it	Format and content of financial accounts intending to give a true and fair view should follow accounting standards and company law.
Nature of information	Financial and non-financial.	Mostly financial.
Time period	Historical and forward-looking.	Mainly an historical record.

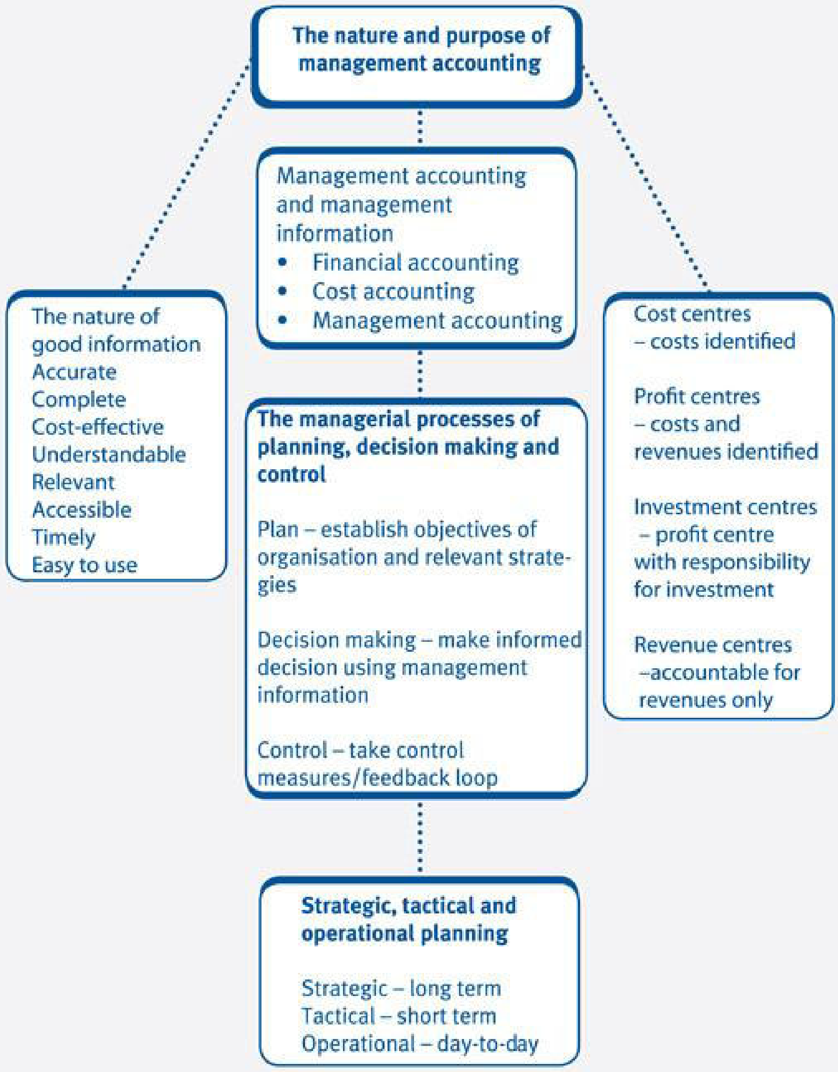
The role of management accounting within an organisation's management information system

The management information system of an organisation is likely to be able to prepare the following:

- annual statutory accounts
- budgets and forecasts
- product profitability reports
- cash flow reports
- capital investment appraisal reports
- standard cost and variance analysis reports
- returns to government departments, e.g. Sales Tax returns.

Management information is generally supplied to management in the form of reports. Reports may be routine reports prepared on a regular basis (e.g. monthly) or they may be prepared for a special purpose (e.g. ad hoc report).

6 Chapter summary



Test your understanding answers

Test your understanding 1

	Planning	Control	Decision making
Preparation of the annual budget for a cost centre	√		√
Revise budgets for next period for a cost centre		√	√
Implement decisions based on information provided			√
Set organisation's objectives for next period	√		√
Compare actual and expected results for a period		√	√

Note that all planning and control functions are part of the decisionmaking process and are therefore identified as being both. The only exception is 'implement decisions based on information provided' which is not part of planning and control, but the one decisionmaking task that there is.

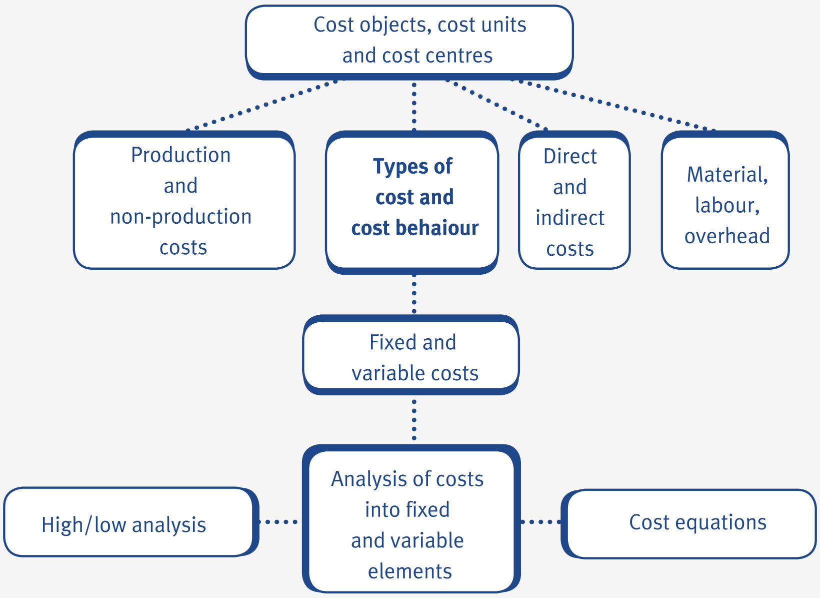
chapter

2

Types of cost and cost behaviour

Chapter learning objectives

- Explain for a manufacturing business, the distinction between production and non-production costs
- Describe, for a manufacturing business, the different elements of production cost - materials, labour and overheads
- Describe, for a manufacturing business, the different elements of non-production cost - administrative, selling, distribution and finance
- Explain the importance of the distinction between production and non-production costs when valuing output and inventories for a business
- Distinguish between, and give examples of, direct and indirect costs in manufacturing and non-manufacturing organisations
- Explain and illustrate the concepts of cost objects, cost units and cost centres for organisations in general
- Describe, using graphs, the following types of cost behaviour and give examples of each: fixed costs; variable costs; stepped fixed costs; and semi-variable costs
- Use high/low analysis to separate the fixed andvariable elements of total costs including situations involving stepped fixed costs and changes in the variable cost per unit
- Explain the structure of linear functions and equations (of the form y = a + bx)



1 Production and non-production costs

Classifying costs

Costs can be classified in a number of different ways.

- Element - costs are classified as materials, labour or expenses (overheads).
- Nature - costs are classified as being direct or indirect.
- Behaviour - costs are classified as being fixed, variable,semi-variable or stepped fixed.
- Function - costs are classified as being production ornon-production costs.

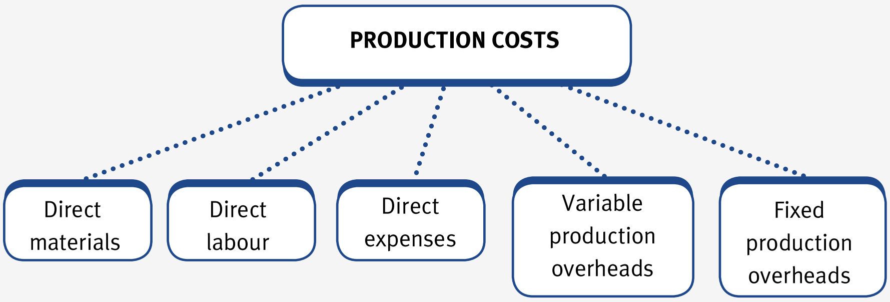
Classification by element

The main cost elements that you need to know about are materials, labour and expenses.

- Materials - all costs of materials purchased for production ornon-production activities. For example, raw materials, components, cleaning materials, maintenance materials and stationery.
- Labour costs - all staff costs relating to employees on the payroll of the organisation.
- Expenses - all other costs which are not materials or labour. This includes all bought-in services, for example, rent, telephone,sub-contractors and costs such as the depreciation of equipment.

Classification by function - production costs

Production costs are the costs which are incurred when raw materials are converted into finished goods and part-finished goods (work in progress).



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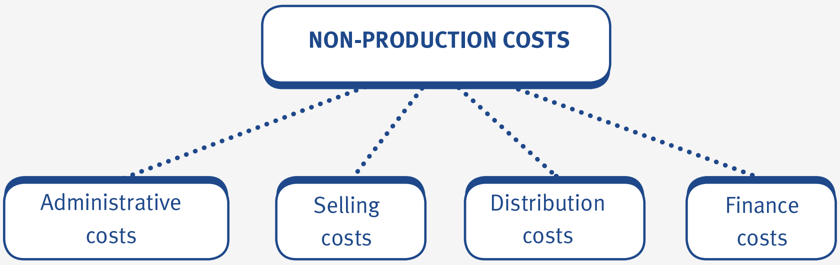
Classification by function - non-production costs

Non-production costs are costs that are not directly associated with the production processes in a manufacturing organisation.



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Distinguishing between production and non-production costs

Once costs have been analysed as being production or non-production costs, management may wish to collect the costs together on a cost card. A cost card (or unit cost card) lists out all of the costs involved in making one unit of a product.

COST CARD - statement of the total cost of one unit of a product

	$
Direct materials	X
Direct labour	X
Direct expenses	X
	—
PRIME COST	XX
Variable production overheads	X
	—
MARGINAL PRODUCTION COST	XX
Fixed production overheads	X
	—
TOTAL PRODUCTION COST	XX
Non-production overheads:	
- Administration	X
- Selling	X
- Distribution	X
	—
TOTAL COST	XX
Profit	X
	—
Sales price	XXX

Expandable text

Test your understanding 1

Cost	**Classification**
Overalls for machine workers	
Cost of printer cartridges in general office	
Salary of factory supervisor	
Salary of payroll supervisor	
Rent of warehouse for storing goods ready for sale	
Loan interest	
Salary of factory security guard	
Early settlement discounts for customers who pay early	
Salary of the Chairman's PA	
Road tax licence for delivery vehicles	
Bank overdraft fee	
Salesmens' commissions	

Complete the following table by classifying each expense correctly.

Classifications

(1) = Production

(2) = Selling

(3) = Distribution

(4) = Administrative

(5) = Finance

2 Direct and indirect costs

Direct costs

Direct costs are costs which can be directly identified with a specific cost unit or cost centre. There are three main types of direct cost:

- direct materials - for example, cloth for making shirts
- direct labour - for example, the wages of the workers stitching the cloth to make the shirts
- direct expenses - for example, the cost of maintaining the sewing machine used to make the shirts.

The total of direct costs is known as the prime cost.

Indirect costs

Indirect costs are costs which cannot be directly identified with a specific cost unit or cost centre. Examples of indirect costs include the following:

- indirect materials - these include materials that cannot be traced to an individual shirt, for example, cotton
- indirect labour - for example, the cost of a supervisor who supervises the shirtmakers
- indirect expenses - for example, the cost of renting the factory where the shirts are manufactured.

The total of indirect costs is known as overheads.

Test your understanding 2

Identify whether the following costs are materials, labour or expenses and whether they are direct or indirect.

Cost	Materials, labour or expense	Direct or indirect?
The hire of tools or equipment		
Rent of a factory		
Packing materials, e.g. cartons and boxes		
Supervisors' salaries		
Oil for lubricating machines		
Wages of factory workers involved in production		
Depreciation of equipment		

Test your understanding 3

(a) Which of the following would be classed as indirect labour?

A Assembly workers

B A stores assistant in a factory store

C Plasterers in a building company

D An audit clerk in an accountancy firm

(b) Direct costs are:

A costs which can be identified with a cost centre but not a single cost unit

B costs which can be identified with a single cost unit

C costs which can be attributed to an accounting period

D none of the above.

3 Fixed and variable costs

Cost behaviour

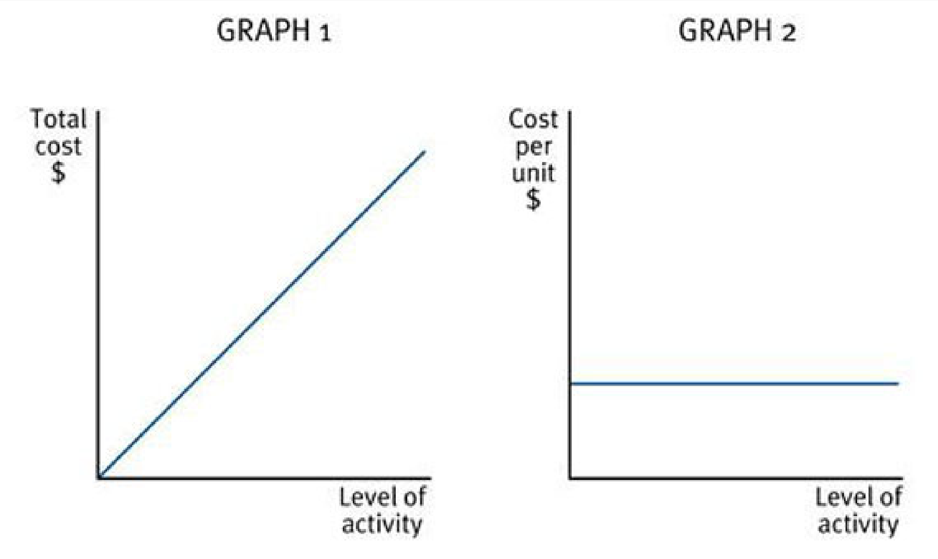
Costs may be classified according to the way that they behave. Cost behaviour is the way in which input costs vary with different levels of activity. Cost behaviour tends to classify costs as one of the following:

- variable cost
- fixed cost
- stepped fixed cost
- semi-variable cost.

Variable costs

Variable costs are costs that tend to vary in total with the level of activity. As activity levels increase then total variable costs will also increase.

- Variable costs can be shown graphically as follows:

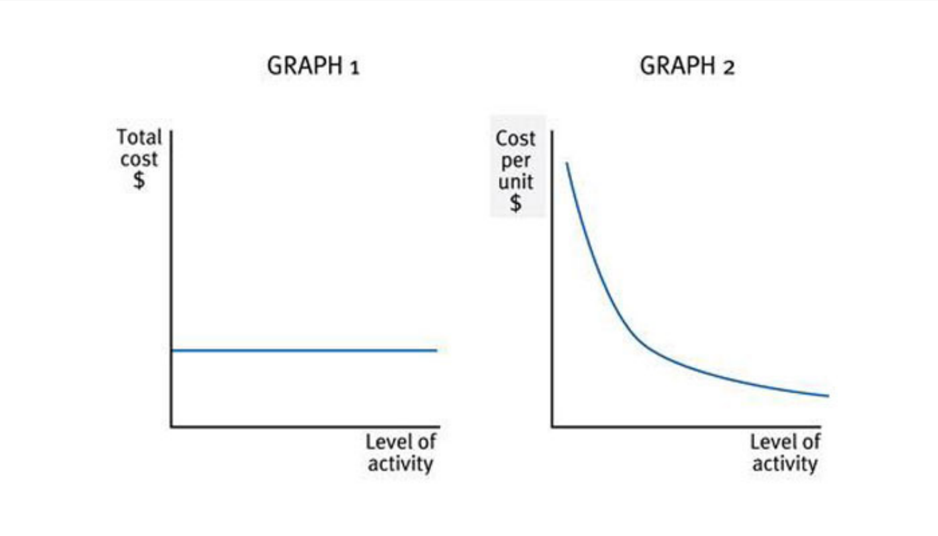


- Note that as total costs increase with activity levels, the cost per unit of variable costs remains constant.
- Examples of variable costs include direct costs such as rawmaterials and direct labour.

Expandable text

Fixed costs

A fixed cost is a cost which is incurred for an accounting period, and which, within certain activity levels remains constant.
Fixed costs can be shown graphically as follows:



- Note that the total cost remains constant over a given level of activity but that the cost per unit falls as the level of activity increases.

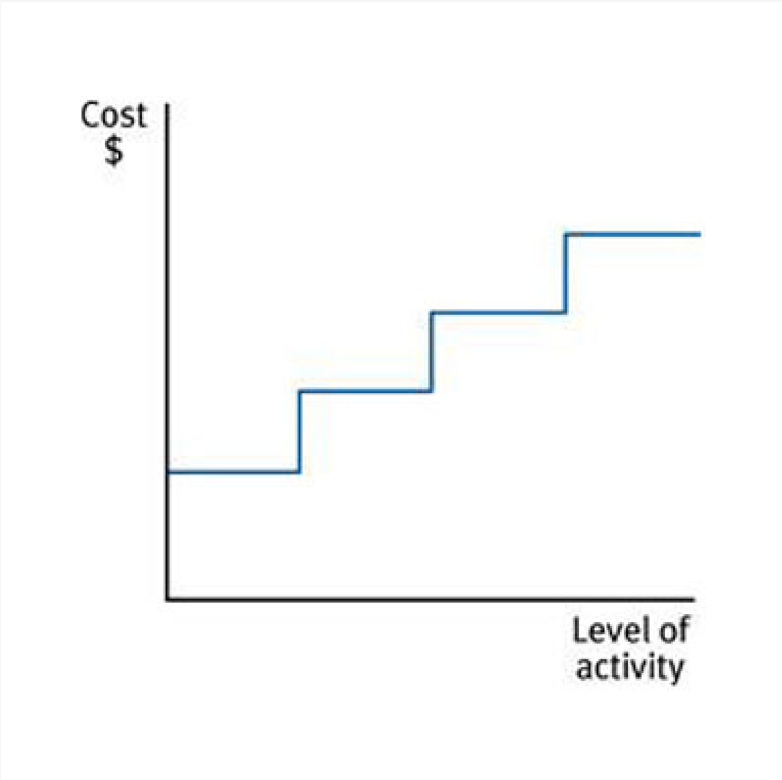
- Examples of fixed costs:
  - rent
  - business rates
  - executive salaries.

Expandable text

Stepped fixed costs

This is a type of fixed cost that is only fixed within certain levels of activity. Once the upper limit of an activity level is reached then a new higher level of fixed cost becomes relevant.

- Stepped fixed costs can be shown graphically as follows:



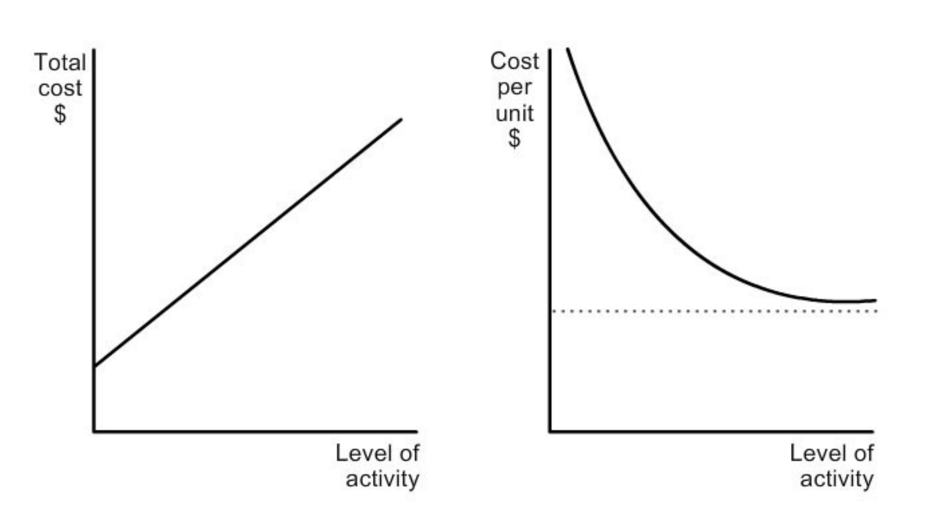
- Examples of stepped fixed costs:
  - warehousing costs (as more space is required, more warehouses must be purchased or rented)
  - supervisors' wages (as the number of employees increases, more supervisors are required).

Expandable text

Semi-variable costs

Semi-variable costs contain both fixed and variable cost elements and are therefore partly affected by fluctuations in the level of activity.

- Semi-variable costs can be shown graphically as follows:



- Examples of semi-variable costs:
  - electricity bills (fixed standing charge plus variable cost per unit of electricity consumed)
  - telephone bills (fixed line rental plus variable cost per call)

Test your understanding 4

Classify the following items of expenditure according to their behaviour i.e. as fixed, variable, semi-variable or stepped fixed costs.

(1) Monthly rent

(2) Council tax charge

(3) Bank loan interest

(4) Petrol

(5) Electricity bill

(6) Telephone bill

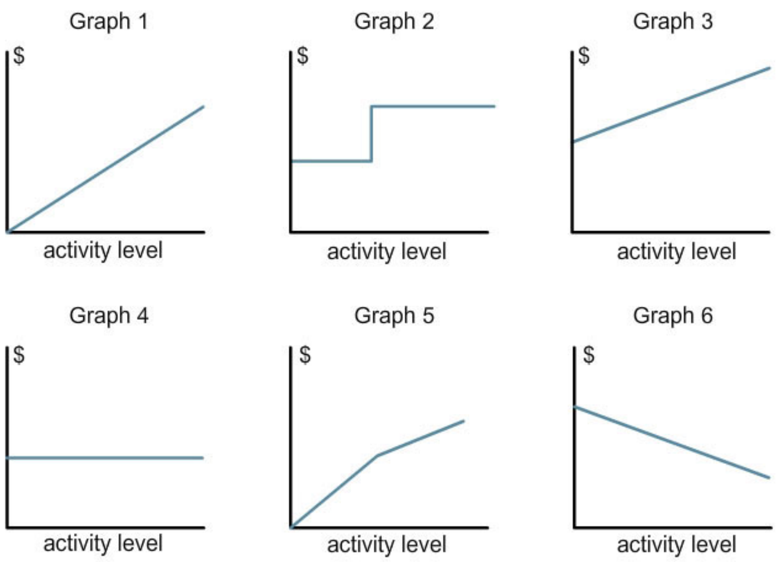
(7) Annual salary

(8) Depreciation of one, two and three factory machines

(9) Raw materials

Test your understanding 5

Study the following graphs and then answer the questions shown below.



A fixed cost is shown in graph

A variable cost is shown in graph

A semi-variable cost is shown in graph

Fixed cost per unit is shown in graph

Variable cost per unit is shown in graph

A stepped fixed cost is shown in graph

4 Analysis of costs into fixed and variable elements

Cost estimation

A number of methods exist for analysing semi-variable costs into their fixed and variable elements. The two main methods are:

- high/low method
- least squares regression - (we will look at this in the next chapter).

High/low method

- Step 1 - select the highest and lowest activity levels, and their associated costs. (Note: do not take the highest and lowest costs)
- Step 2 - find the variable cost per unit

$$\text{Variable cost per unit} = \frac{\text{Cost at high level of activity} - \text{cost at low level of activity}}{\text{High level of activity} - \text{low level of activity}}$$

- Step 3 - find the fixed cost by substitution, using either the high or low activity level.
- Fixed cost = Total cost at activity level - Total variable cost

Expandable text

Illustration 1 - Analysis of costs into fixed and variable elements

Output (Units)	Total cost ($)
200	7,000
300	8,000
400	9,000

Required:

(a) Find the variable cost per unit.

(b) Find the total fixed cost.

(c) Estimate the total cost if output is 350 units.

(d) Estimate the total cost if output is 600 units.

Expandable text

Test your understanding 6

The total costs incurred at various output levels in a factory have been measured as follows:

Output (units)	Total cost ($)
26	6,566
30	6,510
33	6,800
44	6,985
48	7,380
50	7,310

Required:

Using the high/low method, analyse the total cost into fixed and variable components.

High/low method with stepped fixed costs

Sometimes fixed costs are only fixed within certain levels of activity and increase in steps as activity increases (i.e. they are stepped fixed costs).

- The high/low method can still be used to estimate fixed andvariable costs.

- Adjustments need to be made for the fixed costs based on the activity level under consideration.

Illustration 2 - Analysis of costs into fixed and variable elements

An organisation has the following total costs at three activity levels:

Activity level (units)	4,000	6,000	7,500
Total cost	$40,800	$50,000	$54,800

Variable cost per unit is constant within this activity range and there is a step up of 10% in the total fixed costs when the activity level exceeds 5,500 units.

What is the total cost at an activity level of 5,000 units?

A $44,000

B $44,800

C $45,400

D $46,800

Expandable text

High/low method with changes in the variable cost per unit

Sometimes there may be changes in the variable cost per unit, and the high/low method can still be used to determine the fixed and variable elements of semi-variable costs. The variable cost per unit may change because:

- prices may be forecast to increase in future periods

Illustration 3 - Analysis of costs into fixed and variable elements

The following information relates to the manufacture of Product LL in 20X8:

Output (Units)	Total cost ($)
200	7,000
300	8,000
400	8,600

For output volumes above 350 units the variable cost per unit falls by 10%. (Note: this fall applies to all units - not just the excess above 350).

Required:

Estimate the cost of producing 450 units of Product LL in 20X9.

Expandable text

Test your understanding 7

The total costs incurred in 20X3 at various output levels in a factory have been measured as follows:

Output (units)	Total cost ($)
26	6,566
30	6,510
33	6,800
44	6,985
48	7,380
50	7,310

When output is 80 units or more, another factory unit must be rented and fixed costs therefore increase by 100%.

Variable cost per unit is forecast to rise by 10% in 20X4.

Required:

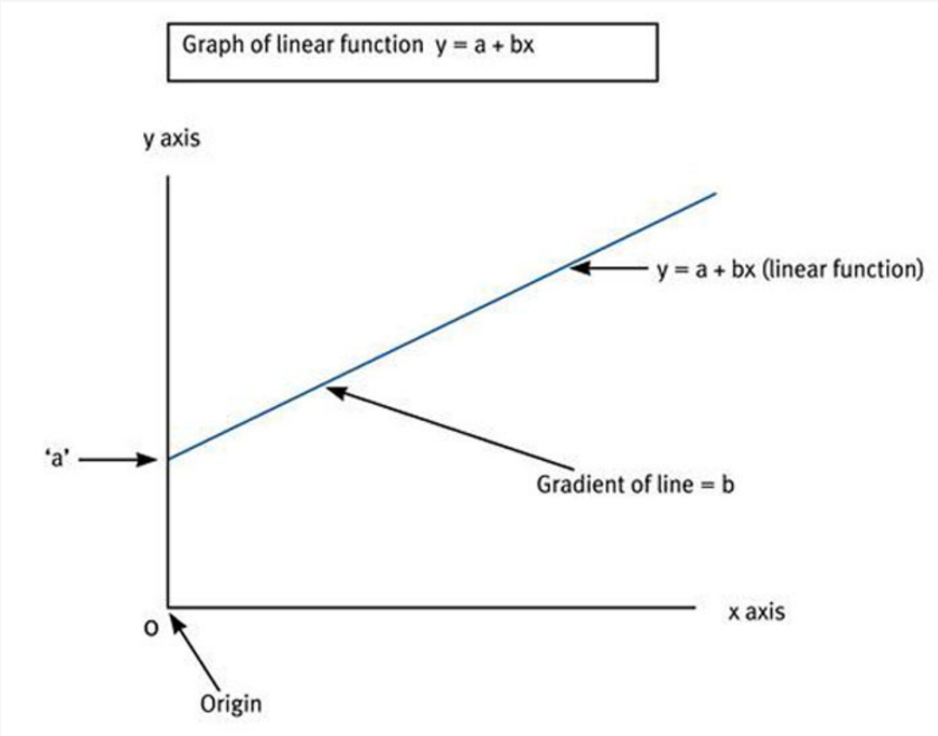
Calculate the estimated total costs of producing 100 units in 20X4.

Expandable text

5 Cost equations

Equation of a straight line

The equation of a straight line is a linear function and is represented by the following equation:



- $y = a + bx$
- 'a' is the intercept, i.e. the point at which the line $y = a + bx$ cuts the y axis (the value of y when $x = 0$).
- 'b' is the gradient/slope of the line $y = a + bx$ (the change in y when x increases by one unit).
- 'x' = independent variable .
- 'y' = dependent variable (its value depends on the value of 'x').

Cost equations

Cost equations are derived from historical cost data. Once a cost equation has been established, like the high/low method, it can be used to estimate future costs. Cost equations have the same formula as linear functions:

- 'a' is the fixed cost per period (the intercept)
- 'b' is the variable cost per unit (the gradient)
- 'x' is the activity level (the independent variable)

- 'y' is the total cost = fixed cost + variable cost (dependent on the activity level)

Suppose a cost has a cost equation of y = $5,000 + 10x, this can be shown graphically as follows:

Graph of cost equation y = 5,000 + 10x

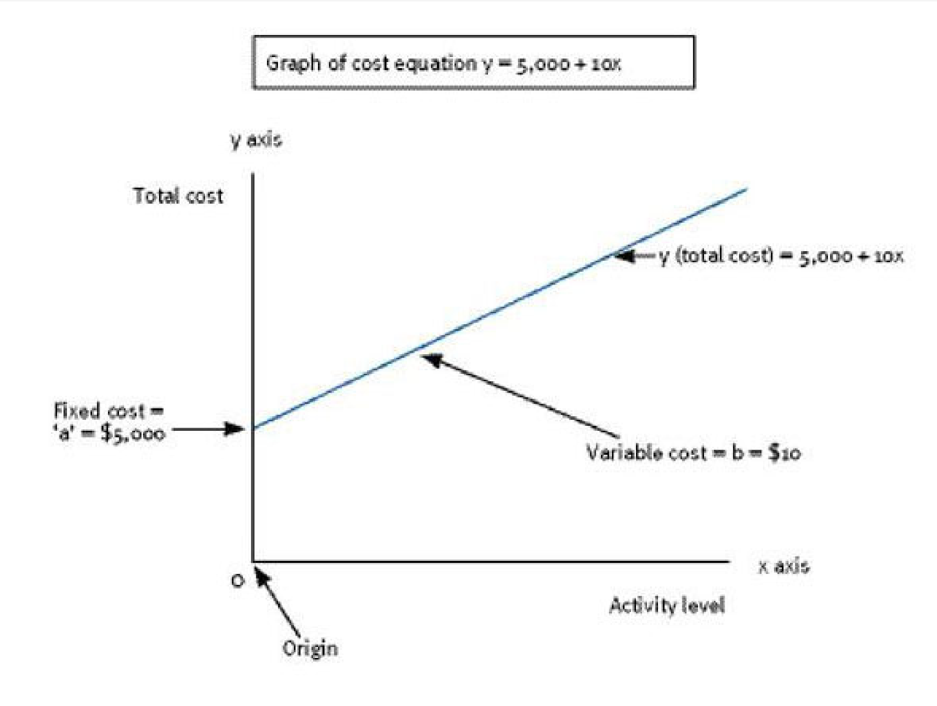


Illustration 4 - Cost equations

If y = 8,000 + 40x

(a) Fixed cost = $

(b) Variable cost per unit = $

(c) Total cost for 200 units = $

Expandable text

Test your understanding 8

Consider the linear function y = 1,488 + 20x and answer the following questions.

(a) The line would cross the y axis at the point

(b) The gradient of the line is

(c) The independent variable is

(d) The dependent variable is

Test your understanding 9

If the total cost of a product is given as:

Y = 4,800 + 8x

(a) The fixed cost is $

(b) The variable cost per unit is $

(b) The total cost of producing 100 units is $

6 Cost objects, cost units and cost centres

Analysing costs

Management will require a variety of different cost summaries, including:

- costs for a particular product - cost unit or cost object
- costs for use in the preparation of external financial reports
- costs for a particular department - cost centre
- costs that may be useful for decision making
- costs that are useful for planning and control.

Cost objects

A cost object is any activity for which a separate measurement of cost is undertaken.

Examples of cost objects:

- cost of a product
- cost of a service (e.g. insurance policy)
- cost of running a department
- cost of running a regional office.

Cost units

A cost unit is a unit of product or service in relation to which costs are ascertained.

Examples of cost units:

- a room (in a hotel)
- a litre of paint (paint manufacturers)
- in-patient (in a hospital).

Cost centres

A cost centre is a production or service location, function, activity or item of equipment for which costs can be ascertained.

Examples of cost centres:

- a department
- a machine
- a project
- a ward (in a hospital).

Cost cards

The following costs are brought together and recorded on a cost card:

- direct materials
- direct labour
- direct expenses
- prime cost (total direct costs)
- variable production overheads
- fixed production overheads
- non-production overheads.

e.g

Illustration 5 - Cost objects, cost units and cost centres

A cost card for a hand-made wooden train set is shown below.

- The cutting and assembly department and the painting department are cost centres.
- One hand-made wooden train set is a cost unit (but may also be classed as a cost object).

Cost card for hand-made wooden train set

		$
Direct materials		
Wood	5m2 @ $2.50 per m2	12.50
Paint	0.1 litre at $10 per litre	1.00
Direct labour		
Cutting and assembly department	0.5 hours at $6.00 per hour	3.00
Painting department	1.0 hours @ $7.00 per hour	7.00
Direct expenses	Licence fee @ $2 per train set	2.00
PRIME COST		25.50
Variable production overheads		
Power for electric saws	0.25 hours @ $2.00 per hour	0.50
MARGINAL PRODUCTION COST		26.00
Fixed production overheads	1.5 labour hours @ $10.00 per labour hour	15.00
TOTAL PRODUCTION COST		41.00

Non-production overheads:		
Administration, selling and distribution	20% of total production cost	8.20
		–––––
TOTAL COST		49.20
Profit	30% of total cost	14.76
		–––––
		63.96

7 Chapter summary

Cost object, cost unit and cost centre

Cost object – activity for which costs can be separately measured

Cost unit – unit of product or service in relation to which costs are ascertained

Cost centre – production or service location, function, activity or item of equipment for which costs can be ascertained

Production and non-production costs

Production costs = prime cost (total direct costs) plus overheads

Non-production costs = admin, selling, distribution and finance

Types of cost behaviour

Direct and indirect costs

Direct costs – directly involved in production

Indirect costs – not directly involved in production

Material, labour, overhead

Fixed and variable costs

Fixed – do not vary with level of activity

Variable – vary with level of activity

Analysis of semi-variable costs into fixed and variable elements

Cost equations
(linear function) y =a + bx

High/low analysis

$$\textbf{Variable cost per unit} = \frac{\text{Cost at high level of activity} - \text{Cost at low level of activity}}{\text{High level of activity} - \text{Low level of activity}}$$

Fixed cost = Total cost at activity level – Total variable cost

Test your understanding answers

Test your understanding 1

Cost	Classification
Overalls for machine workers	1
Cost of printer cartridges in general office	4
Salary of factory supervisor	1
Salary of payroll supervisor	4
Rent of warehouse for storing goods ready for sale	3
Loan interest	5
Salary of factory security guard	1
Early settlement discounts for customers who pay early	2
Salary of the Chairman's PA	4
Road tax licence for delivery vehicles	3
Bank overdraft fee	5
Salesmens' commissions	2

Test your understanding 2

Cost	**Materials, labour or expense**	**Direct or indirect?**
The hire of tools or equipment	Expense	Direct
Rent of a factory	Expense	Indirect
Packing materials, e.g. cartons and boxes	Material	Direct
Supervisors' salaries	Labour	Indirect
Oil for lubricating machines	Material	Indirect
Wages of factory workers involved in production	Labour	Direct
Depreciation of equipment	Expense	Indirect

Test your understanding 3

(a) B Store assistants are not directly involved in producing the output (goods or services) of an organisation.

(b) B This is a basic definition question. Direct costs are costs which can be identified with a single cost unit, or cost centre.

Test your understanding 4

The items of expenditure would be analysed as follows.

(1) Fixed

(2) Fixed

(3) Fixed

(4) Variable

(5) Semi-variable

(6) Semi-variable

(7) Fixed

(8) Stepped fixed

(9) Variable

Note that the depreciation charge for the factory machines (8) is a stepped fixed cost because as activity increases to such a level that a second and third machine are required, the fixed cost will double and then treble.

Test your understanding 5

A fixed cost is shown in graph	4
A variable cost is shown in graph	1
A semi-variable cost is shown in graph	3
Fixed cost per unit is shown in graph	6
Variable cost per unit is shown in graph	4
A stepped fixed cost is shown in graph	2

Test your understanding 6

$$\text{Variable cost per unit} = \frac{\$7,310 - \$6,566}{50 - 26} = \frac{\$744}{24} = \$31 \text{ per unit}$$

Substituting at high activity level:

Total cost	=	$7,310
Total variable cost	= 50 x $31	$1,550
Therefore Fixed cost	=	$5,760

Test your understanding 7

$$\text{Variable cost per unit (20X3)} = \frac{\$7,310 - \$6,566}{50 - 26} = \frac{\$744}{24} = \$31 \text{ per unit}$$

Substituting at high activity level:

Total cost	=	$7,310
Total variable cost	= 50 x $31	$1,550
Therefore Fixed cost (in 20X3)	=	$5,760

Estimated total costs of producing 100 units in 20X4:

Variable cost	= 100 x $31 x 1.1	$3,410
Fixed cost	= $5,760 x 2	$11,520
Total cost	=	$14,930

Test your understanding 8

(a) The line would cross the y axis at the point	1,448
(b) The gradient of the line is	20
(c) The independent variable is	x
(d) The dependent variable is	y

Test your understanding 9

(a) The fixed cost is $	4,800
(b) The variable cost per unit is $	8
(b) The total cost of producing 100 units is $	5,600

Working

Fixed cost = $4,800

Variable cost = 100 x $8 = $800

Total cost = fixed cost + variable cost = $4,800 + $800 = $5,600

chapter

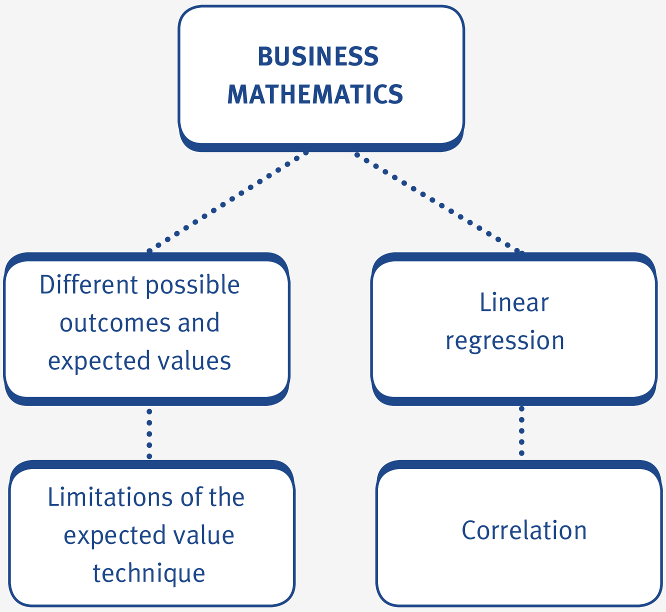
3

Business mathematics

Chapter learning objectives

Upon completion of this chapter you will be able to:

- explain, calculate and demonstrate the use of an expected value in simple decision-making situations
- explain the limitations of the expected value technique, excluding the use of decision trees and conditional profit tables
- establish a linear function using regression analysis and interpret the results including areas where care is needed (extrapolation and the relevant range)
- calculate a correlation coefficient and a coefficient of determination, interpreting and explaining r and r^2.



1 Different possible outcomes and expected values

Expected values

An expected value is a long run average. It is the weighted average of a probability distribution.

- The formula for an expected value is as follows.

  Expected value (EV) = $\sum px$

  Where:

  $\sum$ = 'sum of'

  p = probability of the outcome occurring

  x = the outcome

- Expected values are used in simple decision-making situations in order to decide the best course of action for a company.
- Probabilities are based on an analysis of past data. It is assumed that the past is a good indicator of the future.
- When using expected values to assess projects, accept projects only if the EV is positive. If the EV is negative (i.e. an expected loss) the project should be rejected.

- If deciding between projects which all have positive EVs then the project with the highest expected value should be chosen.

Expandable text

Illustration 1 - Different possible outcomes and expected values

A company has recorded the following daily sales over the last 200 days.

Daily sales (units)	Number of days
100	40
200	60
300	80
400	20

Required:
Calculate the expected sales level in the future.

Expandable text

Test your understanding 1

A company is bidding for three contracts, which are awarded independently of each other. The Board estimates its chances of winning contract X as 50%, of winning contract Y as 1 in 3, and of winning contract Z as 1 in 5. The profits from X, Y and Z are estimated to be $40,000, $60,000 and $100,000 respectively.

The expected value to the company of the profits from all three contracts will be closest to ($000):

A 40

B 60

C 90

D 100

2 Limitations of the expected value technique

There are a number of limitations of using the expected value technique to evaluate decisions.

- Probablities used in expected value calculations are usually based on past data and are therefore likely to be estimates. There is a danger, therefore, that these estimates are likely to be unreliable because they are not accurate.
- Expected values are not always suitable for making one-off decisions. This is because expected values are long-term averages and the conditions surrounding a one-off decision may be difficult to estimate.
- Expected value calculations do not take into account the 'time value of money' and the fact that $1,000 now is worth less than $1,000 in five years' time. EV calcuations are based on the value of money 'today' only.
- Expected values only take into account some of the factors surrounding a decision, and not all of them. For example, they do not look at the decision taker's attitude to risk which will vary from person to person.

3 Linear regression analysis

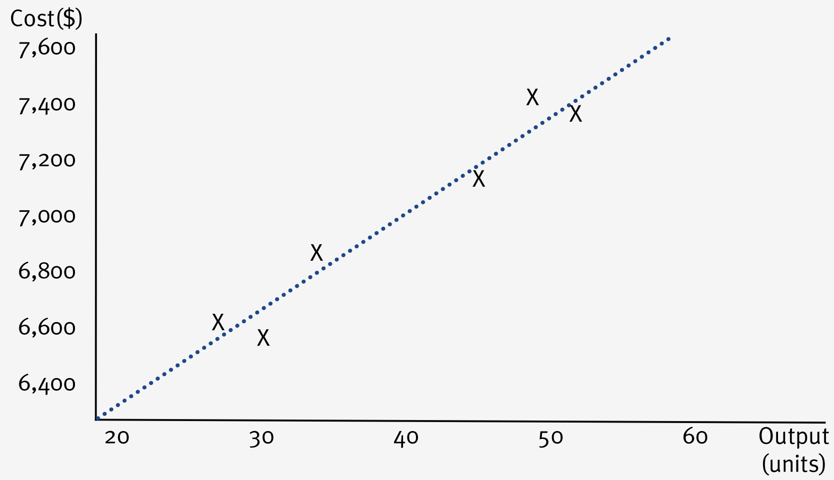
Line of best fit

Consider the following data which relates to the total costs incurred at various output levels in a factory:

Output	**Total cost**
(units)	($)
26	6,566
30	6,510
33	6,800
44	6,985
48	7,380
50	7,310

If the data shown above is plotted on a graph, it will look like this.

Scattergraph showing total costs incurred at various output levels in a factory



- A graph like this is known as a scattergraph (or scatter chart).
- A scattergraph can be used to make an estimate of fixed and variable costs by drawing a 'line of best fit' through the points which represents all of the points plotted.
- The line of best fit is a cost equation (or linear function) of the form y = a + bx where a = fixed costs and b = variable cost per unit.
- Regression analysis can be used to establish the equation of the line of best fit, and therefore, the fixed and variable costs.

Establishing a linear function

Regression analysis, like the high/low method, can be used to predict the linear relationship between two variables (the linear functiony = a + bx).

(1) Linear regression (or regression analysis) calculates the values of 'a' and 'b' from past cost data using the following formulae:

$$a = \frac{\sum y}{n} - \frac{b\sum x}{n}$$

$$b = \frac{n\sum xy - \sum x\sum y}{n\sum x^2 - (\sum x)^2}$$

(1) 'a' is the fixed cost per period (the intercept)

(2) 'b' is the variable cost per unit (the gradient)

(3) 'x' is the activity level (the independent variable)

(4) 'y' is the total cost = fixed cost + variable cost (dependent on the activity level)

(5) ∑ means 'sum of'

(6) n is the sample size

The regression formulae shown above (for calculating values for 'a' and 'b') are provided in your exam. Don't waste valuable time learning them by heart, but do make sure that you know how to use them properly!

Expandable text

Where care is needed

- Regression lines can be used with confidence to calculate intermediate values of variables, i.e. values within the known range ("interpolation").
- Generally speaking, extrapolation must be treated with caution, since once outside the relevant range (range of known values) other factors may influence the situation, and the relationship which has been approximated as linear over a limited range may not be linear outside that range.
- In addition the mathematics will produce an equation even if there is no clear 'cause and effect' relationship between the two sets of figures. The strength of this relationship is explored under 'correlation' below. Particular care must be taken when the number of readings is low.

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Illustration 2 - Linear regression analysis

Bronze recorded the following costs for the past six months.

Month	Activity level	Total cost
	Units	$
1	80	6,500
2	60	6,200
3	72	6,400
4	75	6,500
5	83	6,700
6	66	5,400

Required:

(a) Estimate the fixed costs per month and variable cost per unit using linear regression analysis.

(b) Estimate the total costs for the following activity levels in a month.

(i) 75 units

(ii) 90 units

Expandable text

Test your understanding 2

A company is investigating its current cost structure. An analysis of production levels and costs over the first six months of the year has revealed the following:

Month	**Production level (units)**	**Production cost**
	(000s)	($000)
January	9.125	240
February	10.255	278
March	9.700	256
April	10.500	258
May	11.000	290
June	11.500	300

Required:

(a) Use regression analysis to identify:

(i) Variable cost per unit.

(ii) Monthly fixed costs

(b) It is expected that in July, production will be 12,000 units. Estimate the cost of July's production and comment on the accuracy of your estimate.

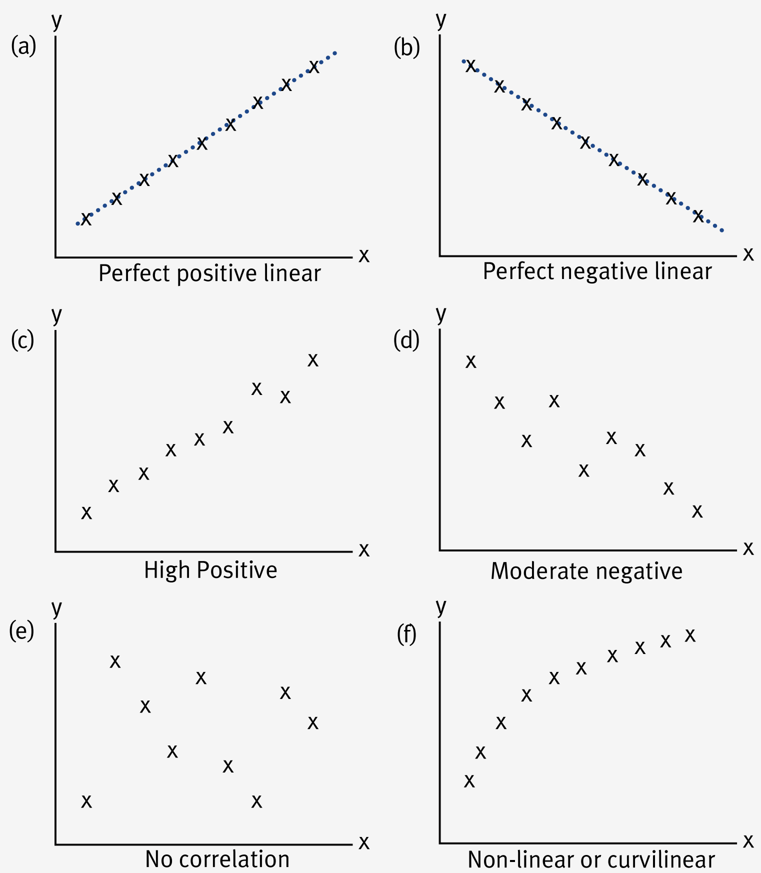
4 Correlation

Correlation

- Correlation measures the strength of the relationship between two variables. In particular, whether movements are related, e.g. does spending more on advertising necessarily result in more sales?
- One way of measuring 'how correlated' two variables are is by drawing a graph (scattergraph or scatter chart) to see if any visible relationship exists. The 'line of best fit' drawn on a scattergraph indicates a possible linear relationship.
- When correlation is strong, the estimated line of best fit should be more reliable.
- Another way of measuring 'how correlated' two variables are is to calculate a correlation coefficient, r, and to interpret the result.

Different degees of correlation

Variables may be either perfectly correlated, partially correlated or uncorrelated.The different degrees of correlation can be shown graphically on scattergraphs as follows.



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The correlation coefficient

An alternative to drawing a graph each time you want to know whether two variables are correlated, and the extent of the correlation if there is any, is to calculate the correlation coefficient (r) and to interpret the result.

- The correlation coefficient, r, measures the strength of a linear relationship between two variables. It can therefore give an indication of how reliable the estimated linear function is for a set of data.
- For example, the linear function of the scattergraph in paragraph 3.1 was estimated to be y = 5,587 + 34.77x. If we were to calculate the associated correlation coefficient we could get some idea of how reliable our estimated linear function was.
- The correlation coefficient can only take on values between –1 and +1.

  r = + 1 indicates perfect positive correlation

  r = 0 indicates no correlation
  r = – 1 indicates perfect negative correlation.
- The correlation coefficient is calculated using the following formula:

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{[(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)]}}$$

Note: This formula is given to you in the exam.

Illustration 3 - Correlation

The following table shows the number of units of a good produced and the total costs incurred.

Units produced	Total costs
	$
100	40,000
200	45,000
300	50,000
400	65,000

500	70,000
600	70,000
700	80,000

Required:
Calculate the correlation coefficient for the data given and comment on the result obtained.

Expandable text

Test your understanding 3

If $\sum x = 440$, $\sum y = 330$, $\sum x^2 = 17{,}986$, $\sum y^2 = 10{,}366$, $\sum xy = 13{,}467$ and $n = 11$, then the value of r, the coefficient of correlation, to two decimal places, is:

A 0.98

B 0.63

C 0.96

D 0.59

If you calculate a correlation coefficient with a value >+1 or <-1 go back and check your calculation again as you will have got the wrong answer. Remember - the value of the correlation coefficient can only lie in the range -1 to +1.

Coefficient of determination

The coefficient of determination is the square of the correlation coefficient, and so is denoted by r^2.

- The coefficient of determination is a measure of how much of the variation in the dependent variable is 'explained' by the variation of the independent variable.
- The variation not accounted for by variations in the independent variable will be due to random fluctuations, or to other specific factors that have not been identified in considering the two-variable problem.
- For example, if $r = 0.98$, $r^2 = 0.96$ or 96%.

- This means that 96% of the variation in the dependent variable (y) is explained by variations in the independent variable (x). This would be interpreted as high correlation.
- For example, the linear function y = 5,587 + 34.77x is plotted (as a 'line of best fit') in paragraph 3.1. The associated correlation coefficient was calculated to be 0.957, and the coefficient of determination, r^2, is therefore:

$$r^2 = 0.957^2 = 0.92 \text{ or } 92\%$$

- This means that 92% of the variations in total costs is explained by variations in the level of activity. The other 8% of variations in total costs are assumed to be due to random fluctuations.

Note: With both correlation coefficients and coefficients of determination there is the possibility that the calculated figure does not indicate true correlation. 'Spurious' correlation can occur when the behaviour of both sets of figures is due to a third factor, e.g. a strengthening currency will make demand for both imported cigars and CD players rise. The correlation between sales of cigars and CD players would be high despite no causal relationship.

5 Chapter summary

Limitations of the expected value technique

- Probabilities based on past data – estimates
- Not always suitable for one-off decisions
- Do not take into account 'time value money'
- Do not take into account all factors surrounding a decision

Different possible outcomes and expected values

An expected value is a long run average. It is the weighted average of a probability distribution.

expected value (ev) = Σpx

Business mathematics

Linear regression

- Used to predict linear relationship (linear cost function) between two variables
- Formulae to calculate 'a' (fixed costs) and 'b' (variable costs) are provided in your exam
- Linear cost function only holds for the 'relevant' range

Correlation

- Correlation measures strength of relationship between two variables
- Measured by correlation coefficient, r (formula given in exam)
- Coefficient of determination, r, measures how much the variation in the dependent variable is 'explained' by the variation in the independent variable

Test your understanding answers

Test your understanding 1

B $\quad EV = 40{,}000 \times 0.5 + 60{,}000 \times \frac{1}{3} + 100{,}000 \times \frac{1}{5} = 60{,}000$

Test your understanding 2

(a)

Month	Production level (units) x (000s)	Production cost y $000	xy	x^2
January	9.125	240	2,190	83.265
February	10.255	278	2,851	105.165
March	9.700	256	2,483	94.090
April	10.500	258	2,709	110.250
May	11.000	290	3,190	121.000
June	11.500	300	3,450	132.250
Totals	62.080	1,622	16,873	646.021

(i) Variable cost = b

$$= \frac{6 \times 16{,}873 - 62.080 \times 1{,}622}{6 \times 646.020 - 62.080^2}$$

$$= \frac{544{,}780}{22{,}197} = \$24.54$$

(ii) Fixed cost = a

$$= \frac{1{,}622}{6} - 24.54$$

(b) The estimated cost of 12,000 units will be given by the linear cost equation:

y = \$16.426 + \$24.54x

y = 16.426 + (24.54 ×12) = \$310.906

It should be noted that since this is outside the range of values for which costs are known it might be inaccurate. For example there may be stepped fixed cost such as an additional supervisor that is required at 11,800 units.

Test your understanding 3

$$r = \frac{11 \times 13{,}467 - 440 \times 330}{\sqrt{[(11 \times 17{,}986 - 440^2)(11 \times 10{,}366 - 330^2)]}} = 0.63$$

chapter

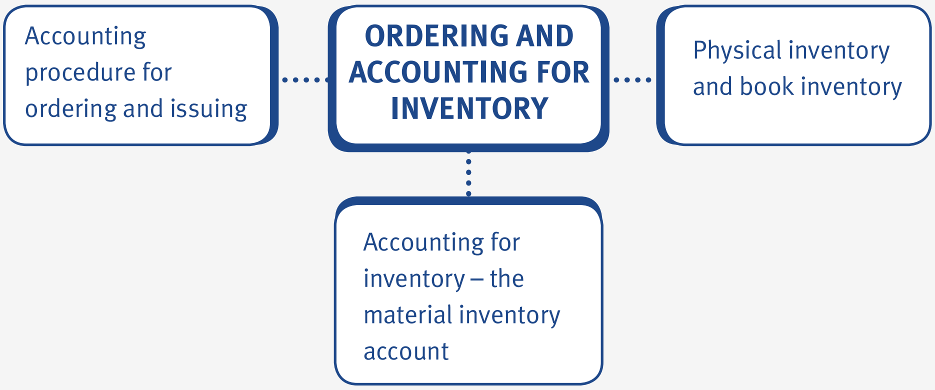
4

Ordering and accounting for inventory

Chapter learning objectives

Upon completion of this chapter you will be able to:

- describe, for a manufacturing business, the different procedures and documents necessary for the ordering, receiving and issuing of materials from inventory
- interpret the entries and balances in the material inventory account for a manufacturing business
- describe the control procedures that can be used in a manufacturing business to monitor physical and 'book' inventory and to minimise discrepancies and losses.



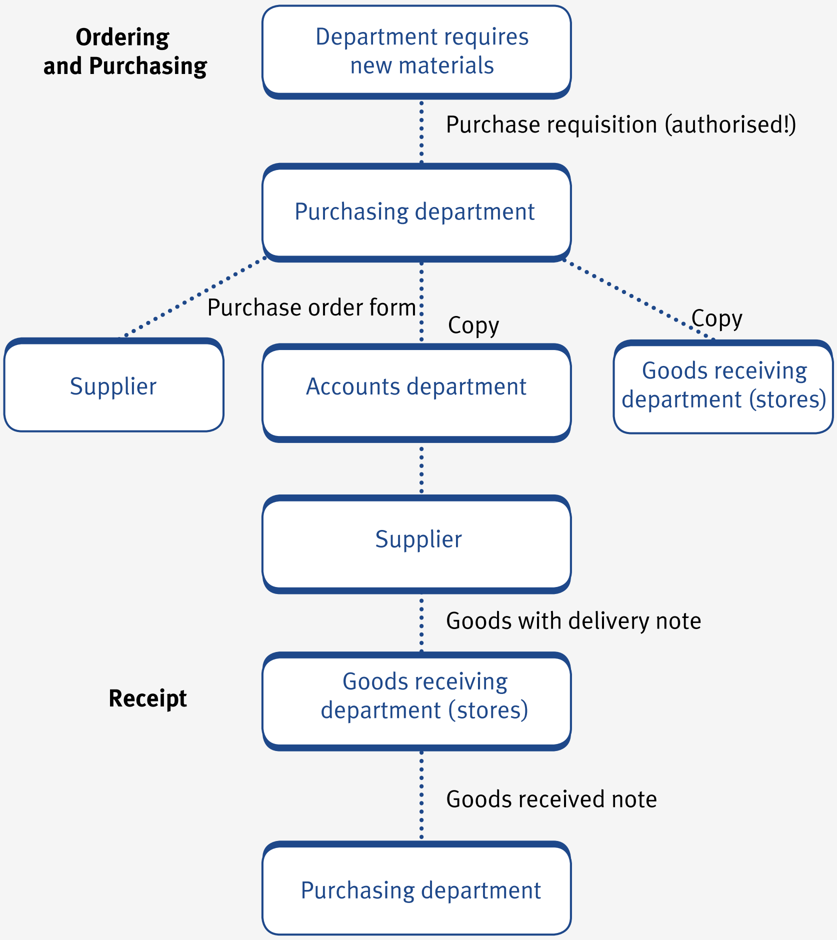
1 Accounting procedures for ordering and issuing inventory

Accounting and control procedures for ordering and issuing inventory include the following functions:

- ordering
- purchasing
- receiving
- issuing
- storing and stocktaking.

Ordering, purchasing and receiving inventory

The procedures for ordering, purchasing and receiving materials are as follows.



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Specimen forms for the ordering, purchasing and receipt of inventory are shown in the illustration below.

Issuing inventory

The accounting procedures for issuing inventory are as follows.

- **Materials requisition notes** are issued to production departments. Their purpose is to authorise the storekeeper to release the goods which have been requisitioned and to update the stores records.
- **Materials returned notes** are used to record any unused materials which are returned to stores. They are also used to update the stores records.

- **Materials transfer notes** document the transfer of materials from one production department to another. They are also used to update the stores records.

e.g

Illustration 1 - Accounting procedures for ordering and issuing

Some specimen documents that are used in the ordering, receiving and issuing of inventory are illustrated as follows.

PURCHASE REQUISITION

Date ....20............ Serial No: ....

Purpose*: inventory/special
capital equipment/(budget reference)
*Delete as appropriate

Quantity and units	Description	Material code	Job or dept. code	Delivery required		Purchase order		
				Date	Place	No.	Date	Supplier

Origination department .... Authorisation ....

PURCHASE ORDER

To: ....
....
....

Serial No: ....
Date: ....
Purchase Req. No: ....

Please supply, in accordance with the attached conditions

Quantity	Description	Code	Delivery date	Price	Per

Your quotation ....
To be delivered, carriage paid, to .... Terms
Please quote our Purchase Order number on all correspondence.

For ABC Ltd

....

GOODS RECEIVED NOTE

To: .... Serial No: ....

.... Date issued: ....

Carrier: .... Purchase Order No: ....

Date of delivery: ....

Description	Code	Quantity	Packages	Gross Weight

INSPECTION REPORT			Received by: ....
Quantity passed	Quantity rejected	Remarks	Required by: ....
			Accepted: ....
Inspector	Date		Date: ....

MATERIAL REQUISITION

Charge job/
Cost Centre No: .... Serial No: ....

Date: ....

Code No.	Description	Quantity or weight	Cost office only				
			Rate	Unit	$	$	Stores ledger

Authorised by:	Storekeeper:	Prices entered by:
Received by:	Bin card entered:	Calculations checked:

Test your understanding 1

A goods received note (GRN) provides (tick all that apply):

Information used to update inventory records.

Information to check that the correct price has been recorded on the supplier's invoice.

Information to check that the correct quantity of goods has been recorded on the supplier's invoice.

Information to record any unused materials which are returned to stores.

2 Accounting for inventory – the material inventory account

Material inventory account

Materials held in store are an asset and are recorded in the balance sheet of a company.
Accounting transactions relating to materials are recorded in the material inventory account.

Material inventory account

Debit entries reflect an **increase** in inventory	Credit entries reflect a **decrease** in inventory
• purchases	• issues to production
• returns to stores	• returns to suppliers

e.g

Illustration 2- Accounting for inventory

Material inventory account

	$000		$000
Opening balance (1)	33	Work-in-progress (4)	137
Creditors (2)	146	Materials returned to suppliers (5)	2
		Production overhead account (6)	4
Materials returned to stores (3)	4		
		Income statement (7)	3
		Closing balance (8)	37
	183		183

(1) The opening balance of materials held in inventory at the beginning of a period is shown as a debit in the material inventory account.

(2) Materials purchased on credit are debited to the material inventory account.

(3) Materials returned to stores cause inventory to increase and so are debited to the material inventory account.

(4) Direct materials used in production are transferred to the work-in-progress account by crediting the material inventory account.

(5) Materials returned to suppliers cause inventory levels to fall and are therefore 'credited out' of the material inventory account.

(6) Indirect materials are not a direct cost of manufacture and are treated as overheads. They are therefore transferred to the production overhead account by way of a credit to the material inventory account.

(7) Any material write-offs are 'credited out' of the material inventory account and transferred to the income statement where they are written off.

(8) The balancing figure on the material inventory account is the closing balance of material inventory at the end of a period. It is also the opening balance at the beginning of the next period.

Test your understanding 2

Transaction	**Debit which account?**	**Credit which account?**
Issue materials to production.		
Purchase new materials on credit.		
Materials returned to store from production.		
Materials written off.		
Indirect materials transferred to production overheads.		

Inventory valuation

- When inventory is received into stores from different suppliers and at different prices every week, it is important that it is valued in a consistent way so that closing inventory values and issues from stores can be valued accurately.
- You are probably aware that inventory is usually valued at the lower of cost and net realisable value. However, various methods are used to value closing inventory and issues from stores in management accounting.
- There are three main inventory valuation methods:
  - FIFO
  - LIFO
  - Weighted average cost.
- First In First Out – FIFO assumes that materials are issued out of stock in the order in which they were delivered into inventory.
- Last In First Out – LIFO assumes that materials are issued out of inventory in the reverse order to which they were delivered into inventory.
- Weighted average cost – AVCO values all items of inventory and issues at an average price. The average price is calculated after each receipt of goods. The average price is calculated by dividing the total purchase costs to date by the total units received to date.
- These valuation methods are outside the F2 syllabus.

3 Physical inventory and book inventory

Perpetual inventory

Perpetual inventory is the recording as they occur of receipts, issues and the resulting balances of individual items of inventory in either quantity or quantity and value.

- Inventory records are updated using stores ledger cards and bin cards.
- Bin cards also show a record of receipts, issues and balances of the quantity of an item of inventory handled by stores.
- As with the stores ledger card, bin cards will show materials received (from purchases and returns) and issued (from requisitions).
- A typical stores ledger card is shown below.

STORES LEDGER CARD

Description: .... Unit: .... Location: .... Code: ....
Maximum: .... Minimum: .... Reorder level: .... Reorder quantity: ....

Receipts			Issues			On order		
Date/ref	Quantity	$	Date/ref	Quantity	$	Date/ref	Quantity	$

Stocktaking

The process of stocktaking involves checking the physical quantity of inventory held on a certain date and then checking this balance against the balances on the stores ledger (record) cards or bin cards. Stocktaking can be carried out on a **periodic basis** or a **continuous basis**.

- **Periodic stocktaking** involves checking the balance of every item of inventory on the same date, usually at the end of an accounting period.
- **Continuous stocktaking** involves counting and valuing selected items of inventory on a rotating basis. Specialist teams count and check certain items of inventory on each day. Each item is checked at least once a year with valuable items being checked more frequently.
- Any differences (or discrepancies) which arise between ‘book’ inventory and physical inventory must be investigated.
- In theory any differences, as recorded in the stores ledger or the bin card, must have arisen through faulty recording.

- Once the discrepancy has been identified, the stores ledger card is adjusted in order that it reflects the true physical inventory count.
- Any items which are identified as being **slow-moving** or **obsolete** should be brought to the attention of management as soon as possible.
- Management will then decide whether these items should be disposed of and written off to the income statement.
- Slow-moving items are those inventory items which take a long time to be used up.
- Obsolete items are those items of inventory which have become out of date and are no longer required.

Expandable text

Control procedures to minimise discrepancies and losses

The level of investment in inventory and the labour costs of handling and recording/controlling them is considerable in many organisations. It is for this reason that organisations must have control procedures in place in order to minimise discrepancies and losses.

Problem	**Control procedure**
Ordering goods at inflated prices.	Use of standard costs for purchases. Quotation for special items.
Fictitious purchases.	Separation of ordering and purchasing. Physical controls over materials receipts, usage and inventory.
Shortages on receipts.	Checking in all goods inwards at gate. Delivery signatures.
Losses from inventory.	Regular stocktaking. Physical security procedures.
Writing off obsolete or damaged inventory which is good.	Control of responsible official over all write-offs.
Losses after issue to production .	Records of all issues.department. Standard usage allowance

4 Chapter summary

ORDERING AND ACCOUNTING FOR INVENTORY

Accounting procedures for ordering and issuing inventory

Order:
Purchase requisition
Purchase order form

Receive:
Delivery note
Goods received note

Issue:
Materials requisition note
Materials returned note
Materials transfer note

Physical inventory and book inventory

Physical inventory checked via stocktaking:

Periodic – Check every item at end of accounting period.

Continuous – Count and value selected items on rotating basis.

Book inventory updated via stores ledger (record) cards and bin cards as inventory received and issued – perpetual inventory system.

Accounting for inventory – the material inventory account

Entries in the material inventory account

Material inventory account debited when inventory increases:

- Purchases
- Materials returned to store

Material inventory account credited when inventory decreases:

- Transfers to WIP (production)
- Materials returned to suppliers
- Indirect materials transferred to production overheads
- Inventory write-offs

Balance in the material inventory account

Closing balance at end of period represents inventory held and may be valued as follows:

- **FIFO**
- **LIFO**
- **AVCO**

Test your understanding answers

Test your understanding 1

√	Information used to update inventory records.
	Information to check that the correct price has been recorded on the supplier's invoice.
√	Information to check that the correct quantity of goods has been recorded on the supplier's invoice.
	Information to record any unused materials which are returned to stores.

Test your understanding 2

Transaction	Debit which account?	Credit which account?
Issue materials to production.	Work-in-progress.	Material inventory account.
Purchase new materials on credit.	Material inventory account.	Creditors.
Materials returned to store from production.	Material inventory account.	Work-in-progress account.
Materials written off.	Income statement.	Material inventory account.
Indirect materials transferred to production overheads.	Production overhead account.	Material inventory account.

chapter

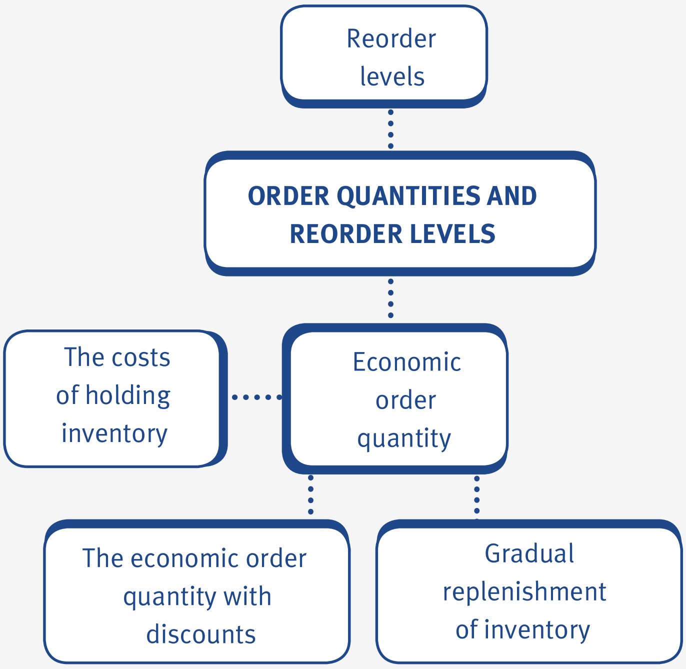
5

Order quantities and reorder levels

Chapter learning objectives

Upon completion of this chapter you will be able to:

- identify and explain the costs of ordering and holding inventory, distinguishing between fixed and variable costs
- from given data, calculate and interpret the economic order quantities
- from given data, calculate and interpret the economic order quantities when quantity discounts are available
- produce calculations to minimise inventory costs when stock is gradually replenished
- describe and apply appropriate methods for establishing reorder levels where demand in the lead time is constant.



1 The costs of holding inventory

Reasons for holding inventory

The main reason that an organisation will hold inventory is in order to make sure that customer demands are met as soon as possible.

- If customer demands are met, customers will be happy.
- If customers are happy, sales and, therefore, profits will increase.
- Holding raw material inventory will prevent hold-ups in the production process of manufacturing companies.

Holding costs

Costs associated with holding inventory are known as **holding costs.**

- Holding costs include the following:

  – interest on capital tied up in inventory

  – cost of storage space

  – cost of insurance.
- **Buffer inventory** is a basic level of inventory held for emergencies and to prevent stock-outs from occurring. It is also known as safety inventory or the minimum inventory level.
- **Holding costs** can be distinguished between fixed holding costs and variable holding costs.

- Fixed holding costs include the cost of storage space and the cost of insurance. Note that the cost of storage space may be a stepped fixed cost if increased warehousing is needed when higher volumes of inventory are held.
- **Variable holding costs** include interest on capital tied up in inventory. The more inventory that is held, the more capital that is tied up.
- Holding costs are often stated as being valued at a certain percentage of the average inventory held.
- Note: **Stock-out costs** are the costs associated with running out of inventory and they include loss of sales, loss of customers (and customer goodwill) and reduced profits.

Ordering costs

Every time an order is placed to purchase materials, an **order cost** is incurred. The costs associated with placing orders are known as **ordering costs** and include **administrative costs** and **delivery costs.**

- **Administrative costs** of placing an order are usually a fixed cost per order. The total administrative costs of placing orders will increase in proportion to the number of orders placed. They therefore exhibit the behaviour of variable costs.
- **Delivery costs** are usually a fixed charge per delivery (order). The total delivery costs will also increase in direct proportion to the number of deliveries in a period, and therefore behave as variable costs.
- If inventory levels are too low, there is a danger that the number of stock-outs will increase and there will therefore be an increase in the number of orders placed.
- An increase in the number of orders will cause a corresponding increase in **ordering costs.**
- It is essential, therefore, to maintain inventory at a level (known as the optimum level) where the total of holding costs, ordering costs and stock-out costs are at a minimum. This is the main objective of stock control.

e.g

Illustration 1 - The cost of holding inventory

A company uses components at the rate of 6,000 units per year, which are bought in at a cost of £1.20 each from the supplier. The company orders 1,000 units each time it places an order and the average inventory held is 500 units. It costs $20 each time to place an order, regardless of the quantity ordered.

The total holding cost is 20% per annum of the average inventory held.

The annual holding cost will be $ [ ]

The annual ordering cost will be $ [ ]

Expandable text

Test your understanding 1

A company has recorded the following details for Component 427 which is sold in boxes of 10 components.

Ordering cost	$32 per order placed
Purchase price	$20 per box of 10 components
Holding cost	10% of purchase price
Monthly demand	1,500 components

Component 427 is currently ordered in batches of 240 boxes at a time. The average inventory held is 120 boxes.

Required:

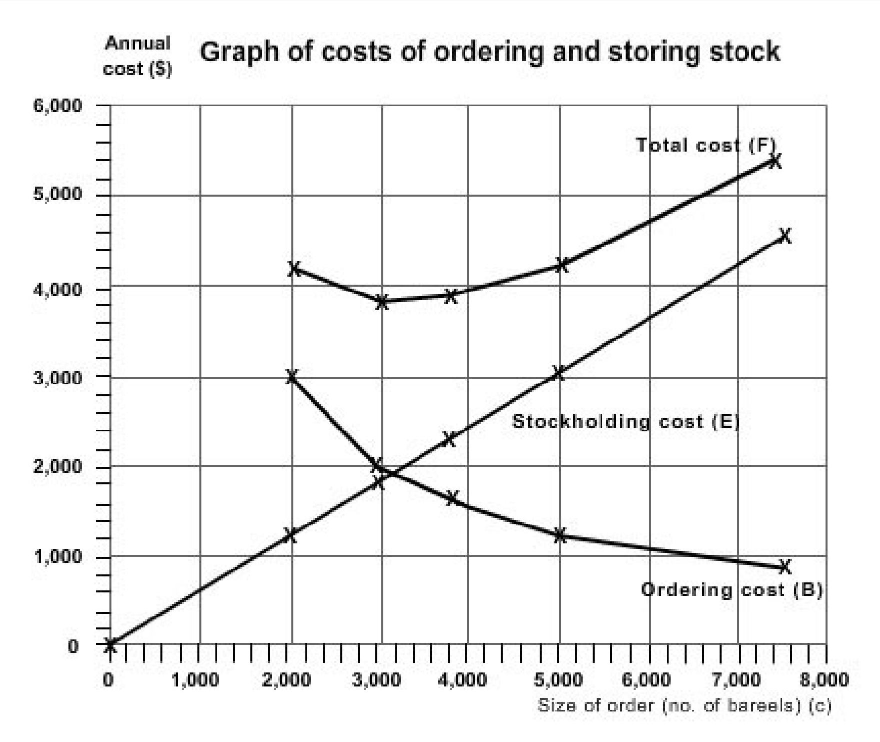
Calculate the annual holding cost and the annual ordering cost for Component 427.

2 The economic order quantity (EOQ)

EOQ

The EOQ is the reorder quantity which minimises the total costs associated with holding and ordering stock (i.e. (holding costs + ordering costs) are at a minimum at the EOQ).

We can estimate the EOQ graphically by plotting holding costs, ordering costs and total costs at different levels of activity.



- Note that the EOQ is found at the point where total costs (holding + ordering) are at a minimum.
- Reading from the graph this is somewhere between 3,000 and 3,200 barrels.
- This is the same point at which holding costs are equal to ordering costs.
- Unfortunately it is only possible to estimate the EOQ by drawing a graph.
- Fortunately there is a formula that allows us to calculate the EOQ more accurately and more speedily.

EOQ formula

The formula for the EOQ (or Q) is as follows:

$$Q = EOQ = \sqrt{\frac{2C_o D}{C_h}}$$

Where:

D = Demand per annum

C_o = Cost of placing one order

C_h = Cost of holding one unit for one year

Q = Reorder quantity (EOQ)

Note that the formula for the EOQ is provided in your exam. You must make sure that you know what the different symbols represent so that you can use the formula correctly.

EOQ assumptions

There are a number of important assumptions and formulae related to the EOQ that you should note.

- Average inventory held is equal to half of the EOQ = EOQ/2.
- The number of orders in a year = Expected annual demand/EOQ.
- Total annual holding cost = Average inventory (EOQ/2) x holding cost per unit of inventory.
- Total annual ordering cost = Number of orders x cost of placing an order.
- There is also a formula that allows us to calculate the Total Annual Costs (TAC) i.e. the total of holding costs and ordering costs.

$$TAC = C_0 \frac{D}{Q} + C_h \frac{Q}{2}$$

Where:

D = Demand per annum

C_o = Cost of placing one order

C_h = Cost of holding one unit for one year

Q = Reorder quantity (EOQ)

Note that the formula for the TAC is not provided in your exam.

Illustration 2 - The economic order quantity(EOQ)

A company uses components at the rate of 500 units per month, which are bought in at a cost of $1.20 each from the supplier. It costs $20 each time to place an order, regardless of the quantity ordered.

The total holding cost is 20% per annum of the value of stock held.

The company should order		components
The total annual cost will be $		

Expandable text

Test your understanding 2

A company is planning to purchase 90,800 units of a particular item in the year ahead. The item is purchased in boxes each containing 10 units of the item, at a price of $200 per box. A safety inventory of 250 boxes is kept.

The cost of holding an item in inventory for a year (including insurance, interest and space costs) is 15% of the purchase price. The cost of placing and receiving orders is to be estimated from cost data collected relating to similar orders, where costs of $5,910 were incurred on 30 orders. It should be assumed that ordering costs change in proportion to the number of orders placed. 2% should be added to the above ordering costs to allow for inflation. Assume that usage of the item will be even over the year.

The order quantity which minimises total costs is		
This will mean ordering the item every		weeks

3 The EOQ with discounts

Quantity discounts

It is often possible to negotiate a quantity discount on the purchase price if bulk orders are placed.

- If a quantity discount is accepted this will have the following effects:
  - The annual purchase price will decrease.
  - The annual holding cost will increase.
  - The annual ordering cost will decrease

- To establish whether the discount should be accepted or not, the following calculations should be carried out.
  - Calculate TAC with the discount.
  - Compare this with the annual costs without the discount (at the EOQ point).

EOQ when quantity discounts are available

The steps involved in calculating the EOQ when quantity discounts are available are as follows:

(1) Calculate the EOQ, ignoring discounts.

(2) If the EOQ is smaller than the minimum purchase quantity to obtain a bulk discount, calculate the total for the EOQ of the annual stockholding costs, stock ordering costs and stock purchase costs.

(3) Recalculate the annual stockholding costs, stock ordering costs and stock purchase costs for a purchase order size that is only just large enough to qualify for the bulk discount.

(4) Compare the total costs when the order quantity is the EOQ with the total costs when the order quantity is just large enough to obtain the discount. Select the minimum cost alternative.

(5) If there is a further discount available for an even larger order size, repeat the same calculations for the higher discount level.

e.g

Illustration 3 -The EOQ with discounts

A company uses components at the rate of 500 units per month, which are bought in at a cost of $1.20 each from the supplier. It costs $20 each time to place an order, regardless of the quantity ordered.
The supplier offers a 5% discount on the purchase price for order quantities of 2,000 items or more. The current EOQ is 1,000 units.
The total holding cost is 20% per annum of the value of stock held.

Required:
Should the discount be accepted?

Expandable text

Test your understanding 3

Watton Ltd is a retailer of beer barrels. The company has an annual demand of 36,750 barrels. The barrels cost $12 each. Fresh supplies can be obtained immediately, but ordering costs and the cost of carriage inwards are $200 per order. The annual cost of holding one barrel in stock is estimated to be $1.20. The economic order quantity has been calculated to be 3,500 barrels.
The suppliers introduce a quantity discount of 2% on orders of at least 5,000 barrels and 2.5% on orders of at least 7,500 barrels.

Required:
Determine whether the least-cost order quantity is still the EOQ of 3,500 barrels.

4 Gradual replenishment of inventory

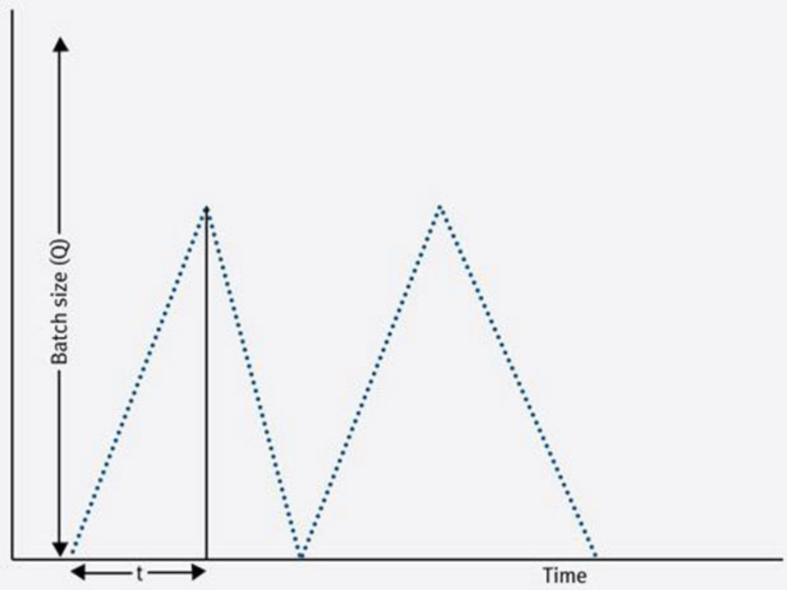
Gradual replenishment of inventory

The situations we have looked at so far have involved inventory levels being replenished immediately when organisations buy inventory from suppliers. Similar problems are faced by organisations who replenish inventory levels gradually by manufacturing their own products internally.

- The decisions faced by organisations that manufacture and store their own products involve deciding whether to produce large batches at long intervals OR produce small batches at short intervals.
- An amended EOQ model is used to help organisations to decide which course of action to take.
- The amended EOQ model is known as the EBQ (economic batch quantity) model.
- As the items are being produced, there is a machine setup cost. This replaces the ordering cost of the EOQ.
- In the EOQ, inventory is replenished instantaneously whereas here, it is replenished over a period of time.
- Depending on the demand rate, part of the batch will be sold or used while the remainder is still being produced.
- For the same size of batch (Q), the average inventory held in the EOQ model (Q/2) is greater than the average in this situation (see diagram below).
- The EBQ model can be shown graphically as follows.

Inventory

(units)



- The maximum inventory level will never be as great as the batch size, because some of the batch will be used up while the remainder is being produced.

The EBQ

The EBQ model is primarily concerned with determining the number of items that should be produced in a batch (compared to the size of an order with the EOQ).

The formula for the EBQ is as follows:

$$\text{Economic batch quantity} = \sqrt{\frac{2C_o D}{C_h\left(1-\frac{D}{R}\right)}}$$

Where:
Q = Batch size
D = Demand per annum
C_h = Cost of holding one unit for one year
C_o = Cost of setting up a batch ready to be produced
R = Annual replenishment rate

Note that the formula for the EBQ is provided in your exam. You must make sure that you know what the different symbols represent so that you can use the formula correctly.

Expandable text

Illustration 4 - Gradual replenishment of inventory

The following is relevant for Item X:

- Production is at a rate of 500 units per week.
- Demand is 10,000 units per annum; evenly spread over 50 working weeks.
- Setup cost is $2,700 per batch.
- Storage cost is $2.50 per unit for a year.

Required:
Calculate the economic batch quantity (EBQ) for Item X.

Expandable text

Test your understanding 4

AB Ltd makes a component for one of the engines that it builds. It uses, on average, 2,000 of these components, steadily throughout the year. The component costs $16 per unit to make and it costs an additional $320 to setup the production process each time a batch of components is made. The holding cost per unit is 10% of the unit production cost. The company makes these components at a rate of 200 per week, and the factory is open for 50 weeks per annum.
Required:
Calculate the EBQ.

5 Reorder levels

Reorder level

Reorder level – when inventory held reaches the reorder level then a replenishment order should be placed.

- Lead time – this is the time expected to elapse between placing an order and receiving an order for inventory.
- Reorder quantity – when the reorder level is reached, the quantity of inventory to be ordered is known as the reorder or EOQ.

- Demand – this is the rate at which inventory is being used up. It is also known as inventory usage.
- If the demand in the lead time is constant, the reorder level is calculated as follows:

Reorder level (when demand in lead time is constant)
=Usage x Lead time

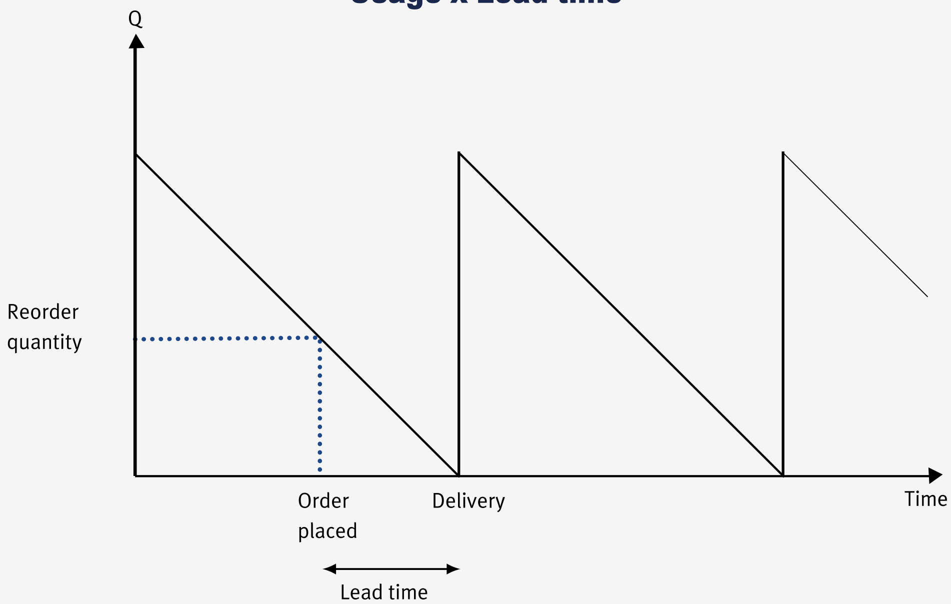


Illustration 5 - Reorder levels

A company uses Component M at the rate of 1,500 per week. The time between placing an order and receiving the components is five weeks. The reorder quantity is 12,000 units.

Required:
Calculate the reorder level.

Expandable text

Test your understanding 5

A national chain of tyre fitters stocks a popular tyre for which the following information is available:

Usage - 175 tyres per day

Lead time - 16 days

Reorder quantity - 3,000 tyres

Based on the data above, at what level of inventory should a replenishment order be issued in order to ensure that there are no stock-outs?

A 2,240

B 2,800

C 3,000

D 5,740

Test your understanding 6

A national chain of tyre fitters stocks a popular tyre for which the following information is available:

Average usage	140 tyres per day
Minimum usage	90 tyres per day
Maximum usage	175 tyres per day
Lead time	10 to 16 days
Reorder quantity	3,000 tyres

Based on the data above, what is the maximum level of inventory possible?

A 2,800

B 3,000

C 4,900

D 5,800

6 Chapter summary

Reorder levels

Reorder level – place new order when inventory reaches reorder level = Maximum lead time

Other control levels – maximum inventory and minimum inventory

ORDER QUANITIES AND REORDER LEVELS

The costs of holding inventory

Interest on captial
Cost of storage
Cost of insurance
Stock-out costs

Economic order quantity

Reorder quanity that minimises total costs associated with holding inventory

$$EOQ = \sqrt{\frac{2C_0D}{C_n}}$$

The costs of ordering inventory

Administration costs
Machine setup costs

The economic order quantity with discounts

Compare total annual costs WITH discount to total annual costs WITHOUT discount. Accept discount if total cost lower with discount than without

Gradual replishment of inventory

Relevant to organisations that manufacture own products. EOQ modified to produce EBQ

$$EBQ = \sqrt{\frac{2C_0D}{C_n(1-D/R)}}$$

Test your understanding answers

Test your understanding 1

Annual holding cost = average inventory held x cost per box x 10%

= 120 x $20 x 10% = $240

$$\text{Annual usage (in boxes)} = \frac{1{,}500}{10} \times 12 \text{ months} = 1{,}800 \text{ boxes}$$

$$\text{Annual ordering cost} = \frac{\text{Annual usage}}{\text{Order size}} \times \$32$$

$$= \frac{1{,}800}{240} \times \$32$$

$$= \$240$$

Test your understanding 2

The order quantity which minimises total costs is	3,487
This will mean ordering the item every	2 weeks

Workings

To avoid confusion this question is best tackled by working in boxes not units.

$C_o = \dfrac{5{,}910}{30} \times 1.02 = \$200.94$

$C_h = 0.15 \times \$200 = \$30$ per box

$D = 90{,}800/10 = 9{,}080$ boxes

$EOQ = \sqrt{\dfrac{2 \times 200.94 \times 9{,}080}{30}} = 349$ boxes

Number of orders per year $= 9{,}080/349 = 26$

26 orders per annum is equivalent to placing an order every 2 weeks (52 weeks / 26 orders).

Test your understanding 3

Order quantity = EOQ of 3,500 barrels	$
Purchase costs (36,750 x $12)	441,000
Annual stockholding costs ($1.20 x 3,500/2)	2,100
Annual ordering costs ($200 x 36,750/3,500)	2,100
Total costs	445,200

Order quantity = 5,000 barrels	$
Purchase costs (36,750 x $12 x 98%)	432,180
Annual stockholding costs ($1.20 x 5,000/2)	3,000
Annual ordering costs ($200 x 36,750/5,000)	1,470
Total costs	436,650

Order quantity = 7,500 barrels	$
Purchase costs (36,750 x $12 x 97.5%)	429,975
Annual stockholding costs ($1.20 x 7,500/2)	4,500
Annual ordering costs ($200 x 36,750/7,500)	980
Total costs	435,455

Total costs are minimised with an order size of 7,500 barrels.

Test your understanding 4

D			=	2,000 units
R	=	200 x 50	=	10,000 units
C_o			=	\$320
C_h	=	10% of \$16	=	\$1.60

$$EBQ = \sqrt{\frac{2C_oD}{C_h(1 - D/R)}} = \sqrt{\frac{2 \times 320 \times 2{,}000}{1.60\,(1 - 2{,}000/10{,}000)}} = 1{,}000 \text{ units}$$

Test your understanding 5

B Reorder level = Usage x Lead time

= 175 x 16

= 2,800 units

Test your understanding 6

C Reorder level = 16 x 175 = 2,800

Maximum inventory level = (Reorder level + reorder quantity) – (minimum usage x minimum lead time)

= (2,800 + 3,000) – (90 x 10)

= 4,900 units

chapter

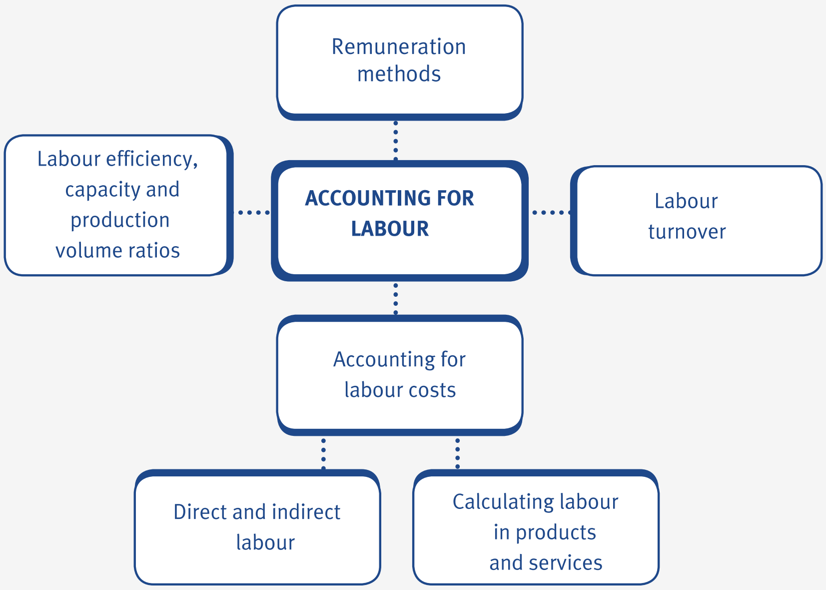
6

Accounting for labour

Chapter learning objectives

Upon completion of this chapter you will be able to:

- calculate direct and indirect costs of labour for both a manufacturing and a service industry
- explain the methods used to relate input labour costs to work done for both a manufacturing and a service industry
- for given data, prepare the journal and ledger entries to record labour costs inputs and outputs, and interpret entries in the labour account
- describe different remuneration methods: time-based systems; piecework systems; individual incentive schemes; group incentive schemes
- calculate the level, and analyse the costs and causes of labour turnover for an organisation in general
- explain and calculate from given data: labour efficiency ratio; capacity ratio; production volume ratios.



1 Direct and indirect labour

Direct and indirect labour costs

Labour is often one of the major expenses of a business. One of the most important distinctions of labour is between **direct** and **indirect** costs.

- Direct labour costs make up part of the prime cost of a product and include the basic pay of direct workers.
- Direct workers are those employees who are directly involved in making an organisation's products.
- Indirect labour costs make up part of the overheads (indirect costs) and include the basic pay of indirect workers.
- Indirect workers are those employees who are **not** directly involved in making products, (for example, maintenance staff, factory supervisors and canteen staff.
- Indirect labour costs also include the following.
  - Bonus payments.
  - Employers' National Insurance Contributions.
  - Idle time (when workers are paid but are not making any products, for example when a machine breaks down).
  - Sick pay.
  - Time spent by direct workers doing 'indirect jobs' for example, cleaning or repairing machines.

Overtime and overtime premiums

When employees work overtime, they receive a **basic pay** element and an **overtime premium**.

- For example, if Fred is paid $8 per hour and overtime is paid at time and a half, when Fred works overtime, he will receive $12 per hour ($8 + $4 (50% x $8)). It is important that his pay is analysed into direct and indirect labour costs.

Overtime pay
$12

BASIC PAY
Direct labour cost = $8

Overtime premium
Indirect labour cost = $4

- Overtime premiums are treated as direct labour costs, if at the specific request of a customer because they want a job to be finished as soon as possible.
- Employees who work night shifts, or other anti-social hours may be entitled to a shift allowance or shift premium. Shift premiums are similar to overtime premiums where the extra amount paid above the basic rate is treated as an indirect labour cost.

e.g

Illustration 1 - Direct and indirect labour

Vienna is a direct labour employee who works a standard 35 hours per week and is paid a basic rate of $12 per hour. Overtime is paid at time and a third. In week 8 she worked 42 hours and received a $50 bonus. Complete the following table.

	Direct labour cost	Indirect labour cost
Basic pay for standard hours $		
Basic pay for overtime hours $		

Overtime premium $		
Bonus $		

Expandable text

Test your understanding 1

A company operates a factory which employed 40 direct workers throughout the four-week period just ended. Direct employees were paid at a basic rate of $4.00 per hour for a 38-hour week. Total hours of the direct workers in the four-week period were 6,528. Overtime, which is paid at a premium of 35%, is worked in order to meet general production requirements. Employee deductions total 30% of gross wages. 188 hours of direct workers' time were registered as idle.

Calculate the following for the four-week period just ended.

Gross wages (earnings) $

Deductions $

Net wages $

Direct labour cost $

Indirect labour cost $

2 Calculating labour in products and services

Determining time spent doing jobs

Methods can include:

- time sheets
- time cards
- job sheets.

Expandable text

Payroll department

The payroll department is involved in carrying out functions that relate input labour costs to the work done.

- Preparation of the payroll involves calculating gross wages from time and activity records.
- The payroll department also calculates net wages after deductions from payroll (PAYE, Employer's National Insurance Contributions and Employee's National Insurance Contributions).
- The payroll department also carries out an analysis of direct wages, indirect wages, and cash required for payment.

3 Accounting for labour costs

Labour costs are an expense and are recorded in an organisation's income statement. Accounting transactions relating to labour are recorded in the labour account.

- The labour account is debited with the labour costs incurred by an organisation. The total labour costs are then analysed into direct and indirect labour costs.
- **Direct labour costs** are credited from the labour account and debited in the work-in-progress (WIP) account. Remember, direct labour costs are directly involved in production and are therefore transferred to WIP before being transferred to finished goods and then cost of sales.
- **Indirect labour costs** are also credited 'out of' the labour account and debited to the production overheads account. It is important that total labour costs are analysed into their direct and indirect elements.

e.g

Illustration 2 - Accounting for labour costs

Labour account

	$000		$000
Bank (1)	80	WIP (2)	60
		Production overheads (3)	
		Indirect labour	14
		Overtime premium	2
		Shift premium	2
		Sick pay	1
		Idle time	1
	80		80

(1) Labour costs incurred are paid out of the bank before they are analysed further in the labour account.

(2) The majority of the labour costs incurred by a manufacturing organisation are in respect of direct labour costs. Direct labour costs are directly involved in production and are transferred out of the labour account via a credit entry to the WIP account as shown above.

(3) Indirect labour costs include indirect labour (costs of indirect labour workers), overtime premium (unless overtime is worked at the specific request of a customer), shift premium, sick pay and idle time. All of these indirect labour costs are collected in the production overheads account. They are transferred there via a credit entry out of the labour account and then debited in the production overheads account.

Test your understanding 2

	Direct workers	Indirect workers	Total
	$	$	$
Basic pay for basic hours	43,000	17,000	60,000
Overtime – basic pay	10,000	4,500	14,500
Overtime – premium	5,000	2,250	7,250
Training	2,500	1,250	3,750
Sick pay	750	250	1,000
Idle time	1,200	-	1,200

The following information is taken from the payroll records of a company.
Required:
Using the information given, complete the labour account shown below.

Labour account

	$		$
	____		____
	____		____

4 Remuneration methods

There are two basic approaches to remuneration, time-related or output-related. The two basic methods are time-based and piecework systems.

Time-based systems

We looked at time-based systems, the most common remuneration method, at the beginning of this chapter.

- Employees are paid a basic rate per hour, day, week or month.

- Basic time-based systems do not provide any incentive for employees to improve productivity and close supervision is often necessary.
- The basic formula for a time-based system is as follows.

Total wages = (hours worked x basic rate of pay per hour) + (overtime hours worked x overtime premium per hour)

Piecework systems

A piecework system pays a fixed amount per unit produced. The basic formula for a piecework system is as follows.

Total wages = (units produced x rate of pay per unit)

Expandable text

Illustration 3 - Remuneration methods

A company operates a differential piecework system and the following weekly rates have been set:

Weekly production	**Rate of pay per unit in this band**
	$
1 to 500 units	0.20
501 to 600 units	0.25
601 units and above	0.55

How much would be paid to the following employees for the week?
Employee A – output achieved = 800 units
Employee B – output achieved = 570 units

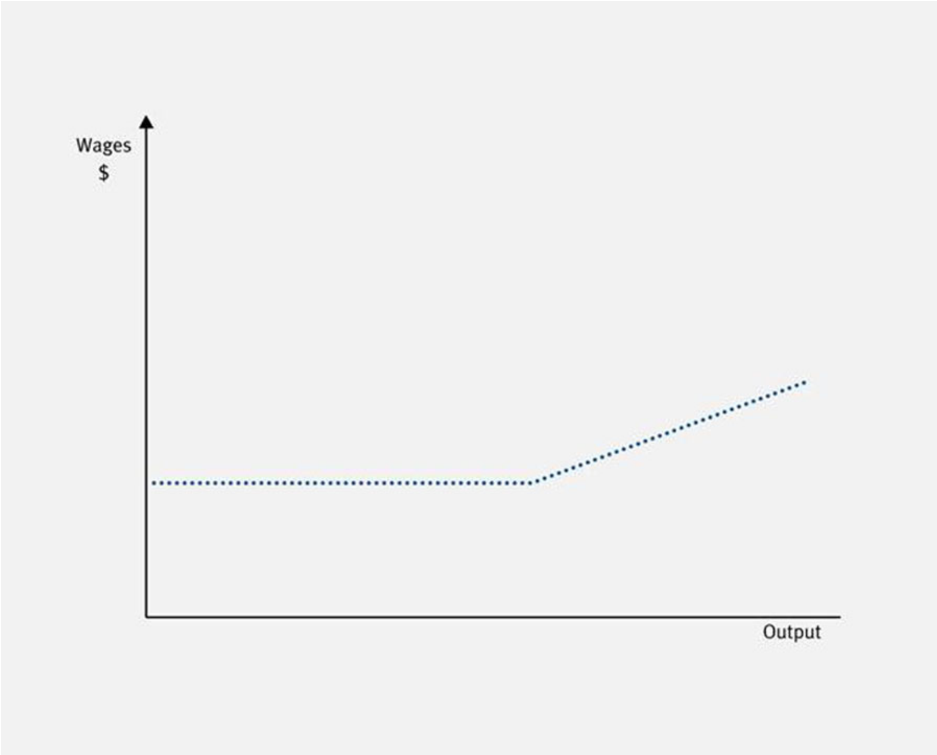
Employee A = $ ☐

Employee A = $ ☐

Expandable text

Test your understanding - 3

The following graph shows the wages earned by an employee during a single day.



Which one of the following remuneration systems does the graph represent?

A Differential piecework

B A flat rate per hour with a premium for overtime working

C Straight piecework

D Piecework with a guaranteed minimum daily wage

Incentive schemes

Incentive schemes can be aimed at individuals and/or groups.

- Many different systems exist in practice for calculating bonus schemes. General rules are as follows:
  - They should be closely related to the effort expended by employees.
  - They should be agreed by employers/employees before being implemented.
  - They should be easy to understand and simple to operate.
  - They must be beneficial to all of those employees taking part in the scheme.
- Most bonus schemes pay a basic time rate, plus a portion of the time saved as compared to some agreed allowed time. These bonus schemes are known premium bonus plans. Examples of such schemes are Halsey and Rowan.
- **Halsey** – the employee receives 50% of the time saved.

$$\text{Bonus} = \frac{\text{Time allowed} - \text{Time taken}}{2} \times \text{Time rate}$$

- **Rowan** – the proportion paid to the employee is based on the ratio of time taken to time allowed.

$$\text{Bonus} = \frac{\text{Time taken}}{\text{Time allowed}} \times \text{Time rate} \times \text{Time saved}$$

- **Measured day work** – the concept of this approach is to pay a high time rate, but this rate is based on an analysis of past performance. Initially, work measurement is used to calculate the allowed time per unit. This allowed time is compared to the time actually taken in the past by the employee, and if this is better than the allowed time an incentive is agreed, e.g. suppose the allowed time is 1 hour per unit and that the average time taken by an employee over the last three months is 50 minutes. If the normal rate is $12/hour, then an agreed incentive rate of (say) $14/hour could be used.
- **Share of production** – share of production plans are based on acceptance by both management and labour representatives of a constant share of value added for payroll. Thus, any gains in value added – whether by improved production performance or cost savings – are shared by employees in this ratio.

e.g

Illustration 4 - Remuneration methods

The following data relate to Job A.
Employee's basic rate = $4.80 per hour
Allowed time for Job A = 1 hour
Time taken for Job A = 36 minutes

Halsey scheme – Total payment for Job A = $ [ ]

Rowan scheme – Total payment for Job B = $ [ ]

Solution

Halsey scheme – Total payment for Job A = $ 3.84

Rowan scheme – Total payment for Job B = $ 4.03

Workings

Halsey — $

Bonus $= \frac{60 - 36}{2} \times \frac{\$4.80}{60}$ — 0.96

Basic rate $= \frac{36}{60} \times \$4.80$ — 2.88

Total payment for Job A — 3.84

Rowan — $

Bonus $= \frac{36}{60} \times \frac{\$4.80}{60} \times 24$ — 1.15

Basic rate $= \frac{36}{60} \times \$4.80$ — 2.88

Total payment for Job A — 4.03

Additional test your understanding

Ten employees work as a group. When production of the group exceeds the standard – 200 pieces per hour – each employee in the group is paid a bonus for the excess production in addition to wages at hourly rates.

The bonus is computed thus: the percentage of production in excess of the standard quantity is found, and one half of the percentage is regarded as the employees' share. Each employee in the group is paid as a bonus this percentage of a wage rate of $5.20 per hour. There is no relationship between the individual worker's hourly rate and the bonus rate.

The following is one week's record:

	Hours worked	**Production**
Monday	90	24,500
Tuesday	88	20,600
Wednesday	90	24,200
Thursday	84	20,100
Friday	88	20,400
Saturday	40	10,200
	480	120,000

During this week, Jones worked 42 hours and was paid $3 per hour basic.

Complete the following.

(1) The bonus rate for the week was $

(2) The total bonus for the week was $

(3) The total pay for Jones for the week was $

Expandable text

5 Labour turnover

In an examination you will be given clear instructions on any bonus scheme in operation. You should follow the instructions given carefully in order to calculate the bonus payable from the data supplied.

Labour turnover is a measure of the proportion of people leaving relative to the average number of people employed.

- Management might wish to monitor labour turnover, so that control measures might be considered if the rate of turnover seems too high, and the business is losing experienced and valuable staff at too fast a rate.
- Labour turnover is calculated for any given period of time using the following formula:

$$\frac{\textbf{Number of leavers who require replacement}}{\textbf{Average number of employees}} \times \mathbf{100}$$

Illustration 5 - Labour turnover

At 1 January a company employed 3,641 employees and at 31 December employee numbers were 3,735. During the year 624 employees chose to leave the company. What was the labour turnover rate for the year?

Expandable text

Test your understanding 4

A company had 4,000 staff at the beginning of 20X8. During the year, there was a major restructuring of the company and 1,500 staff were made redundant and 400 staff left the company to work for one of the company's main competitors. 400 new staff joined the company in the year to replace those who went to work for the competitor.

Required:
Calculate the labour turnover rate for 20X8.

Expandable text

6 Labour efficiency, capacity and production volume ratios

Labour efficiency ratio

Labour is a significant cost in many organisations and it is important to continually measure the efficiency of labour against pre-set targets.

- The labour efficiency ratio measures the performance of the workforce by comparing the actual time taken to do a job with the expected time.
- The labour efficiency ratio is calculated using the following formula:

$$\frac{\textbf{Expected hours to produce output}}{\textbf{Actual hours to produce output}} \times 100\%$$

Labour capacity ratio

Sometimes the workforce is 'idle' through no fault of its own, and cannot get on with productive work. This sometimes happens if machines break down or need to be reset for a new production run.

(1) The labour capacity ratio measures the number of hours spent actively working as a percentage of the total hours available for work (full capacity or budgeted hours).

(2) The labour capacity ratio is calculated using the following formula:

$$\frac{\textbf{Number of hours spent working (active production)}}{\textbf{Total hours available}} \times 100\%$$

Labour production volume ratio

- The labour production volume ratio compares the number of hours expected to be worked to produce actual output with the total hours available for work (full capacity or budgeted hours).
- The labour production volume ratio is calculated using the following formula:

$$\frac{\text{Expected hours to produce actual output}}{\text{Total hours available (budgeted)}} \times 100\%$$

Expandable text

Test your understanding 5

A company budgets to make 40,000 units of Product DOY in 4,000 hours (each unit is budgeted to take 0.1 hours each) in a year.

Actual output during the year was 38,000 units which took 4,180 hours to make.

Required:

Calculate the labour efficiency, capacity and production volume ratios .

Number of hours spent working (active production) x 100%

Total hours available (budgeted)

7 Chapter summary

Remuneration methods

- Time-based systems
- Piecework systems
- Individual incentive schemes
- Group incentive schemes

ACCOUNTING FOR LABOUR

Labour efficiency capacity and production volume ratios

Labour is significant cost in many organisations – important to use these ratios to continually 'measure' how it is doing by reference to effciency, capacity and production volume ratios

Accounting for labour costs

Debit labour account with labour costs incurred

Credit labour account with direct labour – transfer to WIP

Credit labour account with indirect labour – transfer to production overheads account

Labour turnover

Measure of proportion of employees leaving relative to the average number of people employed

Many causes and costs of labour turnover – both avoidable and unavoidable

Direct and indirect labour

Direct labour – makes up part of prime cost and includes basic pay of direct workers

Indirect labour – makes u p part of overheads and includes basic pay of indirect workers

Overtime premiums are treated as overheads unless worked at specific request of customer when treated as direct cost

Calculating labour in products and services

Times records

- Time cards
- Clock cards

Activity records

- Period-related

Test your understanding answers

Test your understanding 1

Gross wages (earnings) $	$26,739.20
Deductions $	$8,021.76
Net wages $	$18,717.44
Direct labour cost $	$25,360
Indirect labour cost $	$1,379.20

Workings

Basic time	= 40 workers x 38 hrs/week x 4 weeks	= 6,080 hrs
Overtime	= Total time – Basic time = 6,528 – 6,080	= 448 hrs
Total wages	= Basic pay + Overtime premium	
	= 6,528 hours at $4.00 per hour	= $26,112.00
	+ 448 hours at $1.40 per hour	= $627.20
Gross wages		$26,739.20
Deductions	= $26,739.20 x 30%	= $8,021.76
Net pay	= $26,739.20 x 70%	= $18,717.44
Productive time	= Total time – Idle time	
	= 6,528 – 188	= 6,340 hours
Direct labour	= 6,340 hours at $4.00 per hour	= $25,360

Indirect labour	= Overtime premium + Idle time costs	
	= $627.20 + $752 (188 hours x $4.00/hr)	= $1,379.20
		———
Gross wages		$26,739.20
		———

Test your understanding 2

Labour account

	$		$
Bank	87,700	WIP (43,000 + 10,000)	53,000
		Production overheads	
		Indirect labour (17,000 + 4,500)	21,500
		Overtime premium	7,250
		Training	3,750
		Sick pay	1,000
		Idle time	1,200
	———		———
	87,700		87,700
	———		———

Test your understanding - 3

D The graph represents a piecework system (as shown by the gentle upward-sloping line) with a guaranteed minimum daily wage (as shown by the horizontal line).

Test your understanding 4

Number of staff at beginning of year = 4,000

Number of staff at end of year = 4,000 – 1,500 – 400 + 400 = 2,500

Labour turnover rate =

$$\frac{\text{Number of leavers who require replacement}}{\text{Average number of employees}} \times 100$$

Average number of employees in the year = $\frac{4{,}000 + 2{,}500}{2} = 3{,}250$

Labour turnover rate = $\frac{400}{3{,}250} \times 100\% = 12.3\%.$

Test your understanding 5

Expected hours to produce output = 38,000 x 0.1 hours = 3,800 standard hours.

Labour efficiency ratio =

$$\frac{\text{Expected hours to produce actual output (std hours)}}{\text{Actual hours to produce output}} \times 100\%$$

$$= 3{,}800/4{,}180 \quad \times 100\% = 91\%$$

Labour capacity ratio =

$$\frac{\text{Number of hours spent working (active production)}}{\text{Total hours available (budgeted)}} \times 100\%$$

$$= 4{,}180/4{,}000 \quad \times 100\% = 104.5\%$$

Production volume ratio =

$$\frac{\text{Expected hours to produce actual output (std hours)}}{\text{Total hours available (budgeted)}} \times 100\%$$

Expected hours to produce actual output (std hours)/Total hours available (budgeted) x 100%

$$= \frac{3{,}800}{4{,}000} \times 100\% = 95\%$$

chapter

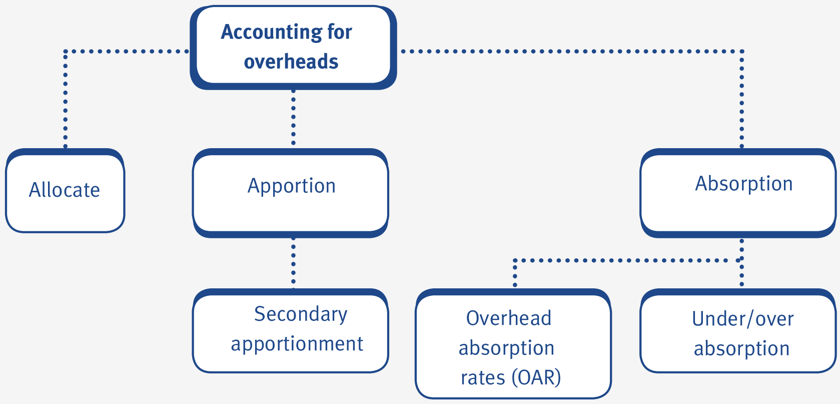
7

Accounting for overheads

Chapter learning objectives

Upon completion of this chapter you will be able to:

- explain the different treatment of direct and indirect expenses for a business
- describe, in overview, the steps involved in determining production overhead absorption rates (OARs) and how such rates are used
- explain, for a business in general, the difference between 'allocation' and 'apportionment' of production overheads
- apportion production overheads to cost centres employing appropriate bases, using data supplied
- reapportion service cost centre costs to production cost centres (using the reciprocal method where service cost centres work for each other), using data supplied
- select, apply and discuss appropriate bases for absorption rates, using data supplied
- prepare journal and ledger entries for manufacturing overheads incurred and absorbed, using data supplied
- calculate and explain the under- and over-absorption of overheads, using data supplied
- apply methods of relating non-production overheads to cost units, using data supplied.



1 Recap of direct and indirect expenses

Direct expenses are expenses that can be directly identified with a specific cost unit or cost centre, for example, the cost of maintaining sewing machines that are used to make shirts. The maintenance expense can be directly related to the cost of producing the shirts.

- Direct expenses are part of the prime cost of a product.

Indirect expenses cannot be directly identified with a specific cost unit or cost centre.

- For example, the cost of renting the factory where shirts are manufactured is classified as an indirect cost because it would be impossible to relate such costs to shirts only, if other clothes, such as dresses and suits were also made in the same factory.
- Indirect expenses are also known as **overheads**.

2 Production overhead absorption

Fixed production overheads

As we have already seen, production overheads are the total of indirect production costs:

Fixed production overheads = indirect materials + indirect labour + indirect expenses

- Fixed production overheads of a factory will include the following costs:
  - heating the factory
  - lighting the factory
  - renting the factory.
- The total cost of a product also includes a share of the fixed production overheads.
- This is because organisations must recover their fixed production overheads and they do this by absorbing a fixed amount into each product that they make and sell.
- One way of recovering fixed production overheads is on a cost per unit basis.

e.g

Illustration 1 - Production overhead absorption

RS Ltd is a manufacturing company producing Product P, which has the following cost card.

		$
Direct labour	2 hrs @ $5 per hour	10
Direct materials	1 kg @ $5 per kg	5
Direct expenses		1
		——
Prime cost		**16**
		——

RS Ltd produces and sells 1,000 units in a month.
Based on past experience, RS Ltd estimates its monthly overheads will be as follows.

	$
Heating	3,000
Power	2,000
Maintenance	500
	———
Total	5,500
	———

The overhead cost allocated to eachunit of Product P is $ [ ]

The cost per unit of Product P is $ [ ]

Expandable text

Test your understanding 1

In addition to producing Product P, RS Ltd now starts to produce another product, Product Q.

RS Ltd plans to make and sell 1,000 units of Product Q in a month, as well as producing 1,000 units of Product P. Due to the increase in production levels, the overheads are likely to increase as follows.

	$
Heating	4 200
Power	2,900
Maintenance	700
Total	7,800

Calculate the overhead cost allocated to each unit of Product P.

Calculate the new cost per unit of Product P.

Absorption costing

Production overheads are recovered by absorbing them into the cost of a product and this process is therefore called absorption costing.

- The main aim of absorption costing is to recover overheads in a way that fairly reflects the amount of time and effort that has gone into making a product or service.
- Absorption costing involves the following stages:
  – allocation and apportionment of overheads
  – reapportionment of service (non-production) cost centre overheads
  – absorption of overheads.

3 'Allocation' and 'apportionment'

Allocation and apportionment of overheads

The first stage of the absorption costing process involves the allocation and apportionment of overheads.

- Allocation involves charging overheads directly to specific departments (production and service).
- If overheads relate to more than one specific department, then they must be apportioned (shared) between these departments using a method known as apportionment.
- Overheads must be apportioned between different production and service departments on a fair basis.

Bases of apportionment

There are no hard and fast rules for which basis of apportionment to use except that whichever method is used to apportion overheads, it must be fair. Possible bases of apportionment include the following:

- floor area – for rent and rates overheads
- net book value (NBV) of fixed assets – for depreciation and insurance of machinery
- number of employees – for canteen costs.

e.g

Illustration 2 - 'Allocation' and 'apportionment'

LS Ltd has two production departments (Assembly and Finishing) and two service departments (Maintenance and Canteen).

The following are budgeted costs for the next period:

Indirect materials	- $20,000
Rent	- $15,000
Electricity	- $10,000
Machine depreciation	- $5,000
Indirect labour	- $16,520
Direct labour	- $125,000

The following information is available:

	Assembly	Finishing	Maintenance	Canteen
Area (sq metres)	1,000	2,000	500	500
kW hours consumed	2,750	4,500	1,975	775
Machine value ($)	45,000	35,000	11,000	9,000
Staff	20	30	10	2
Direct labour hours	3,175	3,800	-	-
Indirect materials budget ($)	7,000	8,000	3,000	2,000
Indirect labour budget ($)	1,600	2,220	11,200	1,500

Required:

Complete the extract from the overhead analysis sheet shown below.

Overhead analysis sheet

Overhead	Basis of apportionment	Assembly $	Finishing $	Maintenance $	Canteen $	Total $
Indirect materials						
Rent						
Direct labour						

Expandable text

Test your understanding 2

Using the information from Illustration 2 above, complete the extract of the overhead analysis sheet shown below.

Overhead	Basis of apportion-ment	Assembly $	Finish-ing $	Mainte-nance $	Canteen $	Total $
Electricity						
Machine depreciation						
Indirect labour						

4 Reapportionment of service cost centre costs to production cost centres

'Reapportionment'

Service cost centres (departments) are not directly involved in making products and therefore the fixed production overheads of service cost centres must be shared out between the production cost centres (departments) using a suitable basis.

- Examples of service cost centres are as follows:
  - stores
  - canteen
  - maintenance department
  - payroll department.
- The basic method of reapportionment (illustrated below) is used when one service department does work for other, but not where services are reciprocated, (e.g. canteen does work for maintenance **and** maintenance does work for the canteen).
- The reciprocal method of reapportionment is used when two service departments do work for each other.

Illustration 3 - Reapportionment of service cost centre costs

The total overheads allocated and apportioned to the production and service departments of LS Ltd are as follows:

Assembly = $17,350

Finishing = $23,970

Maintenance = $18,600

Canteen = $6,600

A suitable basis for sharing out the maintenance costs is the time spent servicing equipment. The amount of time spent by the maintenance department servicing equipment in the Assembly and Finishing departments has been analysed as follows:

Assembly 60%

Finishing 40%

The Canteen department's overheads are to be reapportioned on the basis of the number of employees in the other three departments.

	Assembly	Finishing	Maintenance	Canteen
Number of employees	20	30	10	2

Required:

Complete the overhead analysis sheet below.

Overhead	Basis of apportion-ment	Assembly $	Finish-ing $	Mainte-nance $	Canteen $	Total $
Total from above						
Reapportion canteen						
Sub-total						
Reapportion maintenance						
Total						

Expandable text

Reciprocal reapportionment

Reciprocal reapportionment (or the repeated distribution method) is used where service cost centres (departments) do work for each other.

- It involves carrying out many reapportionments until all of the service departments' overheads have been reapportioned to the production departments.
- The following illustration will demonstrate the reciprocal method of reapportionment.

e.g

Illustration 4 - Reapportionment of service cost centre costs

The total overheads allocated and apportioned to the production and service departments of LS Ltd are as follows.

Assembly = $17,350

Finishing = $23,970

Maintenance = $18,600

Canteen = $6,600

A suitable basis for sharing out the maintenance costs is the time spent servicing equipment. The amount of time spent by the maintenance department servicing equipment in the other three departments has been analysed as follows.

Assembly 50%

Finishing 40%

Canteen 10%

The Canteen department's overheads are to be reapportioned on the basis of the number of employees in the other three departments.

	Assembly	Finishing	Maintenance	Canteen
Number of employees	20	30	10	2

Complete the overhead analysis sheet below and reapportion the service departments' overheads to the production departments.

Overhead	Assembly $	Finishing $	Mainte-nance $	Canteen $	Total $
Total from above	20	30	10	2	
Reapportion canteen					
Reapportion maintenance					

Reapportion canteen					
Reapportion maintenance					
Reapportion canteen					
Reapportion maintenance					
Total					

Expandable text

Test your understanding 3

A company has three production departments, Alpha, Beta and Gamma, and two service departments, Maintenance (M) and Payroll (P). The following table shows how costs have been allocated and the relative usage of each service department by other departments.

	Production			**Service**	
Department	**Alpha**	**Beta**	**Gamma**	**M**	**P**
Costs	$3,000	$4,000	$2,000	$2,500	$2,700
Proportion M (%)	20	30	25	–	25
Proportion P (%)	25	25	30	20	–

Required

Complete the overhead analysis sheet below and reapportion the service department overheads to the production departments using the reciprocal method.

Overhead	**Alpha $**	**Beta $**	**Gamma $**	**M $**	**P $**
Total overheads					
Reapportion M					
Reapportion P					
Reapportion M					
Reapportion P					

Reapportion M					
Reapportion P					
Total					

5 Appropriate bases for absorption rates

Bases of absorption

- Overheads can also be absorbed into cost units using the following absorption bases:
  - machine-hour rate (when production is machine intensive)
  - labour-hour rate (when production is labour intensive)
  - percentage of prime cost
  - percentage of direct wages.
- The overhead absorption rate (OAR) may be calculated as follows:

$$\textbf{OAR} = \frac{\textbf{Total production overhead}}{\textbf{Total of absorption basis}}$$

- The absorption basis is most commonly units of a product, labour hours, or machine hours.

Departmental OARs

It is usual for a product to pass through more than one department during the production process. Each department will normally have a separate departmental OAR.

- For example, a machining department will probably use a machine-hour OAR.
- Similarly, a labour-intensive department will probably use a labour-hour OAR.
- An alternative to a departmental OAR is what is termed a blanket OAR.
- With blanket OARs, only one absorption rate is calculated for the entire factory regardless of the departments involved in production.
- Blanket OARs are also known as single factory-wide OARs.

Predetermined OARs

Production overheads are usually calculated at the beginning of an accounting period in order to determine an OAR for products before they are sold to customers.

- This means that budgeted (or expected) figures must be used for production overheads and activity levels (machine hours, labour hours).
- The predetermined OAR is calculated as follows.

$$\textbf{OAR} = \frac{\textbf{Budgeted overheads}}{\textbf{Budgeted level of activity}}$$

Illustration 5 - Appropriate bases for absorption rates

Ballard Ltd makes three products A, B and C. Each passes through two departments: Machining and Assembly.

Labour hours used in each department by each product

	Machining	**Assembly**
Product A	1 hr	1 hr
Product B	2 hrs	1/2 hr
Product C	None	4 hrs

Production is expected to be as follows:

Product A	1,000 units
Product B	2,000 units
Product C	500 units

Overheads are budgeted as follows:

Machining	**Assembly**
$100,000	$150,000

Complete the following.

Machining department OAR per hour = $	
Assembly department OAR per hour = $	
Blanket OAR per hour = $	
Overhead absorbed by Product B using a separate departmental overhead rate	

Expandable text

Test your understanding 4

The Major Gnome Manufacturing Company has two departments – Moulding and Painting – and uses a single production OAR based on direct labour hours. The budget and actual data for Period 6 are given below:

	Direct wages hours $	**Labour hours**	**Machine**	**Production overhead $**
Budget				
Moulding	24,000	4,000	12,000	180,000
Painting	70,000	10,000	1,000	100,000
	94,000	14,000	13,000	280,000
Actual				
Moulding	30,000	5,000	14,000	200,000
Painting	59,500	8,500	800	95,000
	89,500	13,500	4,800	295,000

During Period 6, a batch of Pixie Gnomes was made, with the following costs and times:

	Direct wages hours $	Labour hours	Machine
Moulding	726	120	460
Painting	2,490	415	38
	3,216	535	498

The direct material cost of the batch was $890.

Complete the following.

(a)

Using a single blanket OAR based on labour hours, the cost of the batch of Pixie Gnomes is $

It has been suggested that appropriate departmental OARs may be more realistic.

(b) The OAR in:

(i) the moulding department is $

(ii) the painting department is $

(c) Using departmental OARs, the cost of the batch of Pixie Gnomes is $

6 Under- and over-absorption of overheads

Under- and over-absorption of overheads

You will remember that the predetermined OAR is calculated using budgeted (estimated) figures as follows:

$$\textbf{OAR} = \frac{\textbf{Budgeted overheads}}{\textbf{Budgeted level of activity}}$$

- If either or both of the estimates for the budgeted overheads or the budgeted level of activity are different from the actual results for the year then this will lead to one of the following:
  – under-absorption (recovery) of overheads
  – over-absorption (recovery) of overheads.

Absorption of overheads

At the end of an accounting period, the overheads absorbed will be calculated as follows.

Overheads absorbed = predetermined OAR × actual level of activity

- If at the end of this period, the overheads absorbed are greater than the actual overheads, then there has been an over-absorption of overheads.
- If, on the otherhand, the overheads absorbed are less than the actual overheads, then there has been an under-absorption of overheads.
- Under-absorption is sometimes referred to as under-recovery of overheads and over-absorption is sometimes referred to as over-recovery of overheads.

e.g

Illustration 6 - Under- and over-absorption of overheads

The following data relate to Lola Ltd for Period 8.

	Budget	Actual
Overheads	$80,000	$90,000
Labour hours worked	20,000	22,000

Overheads were under/over- absorbed by (delete as appropriate) $ ______

Expandable text

Test your understanding 5

The following data relate to Lola Ltd for Period 9.

	Budget	**Actual**
Overheads	$148,750	$146,200
Machine hours	8,500	7,928

Overheads were under-absorbed by: $

Working backwards

Sometimes you may be given information relating to the actual under- or over-absorption in a period and be expected to calculate the budgeted overheads or the actual number of hours worked.

Expandable text

Illustration 7 - Under and overabsorption of overheads

A business absorbs its fixed production overhead on the basis of direct labour hours. The budgeted direct labour hours for week 24 were 4,200. During that week 4,050 direct labour hours were worked and the production overhead incurred was $16,700. The overhead was under-absorbed by $1,310.

The budgeted fixed overhead for the week (to the nearest $10) was $

Expandable text

Test your understanding 6

A business absorbs its fixed overheads on the basis of machine hours worked. The following figures are available for the month of June:

Budgeted fixed overhead	$45,000
Budgeted machine hours	30,000
Actual fixed overhead	$49,000

If there was an over-absorption of overhead of $3,500 how many machine hours were worked in the month?

Expandable text

7 Relating non-production overheads to cost units

Non-production overheads

We have already seen how non-production overheads can be included in the total cost of a product.

- Non-production overheads include administration, distribution and selling overheads.
- Some companies may wish to relate non-production overheads to cost units in the same way that production overheads are allocated and apportioned to cost units.
- Relating non-production overheads to cost units is important in the calculation of a 'fair' profit for a product.

Relating non-production overheads to cost units

There are two main methods of relating non-production overheads to cost units:

- **Method 1**: Find a basis of apportioning the non-production overheads to cost units that closely corresponds to the non-manufacturing overhead, for example, direct labour hours or direct machine hours.
- **Method 2**: Allocate non-production overheads on the basis of the abililty that the product has to carry the cost, for example, as a proportion of production cost. This is the most commonly used method.

Illustration 8 - Relating non-production overheads to cost units

A company manufactures Product X which has a production cost of $25 per unit. The company has budgeted production costs of $400,000 for the manufacture of Product X and budgeted non-production overheads of $80,000 associated with the production of Product X.

Required:
Calculate the total cost of Product X if non-production overheads are related to cost units on the basis of a proportion of production costs of Product X.

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Test your understanding 7

A company makes two products for which the following information for a period is relevant:

Budgeted production and sales – Product X	10,000 units
Budgeted production and sales – Product Y	10,000 units
Selling overheads	$100,000
Distribution overheads	$80,000
Budgeted sales revenue – Product X	$450,000
Budgeted sales revenue – Product Y	$550,000
Budgeted cost of production – Product X	$300,000
Budgeted cost of production – Product Y	$100,000

Selling overheads are absorbed on the basis of sales value and distribution overheads are absorbed as a percentage of production costs.

Calculate the non-production overheads to be absorbed by Product X.

There is no hard and fast rule as to how to relate non-production overheads to cost units. It is usually a matter of using the information given in the question and your common sense. For example, if you are given information regarding sales value or sales revenue, then this may be a good basis on which to apportion selling overheads.

8 Journal and ledger entries for manufacturing overheads

Production overheads account

The direct costs of production (materials, labour and expenses) are debited in the work-in-progress (WIP) account. Indirect production costs are collected in the production overheads account.

- Non-production overheads are debited to one of the following:
- – administration overheads account
- – selling overheads account
- – distribution overheads account
- – finance overheads account.
- Production overheads, as you will remember, include rent and rates, indirect materials and indirect labour costs.
- Absorbed production overheads are credited to the production overheads account.
- Any difference between the actual and absorbed overheads is known as the under- or over-absorbed overhead and is transferred to the income statement at the end of an accounting period.

e.g

Illustration 9 - Journal and ledger entries for manufacturing

Production overheads

	$000		**$000**
Labour	20	WIP (1)	108
Expenses	92	Under-absorption (Bal. figure) (2)	9
Stores	5		
	117		117

Over/under-absorption of overheads

	$000		$000
Production overheads (2)	9	Income statement (3)	9
	——		——
	9		9
	——		——

(1) Production overheads are usually absorbed on the basis of a percentage of the cost of direct labour or some other basis. The absorbed overheads are 'credited out' of the production overheads account and transferred to the WIP account where they are added to the cost of production, and hence the cost of sales.

(2) The production overheads account acts as a collecting place for all the indirect costs of a production process. All the costs are debited to this account. Once the amount of overheads absorbed has been calculated, they are transferred to the WIP account. The difference between the overheads absorbed and the overheads actually incurred is known as the under- or over-absorbed overhead. This is a balancing figure and is transferred to the Over/under-absorption of overheads account.

(3) At the end of an accounting period, the balance on the Over/under-absorption account is transferred to the income statement where it is written off (under-absorbed overhead) or increases profit (over-absorbed overhead).

Test your understanding 8

Transaction	**Debit which account?**	**Credit which account?**
Indirect materials issued from stores		
Indirect wages analysed in the labour account		
Indirect expenses purchased (cash)		
Production overheads absorbed into the cost of production		
Direct materials issued from stores		

9 Chapter summary

Recap of direct and indirect expenses

- **Direct expenses are part of the prime cost of a product**
- **Indirect expenses are known as overheads**

Accounting for overheads

Journal and ledger entries for manufacturing overheads

Relating non-production overheads to cost units

Allocation and apportionment

- Allocation charges overheads directly to specific departments
- If overheads relate to more than one specific department they must be apportioned (shared) between departments
- Apportioned fixed production overheads include: rent, rates, heating and electricity costs

Reapportionment of service cost centre costs to production cost centres

Reapportionment involves sharing out the fixed production overheads of service cost centres betweenp roduction cost centres

Basic method – one service department does work for another, but not vice versa

Production overhead absorption

Reciprocal method – both service departments do work for each other

Appropriate bases for absorption

- Machine-hour rate
- Labour-hour rate
- Percentage of prime cost
- Percentage of direct wages

Under- and over-absorption of overheads

If either or both of the estimates for budgeted overheads or budgeted level of activity are different from actual results then this will lead to under- or over-absorption (recovery) of overheads

Test your understanding answers

Test your understanding 1

The overhead cost allocated to each unit of Product P is	$	3.90
The cost per unit of Product P is now	$	19.90

Workings

The overhead cost per unit is now $\dfrac{\$7,800}{2000}$ = $3.90 per unit

The cost per unit of Product P is now:

	$
Prime cost	16.00
Overheads	3.90
Total	19.90

Test your understanding 2

Overhead	Basis of apportionment	Assembly $	Finishing $	Mainten-ance $	Canteen $	Total $
Electricity	kW Hours (W1)	2,750	4,500	1,975	775	10,000
Machine depreciation	Machine value (W2)	2,250	1,750	550	450	5,000
Indirect labour	Allocated(W3)	1,600	2,220	11,200	1,500	16,520

Workings

(W1)

Electricity is apportioned to all departments on the basis of kW hours.

Total electricity costs	= $10,000
Total kW hours consumed	= (2,750 + 4,500 + 1,975 + 775)= 10,000 kW hours
Apportioned to Finishing department	= 4,500/10,000 x $10,000 = $4,500

(W2)

Machine depreciation is apportioned to all departments on the basis of machine value.

Total machine depreciation costs	= $5,000
Total machine value	= $(45,000 + 35,000 + 11,000 + 9,000) = $100,000
Apportioned to Maintenance department	=11,000/100,000 x $5,000 = $550

(W3)

Indirect labour costs are allocated directly to all departments based on the indirect labour budget for each department.

Test your understanding 3

Overhead	**Alpha $**	**Beta $**	**Gamma $**	**M $**	**P $**
Total overheads	3,000	4,000	2,000	2,500	2,700
Reapportion M	500 (20%)	750 (30%)	625 (25%)	(2,500)	625 (25%)
					3,325
Reapportion P	831 (25%)	831 (25%)	998 (30%)	665 (20%)	(3,325)
Reapportion M	133 (20%)	200 (30%)	166 (25%)	(665)	166 (25%)
Reapportion P	41 (25%)	42 (25%)	50 (30%)	33 (20%)	(166)
Reapportion M	7 (20%)	10 (30%)	8 (25%)	(33)	8 (25%)
Reapportion P	3 (25%)	2 (25%)	3 (30%)		(8)
Total	4,515	5,835	3,850		

Test your understanding 4

(a)

Using a single blanket OAR based on labour hours, the cost of the batch of Pixie Gnomes is	$	14,806

It has been suggested that appropriate departmental OARs may be more realistic.

(b) The OAR in:

(i) the moulding department is	$	15
(ii) the painting department is	$	10

(c)

Using departmental absorption rates, the cost of the batch of Pixie Gnomes is	$	15,156

Workings

$$\text{Blanket OAR} = \frac{\$280{,}000}{14{,}000} = \$20 \text{ per labour hour}$$

	$
Direct materials	890
Direct labour	3,216
Overheads (535 hours @ $20 per hour)	10,700
TOTAL COST	14,806

	(i) Moulding	**(ii) Painting**
Budgeted overheads	$180,000	$100,000
Budgeted hours	12,000 machine hours	10,000 labour hours
OAR	$15 per machine hour	$10 per labour hour

(c) Cost of a batch of Pixie Gnomes using separate departmental OARs

	$
Direct materials	890
Direct labour	3,216
Moulding overheads (460 x $15)	6,900
Painting overheads (415 x $10)	4,150
TOTAL COST	15,156

Test your understanding 5

Overheads were under-absorbed by: $ 7,460

OAR = $\frac{\$148,750}{8,500}$ = $17.50 per machine hour

Overhead absorbed = 7,928 × $17.50 = $138,740

Actual overhead = $146,200

Under-absorbed overhead = $7,460

Test your understanding 6

35,000 machine hours were worked in the month.

Workings

$$\text{OAR} = \frac{\$45{,}000}{30{,}000} = \$1.50 \text{ per hour}$$

Actual overhead	$49,000
Over-absorbed overhead	$3,500
Absorbed overhead	$52,500

$$\text{Machine hours worked} = \frac{\text{Overheads absorbed}}{\text{Overhead absorption rate}}$$

$$= \frac{\$52{,}000}{\$1.50}$$

$$= 35{,}000 \text{ hours}$$

Test your understanding 7

$$\text{Selling overheads - Product X} = \frac{\text{Budgeted sales revenue Product X}}{\text{Total budgeted sales revenue}} = \frac{\$450{,}000}{\$1{,}000{,}000}$$

= 45% of selling overheads
= 45% of $100,000 = $45,000

Distribution overheads are absorbed as a percentage of production costs.

Total production costs = $(300,000 + 100,000) = $400,000

$$\text{Distribution overheads related to Product X} = \frac{\$300{,}000}{\$400{,}000} \times \$80{,}000 = \$60{,}000$$

Budgeted production = 10,000 units, therefore budgeted selling overhead per unit = $60,000/10,000 = $6.00 per unit.

Test your understanding 8

Transaction	Debit which account?	Credit which account?
Indirect materials issued from stores	Production overheads account	Material inventory account
Indirect wages analysed in the labour account	Production overheads account	Labour account
Indirect expenses purchased (cash)	Production overheads account	Bank
Production overheads absorbed into the cost of production	WIP account	Production overheads account
Direct materials issued from stores	WIP account	Material inventory account

chapter

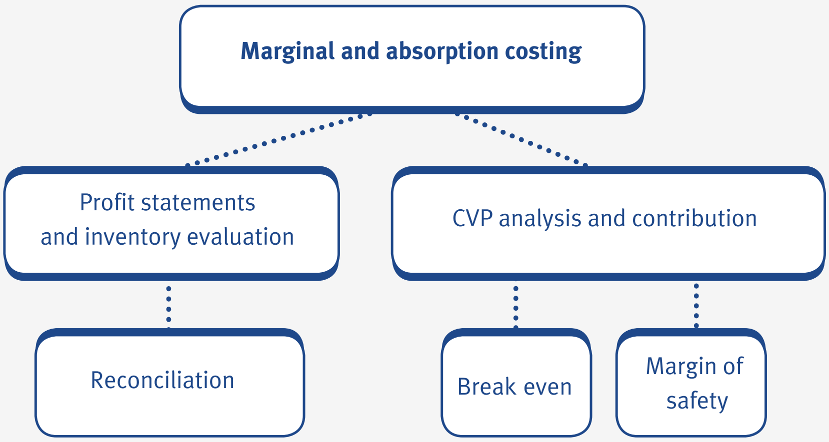
8

Marginal and absorption costing

Chapter learning objectives

Upon completion of this chapter you will be able to:

- explain the importance of, and apply the concept of, contribution using data supplied
- calculate and discuss the effect of absorption and marginal costing on inventory valuation and profit determination using data supplied
- describe the advantages and disadvantages of absorption and marginal costing for a manufacturing business
- understand and use the concept of a contribution to sales ratio, carrying out calculations using supplied data
- calculate and interpret a breakeven point from information supplied
- calculate and interpret a margin of safety from information supplied
- understand and use the concepts of target profit or revenue calculating, from supplied data, the volumes of sales needed to reach the targets
- identify, on supplied charts the elements in: traditional breakeven charts; contribution breakeven charts; profit/volume (P/V) charts
- apply cost/volume/profit (CVP) analysis to single product situations using supplied data.



1 The concept of contribution

Marginal costing

Marginal costing is an accounting system in which variable costs are charged to cost units and fixed costs for the period are written off in full to the income statement.

- Marginal costing is an alternative costing system to absorption costing.
- The marginal cost of a unit of product is the total of the variable costs of the product (i.e. direct materials, direct labour and variable overheads).
- The marginal cost of a product is therefore the additional cost of producing an extra unit of that product.

The contribution concept

The contribution concept lies at the heart of marginal costing. Contribution can be calculated as follows.

Contribution = Sales price – Variable costs

e.g

Illustration 1 The concept of contribution

The following information relates to a company that makes a single product, a specialist lamp, which is used in the diamond-cutting business.

The cost card for the lamp is as follows.

	$	$
Sales price		600
Direct materials	200	
Direct labour	150	
Direct expenses	Nil	
	———	
Prime cost	350	
Variable production overheads	50	
Fixed production overheads	100	
	———	
Total cost	500	

Fixed costs have been estimated to be $120,000 based on a production level of 1,200 lamps.

Let us look at the costs and revenues involved when different volumes of lamps are sold.

		Sales of 1,200 lamps		**Sales of 1,500 lamps**
		$		**$**
Sales revenue		720,000		900,000
Direct materials	240,000		300,000	
Direct labour	180,000		225,000	
Direct expenses	Nil		Nil	
	———		———	
Prime cost	420,000		525,000	
Variable production overheads	60,000		75,000	
Marginal cost of production		**480,000**		**600,000**
CONTRIBUTION		**240,000**		**300,000**
Fixed production overheads		120,000		120,000

Total profit	120,000	180,000
Contribution per unit	**200**	**200**
Profit per unit	100	120

- We can see that the profit per lamp has increased from **$100** when 1,200 lamps are sold to **$120** when 1,500 lamps are sold.
- This is because all of the variable costs (direct materials, direct labour, direct expenses and variable overheads) have increased but the fixed costs have remained constant at $120,000.

Based on what we have seen above, the idea of profit is not a particularly useful one as it depends on how many units are sold. For this reason, the contribution concept is frequently employed by management accountants.

- Contribution gives an idea of how much 'money' there is available to 'contribute' towards paying for the overheads of the organisation.
- At varying levels of output and sales, contribution per unit is constant.
- At varying levels of output and sales, profit per unit varies.
- **Total contribution = Contribution per unit x Sales volume.**
- **Profit = Total contribution – Fixed overheads**.

Marginal costing and the decision-making process

Marginal costing (and therefore the contribution concept) is widely used in the decision-making process.

- The study of marginal costing and decision making is very important in management accounting. It involves the following topics which are relevant to your study of this paper:
  - CVP analysis
  - relevant costing
  - limiting factor analysis.
- Whilst studying these topics you must keep 'the contribution concept' at the front of your mind.

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Test your understanding 1

Buhner Ltd makes only one product, the cost card of which is:

	$
Direct materials	3
Direct labour	6
Variable production overhead	2
Fixed production overhead	4
Variable selling cost	5

The selling price of one unit is $21.

Budgeted fixed overheads are based on budgeted production of 5,000 units. Opening inventory was 1,000 units and closing inventory was 4,000 units.

Sales during the period were 3,000 units and actual fixed production overheads incurred were $25,000.

(a) Calculate the total contribution earned during the period.

(b) Calculate the total profit or loss for the period.

2 The effect of absorption and marginal costing on inventory valuation and profit determination

Absorption and marginal costing

Marginal costing values inventory at the total variable production cost of a unit of product.

Absorption costing values inventory at the full production cost of a unit of product.

- Inventory values will therefore be different at the beginning and end of a period under marginal and absorption costing.
- If inventory values are different, then this will have an effect on profits reported in the income statement in a period. Profits determined using marginal costing principles will therefore be different to those using absorption costing principles.

Absorption costing income statement (statement of comprehensive income)

In order to be able to prepare income statements (statement of comprehensive income) under absorption costing, you need to be able to complete the following proforma.

Absorption costing income statement (statement of comprehensive income)

	$	$
Sales		X
Less Cost of sales:(valued at full production cost)		
Opening inventory	X	
Variable cost of production	X	
Fixed overhead absorbed	X	
	X	
less closing inventory	(X)	
		(X)
		X
(under)/over-absorption		(X) / X
Gross profit		X
Less Non-production costs		(X)
Profit/loss		X

- **Valuation of inventory** – opening and closing inventory are valued at full production cost under absorption costing.
- Under/over-absorbed overhead – an adjustment for under or over absorption of overheads is necessary in absorption costing income statements.

Marginal costing income statement (statement of comprehensive income)

In order to be able to prepare income statements (statement of comprehensive income) under marginal costing, you need to be able to complete the following proforma.

Marginal costing income statement (statement of comprehensive income)

	$	$
Sales		X
Less Cost of sales: (marginal production costs only)		
Opening inventory	X	
Variable cost of production	X	
less Closing inventory	(X)	
		(X)
		X
Less Other variable costs		(X)
Contribution		X
Less fixed costs (actually incurred)		(X)
Profit/loss		X

- Valuation of inventory – opening and closing inventory are valued at full marginal cost under marginal costing.
- Under/over-absorbed overhead – no adjustment for under or over absorption of overheads is needed in marginal costing income statements (statement of comprehensive income). The fixed costs actually incurred are deducted from contribution earned in order to determine the profit for the period.

e.g

Illustration 2 – Effects of absorption and marginal costing

A company commenced business on 1 March making one product only, the cost card of which is as follows:

	$
Direct labour	5
Direct material	8
Variable production overhead	2
Fixed production overhead	5
Standard production cost	$20

The fixed production overhead figure has been calculated on the basis of a budgeted normal output of 36,000 units per annum. The fixed production overhead incurred in March was $15,000 each month.

Selling, distribution and administration expenses are:

Fixed	$10,000 per month
Variable	15% of the sales value

The selling price per unit is $35 and the number of units produced and sold were:

	March (units)
Production	2,000
Sales	1,500

Prepare the absorption costing and marginal costing income statements for March.

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Test your understanding 2

Duo Ltd makes and sells two products, Alpha and Beta. The following information is available for period 3:

	Alpha	**Beta**
Production (units)	2,500	1,750
Sales (units)	2,300	1,600
Opening stock (units)	0	0
Financial data:	**Alpha**	**Beta**
	$	**$**
Unit selling price	90	75
Unit cost:		
direct materials	15	12
direct labour	18	12
variable production overheads	12	8
fixed production overheads	30	20
variable selling overheads	1	1

Fixed production overheads for the period were $105,000 and fixed administration overheads were $27,000.

Required:

(a) Prepare an income statement for period 3 based on marginal costing principles.

(b) Prepare an income statement for period 3 based on absorption costing principles.

Reconciling profits reported under the different methods

We have now seen how profits differ under absorption and marginal costing when inventory increases or decreases in a period.

- If inventory levels increase, absorption costing gives the higher profit.
- If inventory levels decrease, marginal costing gives the higher profit.
- If inventory levels are constant, both methods give the same profit.

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Illustration 3 - Effects of absorption and marginal costing

A company commenced business on 1 March making one product only, the cost card of which is as follows.

	$
Direct labour	5
Direct material	8
Variable production overhead	2
Fixed production overhead	5
Full production cost	$20

- Marginal cost of production = $(5 + 8 + 2) = $15
- Full cost of production = $20 (as above)
- Difference in cost of production = $5 which is the fixed production overhead element of the full production cost.
- This means that each unit of opening and closing inventory will be valued at $5 more under absorption costing.

The number of units produced and sold was as follows.

	March (units)
Production	2,000
Sales	1,500

Closing inventory at the end of March is the difference between the number of units produced and the number of units sold, i.e. 500 units (2,000 – 1,500).

- Loss for March under absorption costing = $375 (as calculated in **Illustration 2**).
- Loss for March under marginal costing = $2,875 (as calculated in **Illustration 2**).
- Difference in loss (profits) = $2,875 – $375 = $2,500.

- This difference can be analysed as being due to the fixed overhead held in inventory, i.e. 500 units of inventory 'holding' $5 fixed overhead per unit.
- 500 x $5 = $2,500 which is the difference between the profit in the profit statements under the different costing methods for March.

Test your understanding 3

(a) In a period where opening inventory was 5,000 units and closing inventory was 3,000 units, a company had a profit of $92,000 using absorption costing. If the fixed overhead absorption rate was $9 per unit, calculate the profit using marginal costing.

(b) When opening inventory was 8,500 litres and closing inventory was 6,750 litres, a company had a profit of $62,100 using marginal costing. The fixed overhead absorption rate was $3 per litre. Calculate the profit using absorption costing.

3 The advantages and disadvantages of absorption and marginal costing

Advantages and disadvantages of absorption and marginal costing

Advantages of marginal costing	Advantages of absorption costing
1 Contribution per unit is constant unlike profit per unit which varies with changes in sales volumes. 2 There is no under or over absorption of overheads (and hence no adjustment is required in the income statement). 3 Fixed costs are a period cost and are charged in full to the period under consideration. 4 Marginal costing is useful in the decision-making process. 5 It is simple to operate.	• Absorption costing includes an element of fixed overheads in inventory values (in accordance with SSAP 9). • Analysing under/over absorption of overheads is a useful exercise in controlling costs of an organisation. • In small organisations, absorbing overheads into the costs of products is the best way of estimating job costs and profits on jobs.

- The main disadvantages of marginal costing are that closing inventory is not valued in accordance with SSAP 9 principles and that fixed production overheads are not 'shared' out between units of production, but written off in full instead.
- The main disadvantages of absorption costing are that it is more complex to operate than marginal costing and it does not provide any useful information for decision making (like marginal costing does).

4 Contribution to sales ratios and breakeven points

CVP analysis

CVP analysis makes use of the contribution concept in order to assess the following measures for a single product:

- contribution to sales (C/S) ratio
- breakeven point
- margin of safety
- target profit.

C/S ratio

The C/S ratio of a product is the proportion of the selling price that contributes to fixed overheads and profits. It is comparable to the gross profit margin. The formula for calculating the C/S ratio of a product is as follows:

$$\text{C/S ratio} = \frac{\text{Contribution per unit}}{\text{Selling price per unit}} \quad \text{or} \quad \frac{\text{Total contribution}}{\text{Total sales revenue}}$$

The C/S ratio is sometimes referred to as the P/V ratio.

Illustration 4 - Contribution to sales ratios and breakeven points

The following information relates to Product J.

Selling price per unit	$20
Variable cost per unit	$12
Fixed costs	$100,000

Calculate the contribution to sales ratio.

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Breakeven point

The breakeven point is the point at which neither a profit nor a loss is made.

- At the breakeven point the following situations occur.

  Total sales revenue = Total costs, i.e. Profit = 0

  or

  Total contribution = Fixed costs, i.e. Profit = 0

- The following formula is used to calculate the breakeven point in terms of numbers of units sold.

$$\textbf{Breakeven point (in terms of numbers of units sold)} = \frac{\textbf{Fixed costs}}{\textbf{Contribution per unit}}$$

(1) It is also possible to calculate the breakeven point in terms of sales revenue using the C/S ratio. The equation is as follows:

$$\textbf{Breakeven point (in terms of sales revenue)} = \frac{\textbf{Fixed costs}}{\textbf{C/S ratio}}$$

Illustration 5 - Contribution to sales ratios and breakeven points

The following information relates to Product K.

Selling price per unit	\$20
Variable cost per unit	\$12
Fixed costs	\$100,000

Required:

(a) Calculate the breakeven point in terms of numbers of units sold.

(b) Calculate the breakeven point in terms of sales revenue.

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5 Margin of safety and target profits

Margin of safety

The margin of safety is the amount by which anticipated sales (in units) can fall below budget before a business makes a loss. It can be calculated in terms of numbers of units or as a percentage of budgeted sales.

The following formulae are used to calculate the margin of safety:

Margin of safety (in terms of units) = Budgeted sales – Breakeven point sales

Margin of safety (as a % of budgeted sales) =

$$\frac{\textbf{Budgeted sales – Breakeven sales}}{\textbf{Budgeted sales}} \times 100\%$$

Illustration 6 - Margin of safety and target profits

Arrow Ltd manufactures Product L to which the following information relates.

Selling price per unit	$20
Variable cost per unit	$12
Fixed costs	$100,000

Budgeted sales for the period are 16,000 units.

Required:

(a) Calculate the margin of safety in terms of units.

(b) Calculate the margin of safety as a % of budgeted sales.

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Target profit

Sometimes an organisation might wish to know how many units of a product it needs to sell in order to earn a certain level of profit (or target profit). This can be calculated by using the following formula.

$$\textbf{Sales volume to achieve a target profit} = \frac{\textbf{(Fixed costs + Required profit)}}{\textbf{Contribution per unit}} = \frac{\textbf{Target contribution}}{\textbf{Contribution per unit}}$$

Illustration 7 - Margin of safety and target profits

Arrow Ltd manufactures Product L and wishes to achieve a profit of $20,000. The following information relates to Product L.

Selling price per unit $20

Variable cost per unit $12

Fixed costs $100,000

Required:

Calculate the sales volume required to achieve a profit of $20,000.

Expandable text

Test your understanding 4

The following information relates to Product Alpha.

Selling price per unit $100

Variable cost per unit $56

Fixed costs $220,000

Budgeted sales are 7,500 units.

Required:

(a) Calculate the C/S ratio.

(b) Calculate the breakeven point in terms of units sold.

(c) Calculate the breakeven point in terms of sales revenue.

(d) Calculate the unit sales required to achieve a target profit of $550,000.

(e) Calculate the margin of safety (expressed as a percentage of budgeted sales).

6 Breakeven charts and P/V charts

Breakeven charts and P/V charts

The measures that we have calculated above (breakeven point, margin of safety and so on) can also be determined by drawing and interpreting the following graphs:

- traditional breakeven charts
- contribution breakeven charts
- P/V charts.

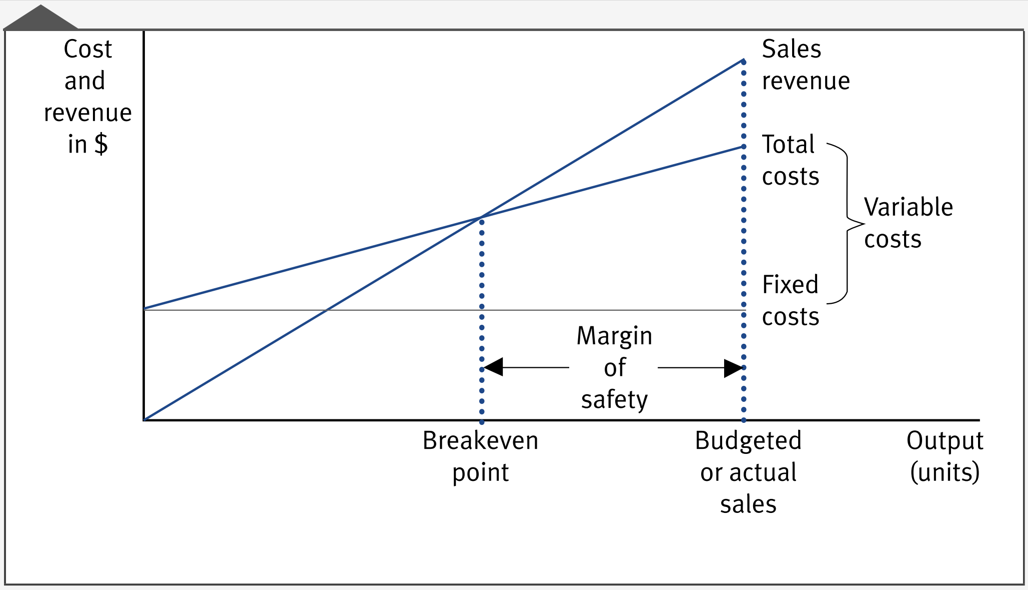
It is important that you are able to identify the points that represent the margin of safety, breakeven point, total contribution, fixed costs, variable costs, total costs and budgeted sales on the graphs that we are going to study in this chapter.

Traditional breakeven charts

The traditional breakeven chart plots total costs and total revenues at different levels of output.

Illustration 8 - Breakeven charts and P/V charts

Traditional breakeven chart



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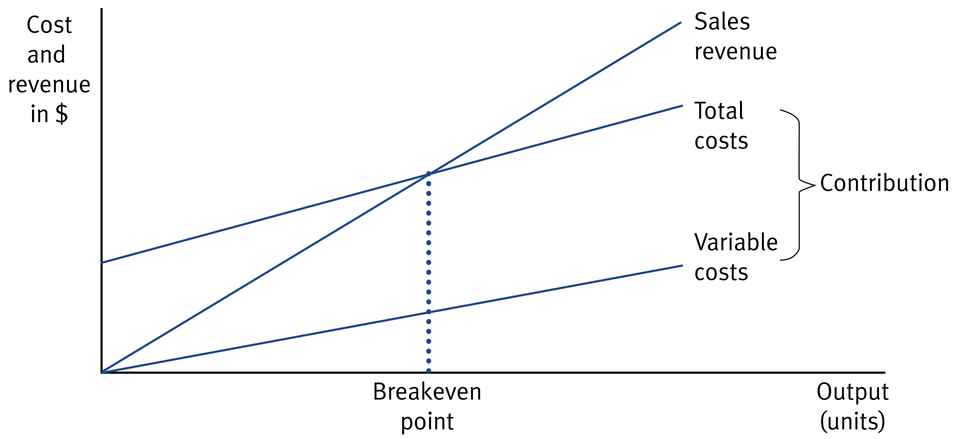
Contribution breakeven charts

A variation on the traditional breakeven chart is the contribution breakeven chart. The main differences between the two charts are as follows.

- The traditional breakeven chart shows the fixed cost line whereas the contribution chart shows the variable cost line.
- Contribution can be read more easily from the contribution breakeven chart than the traditional breakeven chart.

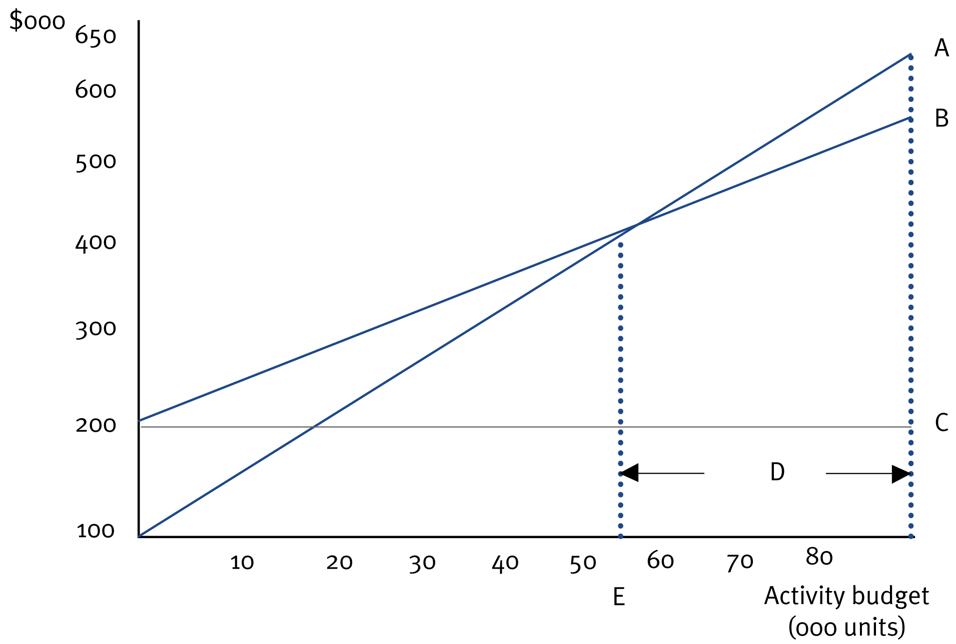
Illustration 9 - Breakeven charts and P/V charts

Contribution breakeven chart



Test your understanding 5

Study the traditional breakeven chart shown below.



(a) What does line A represent?

(b) What does line B represent?

(c) What does line C represent?

(d) What does area D represent?

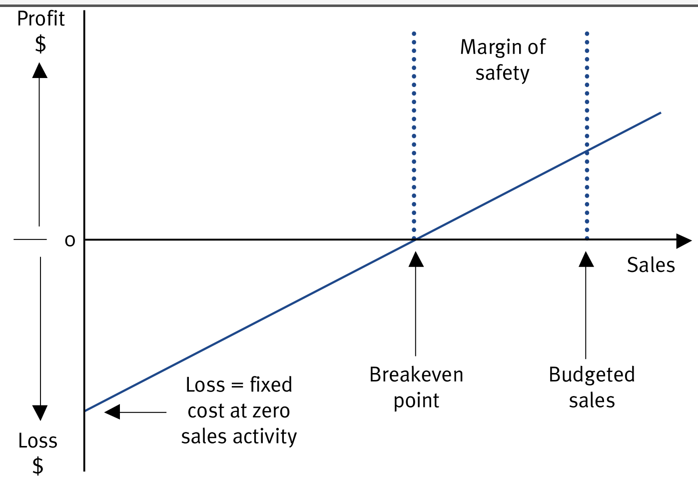
(e) What does point E represent?

P/V charts

Breakeven charts usually show both costs and revenues over a certain level of activity. They do not highlight directly the amounts of profits or losses at the various levels of activity. However, a P/V chart clearly identifies the net profit or loss at different levels of activity.

Illustration 10 - Breakeven charts and P/V charts

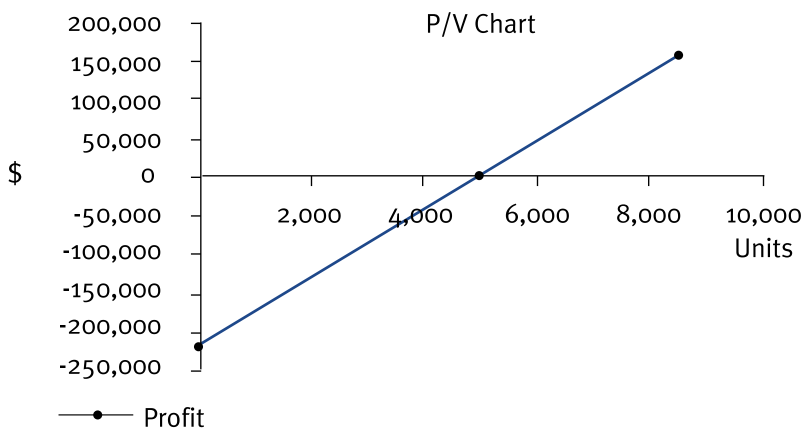
P/V chart



The chart above shows the amount of net profit or loss at different levels of (sales) activity. Make sure that you are able to identify the points shown on the graph above in an examination.

Test your understanding 6

Study the P/V chart shown below.



(a) Estimate the number of units required to break even.

(b) Estimate the fixed costs.

7 Chapter summary

MARGINAL AND ABSORBTION COSTING

The effect of absorption and marginal costing on inventory value and profit determination

Inventory up:
AC → MC profits

Inventory down:
AC ← MC profits

The concept of contribution

Contribution = Sales Price – Variable costs

Widely used in decision-making process, e.g. CVP analysis

The advantages and disadvantages of absorption and marginal costing

Each costing method has a number of advantages and disadvantages

CVP analysis

Contribution to sales ratios and breakeven points

Breakeven point (units)

$$\frac{\text{Fixed cost}}{\text{Contribution per unit}}$$

Breakeven point (revenue)

$$\frac{\text{Fixed cost}}{\text{C/S ratio}}$$

C/S ratio

$$\frac{\text{Contribution per unit}}{\text{Selling price per unit}} \text{ or } \frac{\text{Total contribution}}{\text{Total sales revenue}}$$

Breakeven charts and P/V charts

Traditional breakeven charts

Contribution breakeven charts

P/V charts

Margin of safety and target profits

Margin of safety (in terms of units):
Budgeted sales – Breakeven point sales

MOS may also be stated as a percentage of budgeted sales

Target profit:

$$\frac{\text{Fixed costs+Required profit}}{\text{Contribution per unit}}$$

Test your understanding answers

Test your understanding 1

(a) Total variable costs = $(3 + 6 + 2 + 5) = $16

Contribution per unit (selling price less total variable costs) = $21 – $16 = $5

Total contribution earned = 3,000 x $5 = $15,000

(b) Total profit/(loss) = Total contribution – Fixed production overheads incurred

= $(15,000 – 25,000)

= $(10,000)

Test your understanding 2

(a)

	Alpha		Beta		Total
	$000	$000	$000	$000	$000
Sales		207		120	
Opening inventory	–		–		
Variable production cost (2500 x 45; 1750 x 32)	112.5		56		
Closing inventory (200 x 45; 150 x 32)	(9)		(4.8)		
		(103.5)		(51.2)	
		103.5		68.8	
Variable selling costs		(2.3)		(1.6)	
Contribution		101.2		67.2	168.4p
Fixed production costs					(105)
Fixed administration costs					(27)
Profit					36.4

(b)	Alpha		Beta		Total
	$000	$000	$000	$000	$000
Sales		207		120	
Opening inventory	–		–		
Full production costs (2500 x 75; 1750 x 52)	187.5		91		
Closing inventory (200 x 75; 150 x 52)	(15)		(7.8)		
		(172.5)		(83.2)	
		34.5		36.8	71.3
Over-absorbed overhead (working)					5
Gross profit					76.3
Less: non-production overheads					
Less: variable selling overheads					(3.9)
Less: fixed administration overheads					(27.0)
					45.4

Working

	$
Overhead absorbed = (2,500 x $30) + (1,750 x $20)	110,000
Overheads incurred =	105,000
Over-absorbed overhead	5,000

Test your understanding 3

(a)

Absorption costing profit =	$92,000
Difference in profit = change in inventory x fixed cost per unit = 2,000 x $9 =	$18,000
Marginal costing profit =	$110,000

Since inventory levels have fallen in the period, marginal costing shows the higher profit figure, therefore marginal costing profit will be $18,000 higher than the absorption costing profit, i.e. $110,000.

(b)

Marginal costing profit	$62,100
Difference in profit = change in inventory x fixed cost per unit = (8,500 - 6,750) x $3	$(5,250)
Absorption costing profit	$56,850

Inventory levels have fallen in the period and therefore marginal costing profits will be higher than absorption costing profits. Absorption costing profit is therefore $5,250 less than the marginal costing profit.

Test your understanding 4

(a) Contribution per unit = $(100 – 56) = $44

$$\text{C/S ratio} = \frac{\text{Contribution per unit}}{\text{Selling price per unit}} = \frac{\$44}{\$100} = 0.44$$

(b) Breakeven point in terms numbers of units sold

$$= \frac{\text{Fixed costs}}{\text{Contribution per unit}}$$

$$= \frac{\$220{,}000}{\$44} = 5{,}000 \text{ units}$$

(c) Breakeven point in terms of sales revenue

$$= \frac{\text{Fixed costs}}{\text{C/S ratio}}$$

$$= \frac{\$220{,}000}{0.44} = \$500{,}000 \text{ units}$$

(Proof: breakeven units × selling price per unit = 5,000 × $100 = $500,000)

(d) Unit sales to achieve a target profit

$$= \frac{\text{Fixed costs + Required profit}}{\text{Contribution per unit}}$$

$$= \frac{\$(220{,}000 + 550{,}000)}{\$44} = 17{,}500 \text{ units}$$

(e) Margin of safety (as a % of Budgeted sales)

$$= \frac{\text{Budgeted sales} - \text{Break-even sales}}{\text{Budgeted sales}} \times 100\%$$

$$= \frac{7{,}500 - 5{,}000}{7{,}500} \times 100\%$$

= 33.33%

Test your understanding 5

(a) Line A represents total revenue earned (the total revenue line).

(b) Line B represents total costs (total costs line).

(c) Line C represents fixed costs (fixed cost line).

(d) Area D represents the margin of safety.

(e) Point E represents the breakeven point.

Test your understanding 6

(a) Reading from the graph, the number of units required to breakeven can be estimated as 5,000 units. The breakeven point on a P/V chart is the point at which the profit line cuts the x-axis, i.e. the point at which profit is zero.

(b) The fixed costs can be estimated to be $225,000, i.e. the point at which the profit line cuts the y-axis. This is the point at which costs are incurred even when there is zero sales activity.

chapter

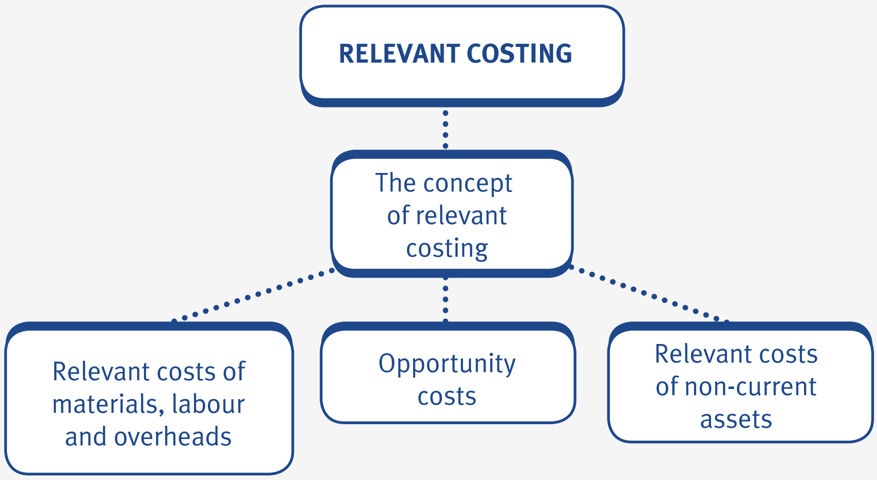
9

Relevant costing

Chapter learning objectives

Upon completion of this chapter you will be able to:

- explain the concept, giving suitable examples, of relevant costing
- calculate, using supplied data, the relevant cost for materials, labour, overheads
- calculate, using supplied data, the relevant costs associated with non-current assets
- explain and apply, using supplied data, the concept of opportunity cost.



1 Relevant costing

The concept of relevant costing

As you already know, the decision-making process involves making a choice between two or more alternatives. Decisions will generally be based on taking the decision that maximises shareholder value, so all decisions will be taken using relevant costs and revenues.

- Relevant costs and revenues are those costs and revenues that change as a direct result of a decision taken.
- Relevant costs and revenues have the following features:
  - They are future costs and revenues.
  - They are cash flows.
  - They are incremental costs and revenues.

Expandable text

Relevant cost terms

Some of the other terms associated with relevant costs and revenues are as follows:

- **Differential costs** are the differences in total costs or revenues between two alternatives.
- **Opportunity cost** is an important concept in decision making. It represents the best alternative that is foregone in taking the decision. The opportunity cost emphasises that decision making is concerned with alternatives and that the cost of taking one decision is the profit or contribution foregone by not taking the next best alternative.

- **Avoidable costs** are the specific costs associated with an activity that would be avoided if that activity did not exist.

In an examination, unless told otherwise, assume that variable costs are relevant costs.

Non-relevant costs

Costs which are not relevant to a decision are known as non-relevant costs and include: sunk costs; committed costs; non-cash flow costs; general fixed overheads; and net book values.

- **Sunk costs** are past costs or historical costs which are not directly relevant in decision making, for example, development costs or market research costs.
- **Committed costs** are future costs that cannot be avoided, whatever decision is taken.
- **Non-cash flow costs** are costs which do not involve the flow of cash, for example, depreciation and notional costs. A notional cost is a cost that will not result in an outflow of cash either now or in the future, for example, sometimes the head office of an organisation may charge a 'notional' rent to its branches. This cost will only appear in the accounts of the organisation but will not result in a 'real' cash expenditure.
- **General fixed overheads** are usually not relevant to a decision. However, some fixed overheads may be relevant to a decision, for example, stepped fixed costs may be relevant if fixed costs increase as a direct result of a decision being taken.
- **Net book values** are not relevant costs because like depreciation, they are determined by accounting conventions rather than by future cash flows.

e.g

Illustration 1 - The concept of relevant costing

A decision has to be made whether to use production method A or B.

The cost figures are as follows.

	Method A		Method B	
	Costs last year $	Expected costs next year $	Costs last year $	Expected costs next year $
Fixed costs	5,000	7,000	5,000	7,000
Variable costs per unit:				
Labour	2	6	4	12
Materials	12	8	15	10

Relevant cost of Method A = $

Relevant cost of Method B = $

Expandable text

Test your understanding -1

A sunk cost is:

A a cost committed to be spent in the current period

B a past cost which is irrelevant for decision making

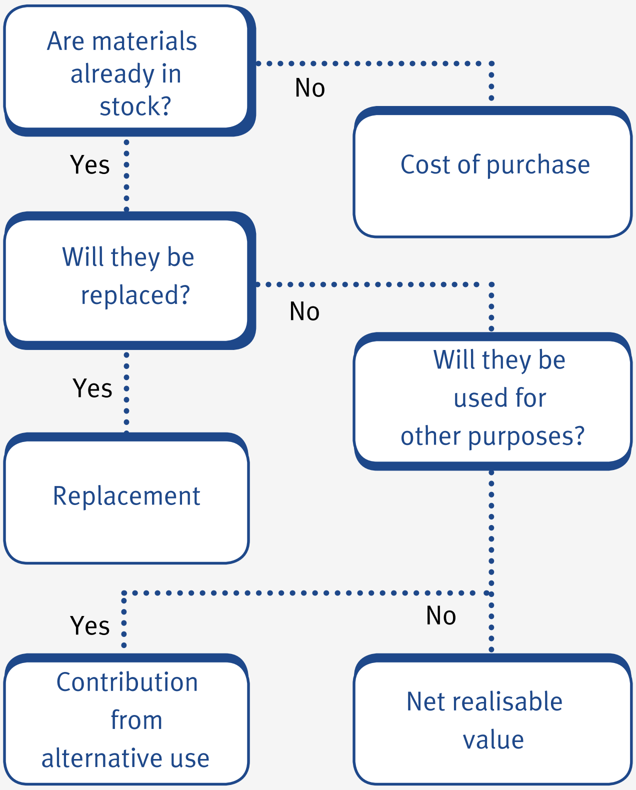
C a cost connected with oil exploration in the North Sea

D a cost unaffected by fluctuations in the level of activity.

2 Relevant costs for materials, labour and overheads

Relevant cost for materials

The relevant cost of materials can be determined by the following decision tree.



Use the diagram shown above when attempting questions on the relevant cost of materials.

e.g

Illustration 2 - Relevant costs for materials, labour and overheads

Z Ltd has 50 kg of material P in inventory that was bought five years ago for $70. It is no longer used but could be sold for $3/kg.

Z Ltd is currently pricing a job that could use 40 kg of material P.

The relevant cost of material P that should be included in the contract is $

Expandable text

Test your understanding-2

A firm is currently considering a job that requires 1,000 kg of raw material. There are two possible situations.

(a) The material is used regularly within the firm for various products. The present inventory is 10,000 kg purchased at $1.80 per kg. The current replenishment price is $2.00 per kg.

The relevant cost per kg is $

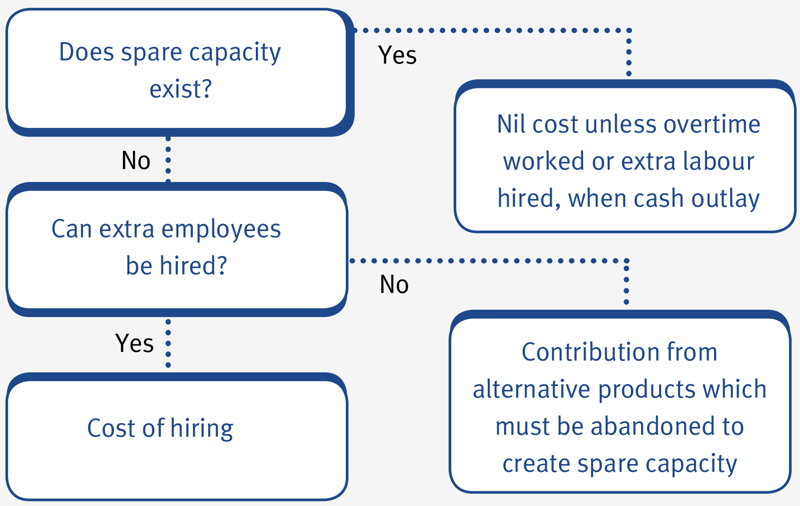
(b) The company has 2,000 kg in inventory, bought 2 years ago for $1.50 per kg, but no longer used for any of the firm's products. The current market price for the material is $2.00, but the company could sell it for $0.80 per kg.

The relevant cost is $

Note: See section 4 for further examples on materials.

Relevant cost for labour

A similar problem exists in determining the relevant costs of labour. In this case the key question is whether spare capacity exists and on this basis a decision tree as shown below can be used to identify the relevant cost.



Use the diagram shown above when doing the following questions on the relevant cost of labour.

Illustration 3 : elevant costs for materials, labour and overheads

(a) 100 hours of unskilled labour are needed for a contract. The company has no surplus capacity at the moment, but additional temporary staff could be hired at $4.50 per hour.

The relevant cost of the unskilled labour on the contract is

$

(b) 100 hours of semi-skilled labour are needed for a contract. There is at the moment roughly 300 hours worth of spare capacity. There is a union agreement that there are no lay-offs. The workers are paid $6.50 per hour.

The relevant cost of the semi-skilled labour on the contract is

$

Expandable text

Test your understanding-3

Z Ltd is pricing a job that involves the use of 20 hours of skilled labour and 50 hours of semi-skilled labour.

The four existing skilled workers are paid $15 per hour with a minimum weekly wage of $450. They are currently working 24 hours a week.

The semi-skilled workforce is currently fully utilised. They are each paid $10 per hour, with overtime payable at time and a half. Additional workers may be hired for $12 per hour.

The relevant labour cost for Z Ltd's job = $

Relevant cost for overheads

In addition to calculating the relevant cost of materials and labour, you may also be required to calculate the relevant cost of overheads.

Relevant cost of overheads

Only those overheads that vary as a direct result of a decision taken are relevant overheads

Illustration 4 - Relevant costs for materials, labour and overheads

JB Ltd absorbs overheads on a machine hour rate, currently $20/hour, of which $7 is for variable overheads and $13 for fixed overheads. The company is deciding whether to undertake a contract in the coming year. If the contract is undertaken, it is estimated that fixed costs will increase for the duration of the contract by $3,200.

Required:

Identify the relevant costs of the overheads for the contract.

Expandable text

Test your understanding-4

Explain why fixed overhead absorption rates are assumed to be irrelevant to a decision and variable overhead absorption rates are assumed to be relevant to the same decision.

3 Relevant costs for non-current assets

Relevant costs of non-current assets

The relevant costs associated with non-current assets, such as plant and machinery, are determined in a similar way to the relevant costs of materials.

- If plant and machinery is to be replaced at the end of its useful life, then the relevant cost is the current replacement cost.
- If plant and machinery is not to be replaced, then the relevant cost is the higher of the sale proceeds (if sold) and the net cash inflows arising from the use of the asset (if not sold).

e.g

Illustration 5 - Relevant costs for non-current assets

A machine which cost $10,000 four years ago has a written-down value of $6,000 and the depreciation to be charged this year is $1,000. It has no alternative use, but it could be sold now for $3,000. In one year's time it will be unsaleable.

Relevant cost of the machine = $

Expandable text

Test your understanding-5

A machine which originally cost $80,000 has an estimated useful life of 10 years and is depreciated at the rate of $8,000 per annum. The net book value of the machine is currently $40,000 and the net sale proceeds if it were to be sold today are $25,000.

Required:

Identify whether the costs associated with the machine are relevant or not. If you think they are not relevant, give reasons.

4 Opportunity costs

The concept of opportunity cost

Opportunity cost represents the best alternative that is foregone in taking a decision. The concept of opportunity cost emphasises that decision making is concerned with alternatives and that the cost of taking one decision is the profit or contribution foregone by not taking the next best alternative.

- If resources to be used on projects are scarce (e.g. labour, materials, machines), then consideration must be given to profits or contribution which could have been earned from alternative uses of the resources.

Expandable Text

Illustartion 6 - Opportunity costs

A new contract requires the use of 50 tons of metal ZX 81. This metal is used regularly on all the firm's projects. There are 100 tons of ZX 81 in inventory at the moment, which were bought for $200 per ton. The current purchase price is $210 per ton, and the metal could be disposed of for net scrap proceeds of $150 per ton. What cost should be charged to the new contract for metal ZX 81?

Expandable text

Test your understanding-6

A new contract requires the use of 50 tons of metal ZX 81. There are 25 tons of ZX 81 in inventory at the moment, which were bought for $200 per ton. The company no longer has any use for metal ZX 81. The current purchase price is $210 per ton, and the metal could be disposed of for net scrap proceeds of $150 per ton. What cost should be charged to the new contract for metal ZX 81?

Illustration 7 - Opportunity costs

A mining operation uses skilled labour costing $4 per hour, which generates a contribution, after deducting these labour costs, of $3 per hour.

A new project is now being considered which requires 5,000 hours of skilled labour. There is a shortage of the required labour. Any used on the new project must be transferred from normal working. What is the relevant cost of using the skilled labour on the project?

Expandable text

Test your understanding -7

A mining operation uses skilled labour costing $4 per hour, which generates a contribution, after deducting these labour costs, of $3 per hour.

A new project is now being considered which requires 5,000 hours of skilled labour. There is a surplus of the required labour which is sufficient to cope with the new project. The workers who are currently idle are being paid full wages. What is the relevant cost of using the skilled labour on the project?

Test your understanding-8

100 hours of skilled labour are needed for a contract. The skilled labour force is working at full capacity at the moment and the workers would have to be taken off production of a different product in order to work on the client's contract. The details of the other product are shown below.

	$/unit
Selling price	60
Direct materials	10
Direct labour	1 hour @ $10/hour
Variable overheads	15
Fixed overheads	15

The skilled workers' pay rate would not change, regardless of which product they worked on.

The relevant cost of the skilled labour on the contract is $

5 Chapter summary

RELEVANT COSTING

The concept of relevant costing

Relevant costs and revenue are:

- Future costs and revenues
- Cash flows
- Incremental costs and revenues

Assume variable costs a relevant unless told otherwise

Relevant costs of materials, labour and overheads

Materials: In stock/replaced/other purpose?

Labour: Spare capacity/extra employees hired? (lost contribution or cost of hiring)

Overheads: Only those that vary as a direct result of a decision taken are relevant overheads

Relevent costs non-current assets

Replaced – Replacement cost

Not replaced – higher of sale proceeds or net cash inflows

Opportunity cost

Opportunity cost represents the best alternative that is foregone in taking a decision. Concept of opportunity cost emphasises that decision making is concerned with alternatives and that the cost of taking one decision is the profit or contribution foregone by not taking the next best.

Test your understanding answers

Test your understanding -1

- B A sunk cost is a past cost and is therefore not relevant for future decisions.

Test your understanding-2

- (a) The relevant cost per kg is $ 2.00
- The material is in inventory and is used regularly by the company. It will therefore be replaced at the current replenishment price of $2.00. The relevant cost per kg is therefore the current replenishment cost of $2.00.
- (b) The relevant cost is $ 800
- After using the material for the job that it is considering, the next best alternative is to sell the material for $0.80 per kg. The relevant cost in this situation is therefore the benefit foregone, i.e. the net realisable value (sale proceeds) of the material in inventory = 1,000 kg x $0.80 = $800.

Test your understanding-3

The relevant labour cost for Z Ltd's job = $	600

- Skilled workers: Minimum weekly wage covers $450/$15 = 30 hours work. Each worker therefore has 6 hours per week spare capacity which is already paid for: 6 x 4 = 24 hours which is sufficient for the job. Relevant cost = $0.
- Semi-skilled workers: Overtime cost = $10 x 11/2 = $15 per hour. It is therefore cheaper to hire additional workers. Relevant cost = 50 hours x $12 = $600.
- Note: see section 4.1 for further examples on labour costs.

Test your understanding-4

Fixed overhead absorption rates are not considered to be relevant costs in decision making because fixed overhead absorption is not a cash flow and does not represent 'real' spending.

Variable overheads, on the other hand, will increase as activity levels increase and will result in additional expenditure being incurred. Such expenditure will represent 'real' cash spending.

Test your understanding-5

	Relevant?	Not relevant – reason
1 $80,000 – original cost	No	Sunk cost
2 $8,000 – annual depreciation	No	Not a cash flow
3 $40,000 – net book value	No	Not a cash flow
4 $25,000 – net sale proceeds	Yes	

Test your understanding-6

The only alternative use for the material held in inventory is to sell it for scrap. To use 25 tons on the contract is to give up the opportunity of selling it for:

25 x $150 = $3,750

The organisation must then purchase a further 25 tons, and assuming this is in the near future, it will cost $210 per ton.

The contract must be charged with:

	$
25 tons @ $150	3,750
25 tons @ $210	5,250
	9,000

Test your understanding -7

How much contribution cash flow is lost if the labour is transferred from doing nothing to the project? Answer: nothing.

The relevant cost is therefore zero.

Test your understanding-8

The relevant cost of the skilled labour on the contract is $ 3,500

How much contribution cash flow is lost if the labour is transferred from normal to working?

Contribution earned fromother product per hour = $(60 – 10 – 10 – 15) = $25

	$
Contribution per hour lost from normal working	25
Add back: labour cost per hour which is not saved	10
	———
Cash lost per labour hour as a result of the labour transfer	35
	———
The contract should be charged with 100 x $35	3,500
	———

chapter

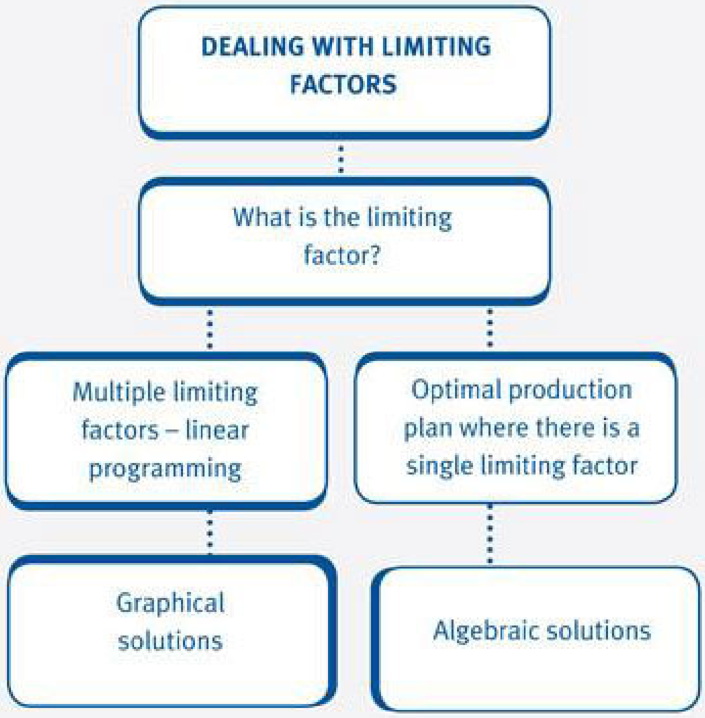
10

Dealing with limiting factors

Chapter learning objectives

Upon completion of this chapter you will be able to:

- identify an organisation's single limiting factor using data supplied
- determine the optimal production plan where an organisation is restricted by a single limiting factor using data supplied
- formulate a linear programming problem involving two variables using data supplied
- determine the optimal solution to a linear programming problem using a graphical approach from a supplied or derived linear programming formulation
- for two supplied simple linear equations, use simultaneous equations to determine where the two lines cross
- use simultaneous equations, where appropriate, in the solution of a linear programming problem.



1 What is a limiting factor?

- A **limiting factor** is a factor that prevents a company from achieving the level of activity that it would like to.
- Limiting factor analysis looks at using the contribution concept to address the problem of scarce resources.
- Scarce resources are where one or more of the manufacturing inputs (materials, labour, machine time) needed to make a product is in short supply.
- Production can also be affected by the number of units of a product that is likely to be demanded in a period (the sales demand).

Illustration 1 What is a limiting factor?

Suppose A Ltd makes two products, X and Y. Both products use the same machine and the same raw material that are limited to 600 hours and $800 per week respectively. Individual product details are as follows.

	Product X	**Product Y**
	per unit	**per unit**
Machine hours	5.0	2.5
Materials	$10	$5

Contribution	$20	$15
Maximum weekly demand	50 units	100 units

Comment on whether machine hours and/or materials are limiting factors

Expandable text

Test your understanding 1

Two products, Alpha and Gamma are made of Material X and require skilled labour in the production process. The product details are as follows:

	Alpha	**Gamma**
	$	**$**
Selling price	10.00	15.00
Variable cost	6.00	7.50
	–––––	–––––
Contribution	4.00	7.50
	–––––	–––––
Material X required per unit	2 kg	4 kg
Skilled labour time required per unit	1 hour	3 hours

The maximum demand per week is 30 units of Alpha and 10 units of Gamma.

The company can sell all the Alphas and Gammas that it can make, but there is a restriction on the availability of both Material X and skilled labour. There are 150 kg of material, and 45 hours of skilled labour, available per week.

Identify the limiting factor.

2 Optimal production plan where there is a single limiting factor

Contribution per unit of limiting factor

In order to decide which products should be made in which order, it is necessary to calculate the contribution per unit of limiting factor (or scarce resource).

$$\text{Contribution per unit of limiting factor} = \frac{\text{Contribution per unit}}{\text{Units of limiting factor required per unit}}$$

Optimal production plan

When limiting factors are present, contribution (and therefore profits) are maximised when products earning the highest amount of contribution per unit of limiting factor are manufactured first. The profit-maximising production mix is known as the optimal production plan.

The optimal production plan is established as follows.

- **Step 1** Calculate the contribution per unit of product.
- **Step 2** Calculate the contribution per unit of scarce resource.
- **Step 3** Rank products.
- **Step 4** Allocate the scarce resource to the highest-ranking product.
- **Step 5** Once the demand for the highest-ranking product is satisfied, move on to the next highest-ranking product and so on until the scarce resource (limiting factor) is used up.

e.g

Illustration 2 - Optimal production plan where there is a single

A company is able to produce four products and is planning its production mix for the following period. Relevant data is given below:

	A	B	C	D
Selling price ($) per unit	19	25	40	50
Labour cost per unit ($)	6	12	18	24
Material cost per unit ($)	9	9	15	16
Maximum demand (units)	1,000	5,000	4,000	2,000

Labour is paid $6 per hour and labour hours are limited to 12,000 hours in the period.

Required:

Determine the optimal production plan and calculate the total contribution it earns for the company.

Expandable text

Test your understanding 2

The following data relates to Products Able and Baker.

		Product	
		Able	**Baker**
Direct materials per unit		$10	$30
Direct labour:			
Grinding	$5 per hour	7 hours per unit	5 hours per unit
Finishing	$7.50 per hour	15 hours per unit	9 hours per unit
Selling price per unit		$206.50	$168.00
Budgeted production		1,200 units	600 units
Maximum sales for the period		1,500 units	800 units

Notes:

(1) No opening or closing inventory is anticipated.

(2) The skilled labour used for the grinding processes is highly specialised and in short supply, although there is just sufficient to meet the budgeted production. However, it will not be possible to increase the supply for the budget period.

Determine the optimal production plan and calculate the total contribution it earns for the company.

3 Multiple limiting factors – linear programming

Linear programming

As we have seen, when there is only one resource that limits the activities of an organisation (other than sales demand), products are ranked in order of contribution per unit of limiting factor in order to establish the optimal production plan.

- When there is more than one limiting factor (apart from sales demand) the optimal production plan cannot be established by ranking products.
- In such situations, a technique known is linear programming is used.

Formulating a linear programming problem

The first stage in solving a linear programming problem is to 'formulate' the problem, i.e. translate the problem into a mathematical formula.

The steps involved in this stage are as follows.

- **Step 1** Define the unknowns, i.e. the variables (that need to be determined).
- **Step 2** Formulate the constraints, i.e. the limitations that must be placed on the variables.
- **Step 3** Formulate the objective function (that needs to be maximised or minimised).

Expandable text

Illustration 3 - Multiple factors – linear programming

A company makes two products, X and Y, and wishes to maximise profit. Information on X and Y is as follows:

	Product X	Product Y
Material kg per unit	1	1
Labour hours per unit	5	10

	$	$
Selling price per unit	80	100
Variable cost per unit	50	50
Contribution per unit	30	50

The company can sell any number of product X, but expects the maximum annual demand for Y to be 1,500 units. Labour is limited to 20,000 hours and materials to 3,000 kg per annum.

Required:

Using the information given, formulate the linear programming problem.

Expandable text

Test your understanding 3

A builder has purchased 21,000 square metres of land on which it is planned to build two types of dwelling, detached and town houses, within an overall budget of $2.1 million.

A detached costs $35,000 to build and requires 600 square metres of land.

A town house costs $60,000 to build and requires 300 square metres of land.

To comply with local planning regulations, not more than 40 buildings may be constructed on this land, but there must be at least 5 of each type.

From past experience the builder estimates the contribution on the detached house to be about $10,000 and on the town house to be about $6,000. Contribution is to be maximised.

Using the information given, formulate the linear programming problem.

4 Graphical solutions

It is possible to solve linear programming problems with two variables by drawing a graph of the constraints and the objective function.

The steps involved in the graphical approach are as follows:

- **Step 1** Define the unknowns, i.e. the variables (that need to be determined).
- **Step 2** Formulate the constraints, i.e. the limitations that must be placed on the variables.
- **Step 3** Formulate the objective function (that needs to be maximised or minimised).
- **Step 4** Graph the constraints and objective function.
- **Step 5** Determine the optimal solution to the problem by reading from the graph.

e.g

Illustration 4 - Graphical solutions

The constraints and the objective function of a linear programming problem concerning how many units of products X and Y to be produced and sold each year have been formulated to be as follows:

Constraint

Materials $x + y \leq 3,000$

Labour hours $5x + 10y \leq 20,000$

Maximum sales $y \leq 1,500$

Non-negativity $x \geq 0$

$y \geq 0$

The objective function is to maximise $30x + 50y$.

Where:

x = number of units of X produced and sold each year

y = number of units of Y produced and sold each year

Required:

Determine the optimal solution to this linear programming problem using a graphical approach.

Expandable text

Test your understanding 4

The constraints and the objective function of a linear programming problem concerning how many detached houses and town houses to be built have been formulated to be as follows:

Constraints:

Area: $600x + 300y \leq 21{,}000$

Cost: $35x + 60y \leq 2{,}100$ (in $000)

Number: $x + y \leq 40$

Minimum: $x \geq 5$

$y \geq 5$

Non-negativity : $x \geq 0$

$y \geq 0$

The objective function is to maximise $10x + 6y$

Where:

x = the number of detached houses

y = the number of town houses.

Determine the optimal solution to this linear programming problem using a graphical approach. (Note: do not draw the iso-contribution line in order to determine the optimal solution.)

5 Algebraic solutions

Using equations to solve linear programming problems

Equations can be used to determine where two lines cross.

- For example, in **Illustration 4**, we established that the optimal solution was at Point C using the graphical method.
- Point C represents the point at which the sales constraint intersects the labour constraint.

Labour constraint $5x + 10y = 20{,}000$ (1)

Materials constraint $x + y = 3{,}000$ (2)

- The basic method is to eliminate one of the two unknowns between the equations.
- This is achieved by adding or subtracting the equations.
- This process is known as solving simultaneous equations.

Illustration 5 Algebraic solutions

Solve the following simultaneous equations.

$5x + 10y = 20{,}000$ (1)

$x + y = 3{,}000$ (2)

Expandable text

Test your understanding 5

Suppose that the optimal solution to a linear programming problem is at the intersection of the following two constraint lines:

(1) $2x + 3y = 1{,}700$

(2) $5x + 2y = 2{,}600$

Determine the values of x and y using simultaneous equations.

Establishing the optimal solution using simultaneous equations

You may consider that the whole process of linear programming would be easier by solving the constraints as sets of simultaneous equations and not bothering with a graph. This is possible and you may get the right answer, but such a technique should be used with caution!

- It is not recommended that you solve sets of simultaneous equations until you have determined graphically which constraints are effective in determining the optimal solution.
- The solutions obtained for linear programming problems using the graphical method rely on the graphs being drawn very clearly so that accurate readings may be taken.
- If it is not easy to read the co-ordinates of the required points from a linear programming graph then the use of simultaneous equations provides a reliable check.

Test your understanding 6

Alfred Ltd is preparing its plan for the coming month. It manufactures two products, the Flaktrap and the Saptrap. Details are as follows.

	Product		
Amount per unit	Flaktrap	Saptrap	Price/wage rate
Selling price ($)	125	165	
Raw material (kg)	6	4	$5/kg
Labour hours			
Skilled	10	10	$3/hour
Semi-skilled	5	25	$3/hour

The company's variable overhead rate is $1 per labour hour (for both skilled and semi-skilled labour). The supply of skilled labour is limited to 2,000 hours per month and the supply of semi-skilled labour is limited to 2,500 hours per month. At the selling prices indicated, maximum demand for Flaktraps is expected to be 150 units per month and the maximum demand for Saptraps is expected to be 80 units per month.

Use simultaneous equations to solve the linear programming problem given above.

6 Chapter summary

DEALING WITH LIMITING FACTORS

What is the limiting factor?

A limiting factor is a factor that prevents a company from achieving the level of activity that it would like to. It is usually sales demand but it might be material, labour, machine capacity and so on.

Multiple limiting factors – linear programming

Step 1 Define the unknowns, i.e. the variables (that need to be declared).
Step 2 Formulate the constraints, i.e. the limitations that must be placed on the variables.
Step 3 Formulate the objective function (that needs to be maximised or minimised).

Optimal production plan where there is a single limiting factor

Make products in order of contribution earned per unit of limiting factor (start with the highest earner).

Contribution per unit of limiting factor =

$$\frac{\text{Contribution per unit}}{\text{Units of limiting factor required per unit}}$$

Graphical solutions

Step 4 Graph the constraints and objective function.
Step 5 Determine the optimal solution to the problem by manipulating the iso-contibution line and reading from the graph or calculating the contribution earned at each point of feasible area.

Algebraic solutions

When it is difficult to read the required points from a linear programming graph, the optimal solution to the problem can be established (or confirmed) by solving the relevant simultaneous equations. Simultaneous equations are used in conjunction with the graphical method in solving linear programming problems, and not on their own.

Test your understanding answers

Test your understanding 1

	Alpha	Gamma	Total
Maximum demand per week (units)	30	10	
Material X required per unit (kg) (W1)	2	4	
Material X required for maximum demand (kg)	60	40	100
Labour time per unit (hours) (W2)	1	3	
Skilled labour time required for maximum demand (hours)	30	30	60

Material X requirement (100 kg) is less than the amount of Material X available in a week (150 kg). Material X does not, therefore, limit the activities of the organisation.

60 hours of skilled labour are required in order to produce the maximum number of units of each product demanded each week. However, only 45 hours of skilled labour are available and so skilled labour is the limiting factor in this situation because it limits the organisation's activities.

Workings

(W1) Alpha = 30 x 2 kg = 60 kg

Gamma = 10 x 4 kg = 40 kg

(W2) Alpha = 30 x 1 hour = 30 hours

Gamma = 10 x 3 hours = 30 hours

Test your understanding 2

	Product			
	Able		Baker	
	$	$	$	$
Selling price		206.50		168.00
Variable costs:				
Direct material	10.00		30.00	
Direct labour: Grinding	35.00		25.00	
Finishing	112.50		67.50	
		(157.50)		(122.50)
Contribution per unit		49.00		45.50

Calculate the contribution per grinding hour (resource that is in short supply) and rank products.

	Able	**Baker**
Contribution per unit	$49.00	$45.50
Grinding hours per unit	7	5
Contribution per grinding hour	$7.00	$9.10
Ranking	2	1

Optimal production plan

Product	Units	Hours used	Hours left(W)	Contribution per unit ($)	Total contribution ($)
Baker	800 (x5)	4,000	7,400	45.50	36,400
Able	1,057 (x7)	7,399	1	49.00	51,793
					88,193

(W) Hours available = 1200 x 7 + 600 x 5 = 11,400

After making 800 Bakers there will be 11,400 – 4000 = 7,400 left.

Test your understanding 3

Step 1 Define the unknowns, i.e. the variables that need to be determined

Let x = the number of detached houses

Let y = the number of town houses

Step 2 Formulate the constraints, i.e. the limitations that must be placed on the variables

Constraint:

Area: $600x + 300y \leq 21{,}000$

Cost: $35x + 60y \leq 2{,}100$ (in \$000)

Number: $x + y \leq 40$

Minimum: $x \geq 5$

$y \geq 5$

Non-negativity : $x \geq 0$

$y \geq 0$

Step 3 Formulate the objective function that needs to be maximised or minimised

The objective is to maximise contribution.

So objective: maximise $10x + 6y$ (in \$000).

Test your understanding 4

The graph of the linear programming problem can be constructed by drawing the following lines:

End points

Area constraint: 600x + 300y = 21,000 (0,70), (35,0)

Cost constraint: 35x + 60y = 2,100 (in $000) (60,0), (0,35)

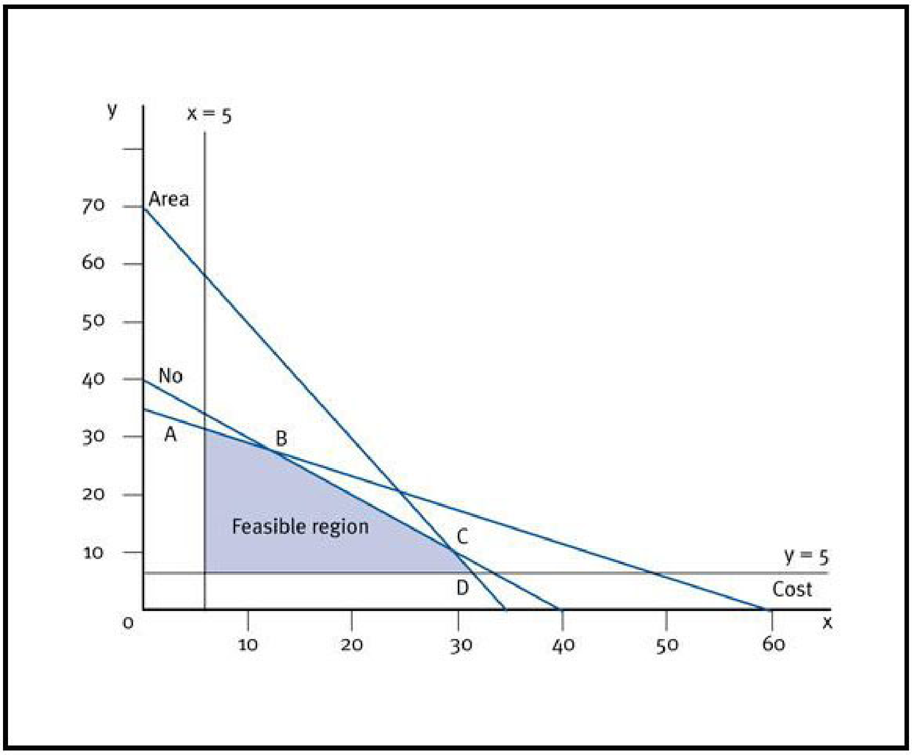
Number constraint: x + y = 40 (0,40), (40,0)

Minimum constraint: x = 5

y = 5
Non-negativity constraints: x = 0

y = 0

Graph of linear programming problem



The feasible region is shown by the shaded area.

Reading from the graph, Point A is at (5,32), Point B is at (12,28), Point C is at (30,10) and Point D is at (32,5). We can calculate the contributions earned at each of these points as follows:

Point	x	y	Contribution($000) (10x + 6y)
A	5	32	242
B	12	28	288
C	30	10	360
D	32	5	350

Maximum contribution is earned at Point C ($360,000) where x = 30 and y = 10.

The objective function is to maximise 10x + 6y.

At Point C, x = 30 and y = 10. If we insert these values into our objective function, maximum contribution is $360,000 [(10 x 30)+ (6 x 10) = $360,000].

The optimal solution to this linear programming problem is therefore: build 30 detached houses, and 10 town houses.

Test your understanding 5

In this example, if we multiply equation (1) by 5 and equation (2) by 2, we will have 10x in both multiplied equations, as follows:

(3)	$10x + 15y$	$= 8,500$
(4)	$10x + 4y$	$= 5,200$

Next, we can get rid of x by subtracting equation (4) from equation (3).

(3)	$10x + 15y$	$= 8,500$
(4)	$10x + 4y$	$= 5,200$
		______
Subtract	$11y$	$= 3,300$
		______
	y	$= 300$

We can establish the value of x by substituting 300 for y in any of the equations.

(1)	$2x + 3\ (300)$	$= 1,700$
	$2x + 900$	$= 1,700$
	$2x$	$= 800$
	x	$= 400$

Therefore x = 400 and y = 300.

Test your understanding 6

Let x be the number of Flaktraps to be produced each month and y be the number of Saptraps to be produced each month.

The constraints are as follows:

Skilled labour	$10x + 10y \leq 2{,}000$
Semi-skilled labour	$5x + 25y \leq 2{,}500$
Flaktrap demand	$x \leq 150$
Saptrap demand	$y \leq 80$
Non-negativity constraints	$x \geq 0$
	$y \geq 0$

The possible production combinations of Flaktraps (x) and Saptraps (y) are shown in the graph below. This gives a feasibility region of OABCDE.

Determining the optimal solution

Having identified the feasible region we must find the optimal solution within this feasible region. We can do this by drawing the objective function, which is to maximise contribution. We are not given the contribution for each product so we must therefore calculate it.

	Flaktrap	**Saptrap**
	$	**$**
Selling price	125	165
Raw materials	30	20
Labour: skilled	30	30
semi-skilled	15	75
Variable overheads	15	35
Contribution	35	5

The objective function is therefore to maximise contribution = $35x + 5y$

Let $350 = 35x + 5y$

When $x = 0$, $y =$	350/5	= 70
When $y = 0$, then $x =$	350/35	= 10

When x = 0, y = 350/5 = 70

When y = 0, x = 350/35 = 10

Plotting and 'sliding' the objective function gives an optimal point at D.

Point D is at the intersection of:

$$x = 150 \text{ and}$$

$$10x + 10y = 2{,}000$$

Substituting for x gives:

$$10(150) + 10y = 2{,}000$$

$$10y = 500$$

$$y = 50$$

The optimal solution is to produce 150 Flaktraps and 50 Saptraps giving a total contribution of [(150 x 35) + (50 x 5)] = $5,500.

chapter

11

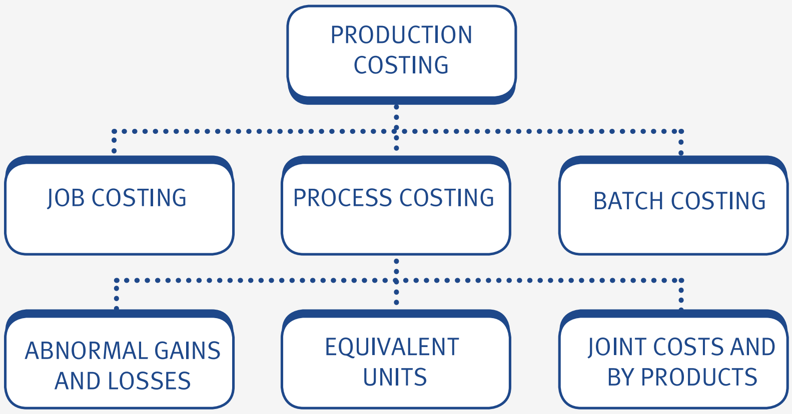
Job, batch and process costing

Chapter learning objectives

Upon completion of this chapter you will be able to:

- describe the characteristics of: job costing, batch costing, process costing and describe situations in which each would be appropriate using suitable examples
- prepare cost records and accounts in job and batch costing situations from information supplied
- for process costing, using a simple numerical example, explain the concepts of normal and abnormal losses and abnormal gains
- calculate the cost per unit of process outputs from data supplied
- prepare process accounts involving normal and abnormal losses and abnormal gains from data supplied
- calculate and explain the concept of equivalent units from data supplied
- apportion process costs between work remaining in process and transfers out of a process using the weighted average method from data supplied (Note: situations involving work-in-progress (WIP) and losses in the same process are excluded)
- apportion process costs between work remaining in process and transfers out of a process using the FIFO method from data supplied
- prepare process accounts in situations where work remains incomplete, from data supplied
- prepare process accounts where losses and gains are identified at different stages of the process, from data supplied
- distinguish between by-products and joint products using appropriate examples

- value by-products and joint products at the point of separation using a variety of ways of splitting pre-separation costs, from data supplied
- show how post-separation costs are treated to value final joint products from data supplied
- prepare process accounts in situations where by-products and/or joint products occur, from data supplied.



1 Different types of production

Costing systems

- Specific order costing is the costing system used when the work done by an organisation consists of separately identifiable jobs or batches.
- Continuous operation costing is the costing method used when goods or services are produced as a direct result of a sequence of continuous operations or processes, for example process and service costing.

Job costing

Job costing is a form of specific order costing and it is used when a customer orders a specific job to be done. Each job is priced separately and each job is unique.

- The main aim of job costing is to identify the costs associated with completing the order and to record them carefully.
- Individual jobs are given a unique job number and the costs involved in completing the job are recorded on a job cost sheet or job card.
- The selling prices of jobs are calculated by adding a certain amount of profit to the cost of the job.

e.g

Illustration 1 – Different types of production

Individual job costs are recorded on a job card similar to the one shown below.

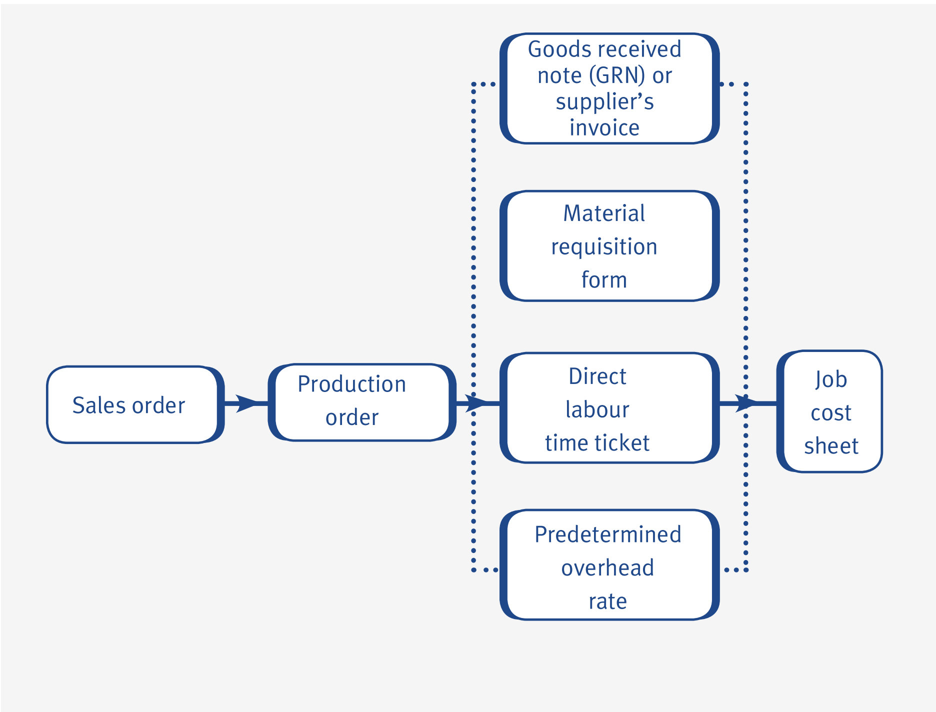
JOB CARD

Customer	Green & Co. Ltd		Job No: 342
Description	Transfer machine	Promised delivery date	3.11.X1
Date commenced	25.9.X1	Actual delivery date	13.11.X1
Price quoted	$2,400		
Despatch note no:	7147		

	Materials estimate $1,250		Labour estimate $100		Overhead estimate $176		Other charges estimate $25	
Date Ref					Hourly rate $11			
20X1 b/f	Cost	Cum $	Hrs	Cum $	Cost $	Cum $	Cost $	Cum $
		1,200	17	110		187		13
6 Nov MR 1714	182	1,382						
7 Nov Consultant's test fee								10
8 Nov MR 1937	19	1,401						
9 Nov MRN								
10 Nov Labour	(26)	1,375						
analysis			5	138		55	242	

Summary	$	Comments
Materials	1,375	
Labour	138	
Overhead	242	
Other charges	23	
	———	
	1,778	
Invoice price		
(invoice number 7147 dated 12.12.X1)	2,400	
	———	
Profit	622	
	———	

The flow of documents in a job costing system is shown as follows:



Batch costing

Batch costing is a form of specific order costing which is very similar to job costing.

1 Within each batch are a number of identical units but each batch will be different.
2 Each batch is a separately identifiable cost unit which is given a batch number in the same way that each job is given a job number.
3 Costs can then be identified against each batch number. For example materials requisitions will be coded to a batch number to ensure that the cc of materials used is charged to the correct batch.
4 When the batch is completed the unit cost of individual items in the batch is found by dividing the total batch cost by the number of items in the batch.

$$\textbf{Cost per unit in batch} = \frac{\textbf{Total production cost of batch}}{\textbf{Number of units in batch}}$$

- Batch costing is very common in the engineering component industry, footwear and clothing manufacturing industries.
- The selling prices of batches are calculated in the same ways as the selling prices of jobs, i.e. by adding a profit to the cost of the batch.

Test your understanding 1

Which of the following are characteristics of job costing?

(i) Homogenous products.

(ii) Customer-driven production.

(iii) Production can be completed within a single accounting period.

A (i) only

B (i) and (ii) only

C (ii) and (iii) only

D (i) and (iii) only

2 Profit calculations for job and batch costing

As we have already seen, the costs of jobs and batches are built up from the cost of materials purchased, cost of wages needed to complete the job or batch production and any direct expenses and overheads.

Once the total production costs of the job or batch have been established, a profit can be calculated and added to the total production costs. There are two main methods of calculating job and batch profits.

- Profit mark-up (or profit on total cost) expresses profit as a percentage of the total production cost of a product.
- Profit margin (or profit on sales), on the other hand, expresses profit as a percentage of the selling price of a product.

e.g

Illustration 2 - Costing for job and batch costing

Jetprint Ltd specialises in printing advertising leaflets and is in the process of preparing its price list. The most popular requirement is for a folded leaflet made from a single sheet of A4 paper. From past records and budgeted figures, the following data has been estimated for a typical batch of 10,000 leaflets:

Artwork	$65
Machine setting	4 hours @ $22 per hour
Paper	$12.50 per 1,000 sheets
Ink and consumables	$40
Printer's wages 4 hours at	$8 per hour

Note: Printer's wages vary with volume.

General fixed overheads are $15,000 per period during which a total of 600 labour hours are expected to be worked.

The firm wishes to achieve 30% profit mark-up when 10,000 leaflets are produced and 30% profit margin when 20,000 leaflets are produced.

Required:

(a) Calculate the selling price for 10,000 leaflets.

(b) Calculate the selling price for 20,000 leaflets.

Expandable text

Test your understanding 2

A business has a job costing system and its aim is to make a net profit of 25% on sales. The cost estimates for Job 264 are as follows:

Direct materials 50 kg @ $4 per kg

Direct labour 30 hours @ $9 per hour

Variable production overhead $6 per direct labour hour

Fixed production overheads are budgeted as $80,000 and are absorbed on the basis of direct labour hours. The total budgeted direct labour hours for the period are 20,000.

Other overheads are recovered at the rate of $40 per job.

Calculate the price that should be quoted for Job 264.

3 Simple process costing

Process costing

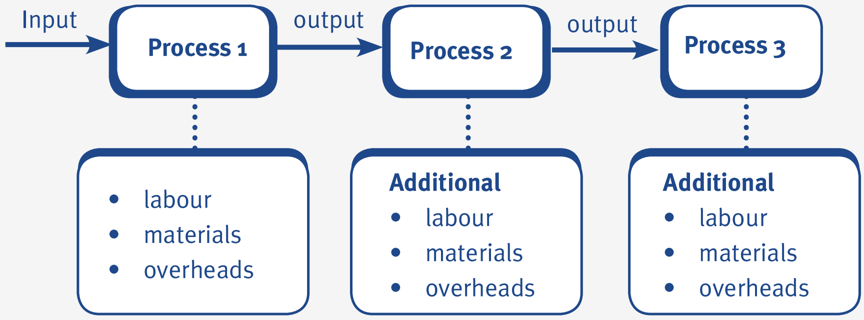
Process costing is a costing method used when mass production of many identical products takes place, for example, the production of bars of chocolate, cans of soup or tins of paint.

- One of the distinguishing features of process costing is that all the products in a process are identical and indistinguishable from each other.

- For this reason, an average cost per unit is calculated for each process.

  $$\textbf{Average cost per unit} = \frac{\textbf{Costs of production}}{\textbf{Expected or normal output}}$$

- Another main feature of process costing is that the output of one process forms the material input of the next process.



- Also, where there is closing work-in-progress (WIP) at the end of one period, this forms the opening WIP at the beginning of the next period.

The details of process costs and units are recorded in a process account which shows (in very general terms) the materials, labour and overheads input to the process and the materials output at the end of the process.

Test your understanding 3

Process 2 Account

	Units	$		Units	$
Input transferred from Process 1	1,000	24,000			
Additional raw materials		5,000			
Direct labour		4,000	Output transferred to Process 3	1,000	
Departmental overheads		3,000			
	1,000			1,000	

Calculate the average cost per unit in Process 2 and complete the Process 2 Account shown above.

4 Process costing with losses and gains

Normal losses

Sometimes in a process, the total of the input units may differ from the total of the output units. This is quite normal and usually happens when there are losses or gains in the process.

- Losses may occur due to the evaporation or wastage of materials and this may be an expected part of the process.
- Losses may sometimes be sold and generate a revenue which is generally referred to as scrap proceeds or scrap value.
- Normal loss is the loss that is expected in a process and it is often expressed as a percentage of the materials input to the process.

e.g

Illustration 3 - Process costing with normal losses

If 1,000 units are input into a process at a cost of $1,800 and normal loss is 10% of input, the average cost per unit is calculated as follows:

$$\textbf{Average cost per unit} = \frac{\textbf{Total cost of inputs}}{\textbf{Units input} - \textbf{Normal loss}}$$

Where:
Total cost of inputs = $1,800
Units input = 1,000 units
Normal loss = 10% of 1,000 units = 100 units

$$\text{Average cost per unit} = \frac{\$1{,}800}{1{,}000 - 100}$$

$$= \frac{\$1{,}800}{900} = \$2 \text{ per unit}$$

Normal loss and scrap value

If normal loss is sold as scrap it is not given a cost but the revenue is used to reduce the input costs of the process.

1 If normal loss is sold as scrap then the formula for calculating the average cost of the units output from a process is modified as follows.

$$\textbf{Average cost per unit} = \frac{\textbf{Total cost of inputs – Scrap value of normal loss}}{\textbf{Units input – Normal loss}}$$

2 If normal loss has a scrap value, it is valued in the process account at this value.

3 If normal loss does not have a scrap value, it is valued in the process accou as $Nil.

e.g

Illustration 4 - Process costing with normal losses and scrap

The following data relates to Process 1.
Materials input – 1,000 units costing $10,000
Labour costs – $8,000
Departmental overheads – $6,000

Normal loss is 4% of input and is sold as scrap for $12 per unit.

Required:

Calculate the average cost per unit in Process 1 and complete the process account and the scrap account.

Expandable text

Abnormal losses and gains

Normal loss is the expected loss in a process. Normal gain is the expected gain in a process. If the loss or the gain in a process is different to what we are expecting (i.e. differs from the normal loss or gain), then we have an abnormal loss or an abnormal gain in the process.

- The costs of abnormal losses and gains are not absorbed into the cost of good output but are shown as losses and gains in the process account.
- Abnormal loss and gain units are valued at the same cost as units of good output.

e.g

Illustration 5 - Process costing with losses and gains

The following data relates to Process 1.

Materials input – 1,000 units costing $10,000

Labour costs – $8,000

Departmental overheads – $6,000

Normal loss is 4% of input and is sold as scrap for $12 per unit.

Actual output = 944 units

Required:

Calculate the average cost per unit in Process 1 and complete the process account, abnormal gains and losses account and the scrap account .

Expandable text

Suggested approach for answering normal loss, abnormal loss/gain questions

(1) Draw the process account, and enter the inputs – units and values.

(2) Enter the normal loss – units and value (if any).

(3) Enter the good output – units only.

(4) Balance the units. Balancing figure is abnormal loss or gain.

(5) Calculate the average cost per unit:

$$\frac{\textbf{Total cost of inputs – Scrap value of normal loss}}{\textbf{Units input – Normal loss}}$$

(6) Value normal loss at scrap value (or at $0 if it does not have a scrap value).

(7) Value the good output and abnormal loss or gain at this average cost per unit.

Test your understanding 4

W&B Ltd produce a breakfast cereal that involves several processes. At each stage in the process, ingredients are added, until the final stage of production when the cereal is boxed up ready to be sold.

In Process 2, W&B Ltd have initiated a quality control inspection. This inspection takes place BEFORE any new ingredients are added in to Process 2. The inspection is expected to yield a normal loss of 5% of the input from Process 1. These losses are sold as animal fodder for $1 per kg.

The following information is for Process 2 for the period just ended:

	Units	**$**
Transfer from Process 1	500 kg	750
Material added in Process 2	300 kg	300
Labour	200 hrs	800
Overheads	–	500
Actual output	755 kg	–

Prepare the process account, abnormal loss and gain account, and scrap account for Process 2 for the period just ended.

5 Work-in-progress (WIP) and equivalent units (EUs)

WIP

At the end of an accounting period there may be some units that have entered a production process but the process has not been completed. These units are called closing WIP (or WIP) units.

- If we assume that there is no opening WIP, then the output at the end of a period will consist of the following:
  - fully-processed units
  - part-processed units (closing WIP).
- Closing WIP units become the opening WIP units in the next accounting period.
- It would not be fair to allocate a full unit cost to part-processed units and so we need to use the concept of EUs which spreads out the process costs of a period fairly between the fully- processed and part-processed units.

Concept of EUs

Process costs are allocated to units of production on the basis of EUs.

- The idea behind this concept is that a part-processed unit can be expressed as a proportion of a fully-completed unit.
- For example, if 100 units are exactly half-way through the production process, they are effectively equal to 50 fully-completed units. Therefore the 100 part-processed units can be regarded as being equivalent to 50 fully-completed units or 50 EUs.

e.g

Illustration 6 - WIP and EUs

For process 1 in ABC Co the following is relevant for the latest period:

Period costs $4,440

Input 800 units

Output 600 fully-worked units and 200 units only 70% complete

There were no process losses.
Required :
Complete the process Account

Expandable text

Different degrees of completion

For most processes the material is input at the start of the process, so it is only the addition of labour and overheads that will be incomplete at the end of the period.

- This means that the material cost should be spread over all units, but conversion costs should be spread over the EUs.
- This can be achieved using an expanded Statement of EUs which separates out the materials and labour costs.
- Note that the term conversion costs is often used to describe the addition of labour and overheads in a process.

Illustration 7 - WIP and EUs

For Process 1 in LJK Ltd the following is relevant for the latest period:

Material costs	500 units @ $8 per unit
Labour	$2,112
Overheads	150% of labour cost

Output: 400 fully-worked units, transferred to Process 2. 100 units only 40% complete with respect to conversion, but 100% complete with respect to materials.

There were no process losses.
Required:
Complete the process Account

Expandable text

Test your understanding 5

A firm operates a process costing system. Details of Process 2 for Period 1 are as follows.

During the period 8,250 units were received from the previous process at a value of $453,750, labour and overheads were $350,060 and material introduced was $24,750.

At the end of the period the closing WIP was 1,600 units which were 100% complete in respect of materials, and 60% complete in respect of labour and overheads. The balance of units was transferred to Finished goods.

There was no opening WIP.

Calculate the cost per EU, the value of finished goods and closing WIP.

Statement of EUs

	Output	Materials		Conversion	
		%	EUs	%	EUs
Fully-worked					
Closing WIP					
Total units					
Costs:					
Total cost					
Cost per EU					

The value of finished goods is $

The value of WIP is $

6 Weighted average costing of production

Until now, we have assumed that there has been no opening WIP. In reality, this is unlikely to be the case.

- Work remaining in process (WIP) and transfers out of a process (fully-completed units) can be valued on different bases: weighted average method and the FIFO method.
- These methods are similar to the valuation methods studied when we looked at materials in an earlier chapter.
- In the weighted average method opening inventory values are added to current costs to provide an overall average cost per unit.
- No distinction is made between units in process at the start of a period and those added during the period.

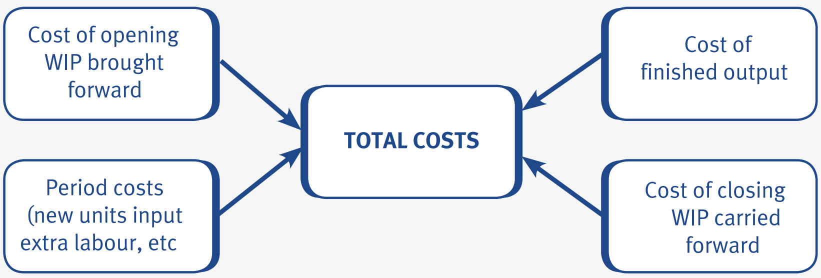


Illustration 8 - Weighted average costing of production

BR Ltd makes a product requiring several successive processes. Details of the first process for August are as follows:

Opening WIP:	400 units
Degree of completion:	
Materials (valued at $19,880)	100 %
Conversion (valued at $3,775)	25 %
Units transferred to Process 2	1,700 units
Closing WIP	300 units
Degree of completion:	
Materials	100 %
Conversion	50 %
Costs incurred in the period:	
Material	$100,000
Conversion	$86,000

There were no process losses.
Required:
Prepare the process account for August using the weighted average method.

Expandable text

Test your understanding 6

A business makes one product that passes through a single process. The business uses weighted average costing. The details of the process for the last period are as follows:

Materials	$98,000
Labour	$60,000
Production overheads	$39,000
Units added to the process	1,000

There were 200 units of opening WIP which are valued as follows:

Materials	$22,000
Labour	$6,960
Production overheads	$3,000

There were 300 units of closing WIP fully complete as to materials but only 60% complete for labour and 50% complete for overheads.

Calculate the following:

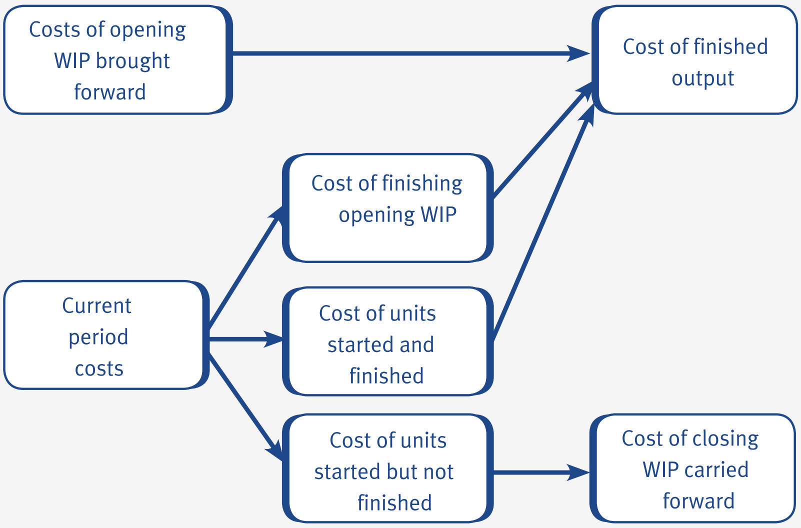
(a) The value of the completed output for the period.

(b) The value of the closing WIP.

7 FIFO costing of production

In the weighted average method opening inventory values are added to current costs to provide an overall average cost per unit. With FIFO, opening WIP units are distinguished from those units added in the period.

- With FIFO it is assumed that the opening WIP units are completed first.
- This means that the process costs in the period must be allocated between:
  - opening WIP units
  - units started and completed in the period (fully-worked units)
  - closing WIP units.
- This also means that if opening WIP units are 75% complete with respect to materials and 40% complete with repect to labour, only 25% 'more work' will need to be carried out with respect to materials and 60% with respect to labour.



- This is a very important point to remember when calculating the EUs of materials and conversion costs and is demonstrated in the following illustration.

e.g

Illustration 9 - FIFO costing of production

BR Ltd makes a product requiring several successive processes. Details of the first process for August are as follows:

Opening WIP:	400 units
Degree of completion:	
Materials (valued at $19,880)	100%
Conversion (valued at $3,775)	25%
Units transferred to Process 2	1,700 units
Closing WIP	300 units
Degree of completion:	
Materials	100%
Conversion	50%
Costs incurred in the period:	
Material	$100,000
Conversion	$86,000
There were no process losses.	

Required:
Prepare the process account for August using the FIFO method.

Expandable text

Expandable text

Test your understanding 7

AXL Ltd operates a process costing system. Details of Process 1 are as follows.

All materials used are added at the beginning of the process. Labour costs and production overhead costs are incurred evenly as the product goes through the process. Production overheads are absorbed at a rate of 100% of labour costs.

The following details are relevant to production in the period:

	Units	Materials	Labour and production overheads
Opening inventory	200	100% complete	75% complete
Closing inventory	100	100% complete	50% complete

Opening inventory

Costs associated with these opening units are $1,800 for materials. In addition $4,000 had been accumulated for labour and overhead costs.

Period costs

Costs incurred during the period were:

Materials	$19,000
Labour costs	$19,000

During the period, 2,000 units were passed to Process 2. There were no losses.

The company uses a FIFO method for valuing process costs.

Calculate the total value of the units transferred to Process 2.

8 Losses made part way through production

In the examples we have looked at so far, the losses have occurred at the end of the process. It is possible that losses (or gains) could be identified part way through a process. In such a case, EUs must be used to assess the extent to which costs were incurred at the time at which the loss was identified.

e.g

Illustration 10 - Losses made part way through production

BLT manufactures chemicals and has a normal loss of 15% of material input. Information for February is as follows:

Material input 200 kg costing $5 per kg

Labour and overheads $4,100

Transfers to finished goods 160 kg

Losses are identified when the process is 40% complete

No opening or closing WIP.

Required:

Prepare the process account for February.

Expandable text

9 Joint and by-products

Introduction

The nature of process costing is such that processes often produce more than one product. These additional products may be described as either joint products or by-products. Essentially joint products are both main products whereas by-products are incidental to the main products.

Joint products

Joint products are two or more products separated in the course of processing, each having a sufficiently high saleable value to merit recognition as a main product.

- Joint products include products produced as a result of theoil-refining process, for example, petrol and paraffin.

- Petrol and paraffin have similar sales values and are therefore equally important (joint) products.

By-products

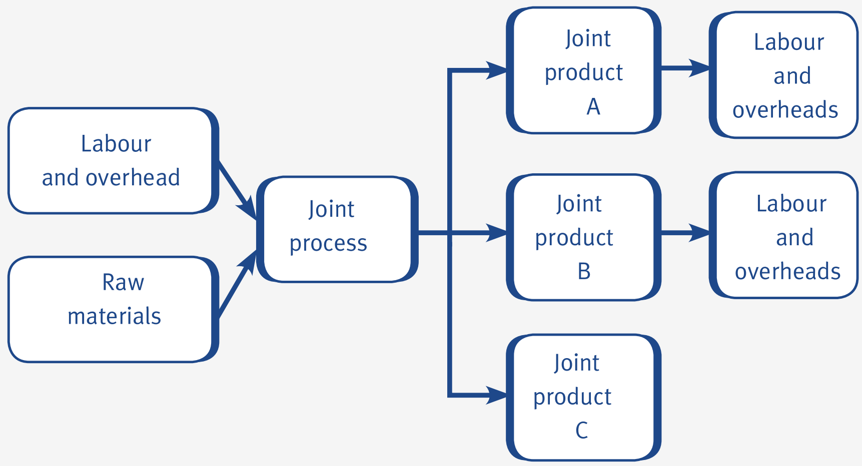
By-products are outputs of some value produced incidentally in manufacturing something else (main products).

- By-products, such as sawdust and bark, are secondary products from the timber industry (where timber is the main or principal product from the process).
- Sawdust and bark have a relatively low sales value compared to the timber which is produced and are therefore classified asby-products.

10 Treatment of joint costs

Accounting treatment of joint products

The distinction between joint and by-products is important because the accounting treatment of joint products and by-products differs.



- Joint process costs occur before the split-off point. They are sometimes called pre-separation costs or common costs.
- The joint costs need to be apportioned between the joint products at the split-off point to obtain the cost of each of the products in order to value closing inventory and cost of sales.
- The basis of apportionment of joint costs to products is usually one of the following:
  - sales value of production (also known as market value)
  - production units
  - net realisable value.

e.g

Illustration 11 - Treatment of joint costs

The following information is relevant for a production process for Period 1:

	$
Direct material cost	10,000
Direct labour cost	5,000
Overheads	3,000
Total cost	18,000

The process produces joint products A and B, which are then sold at the prices given below. The output figure represents all of the output from the process.

	Product A	**Product B**
Units of output	2,000	8,000
Price per unit	$5	$2.50

Required:

Calculate the cost of sales, and gross profit for products A and B assuming:

(i) joint costs are apportioned by market value

(ii) joint costs are apportioned by production units.

Expandable text

Accounting treatment of by-products

As by-products have an insignificant value the accounting treatment is different.

- The costs incurred in the process are shared between the joint products alone. The by-products do not pick up a share of the costs, like normal loss.

- The sales value of the by-product at the split-off point is treated as a reduction in costs instead of an income, again just the same as normal loss.
- If the by-product has no known value at the split-off point but does have a value after further processing, the net income of the by-product is used to reduce the costs of the process

Net income (or net realisable value) =Final sales value – Further processing costs

Illustration 12 - Treatment of joint costs

Process M produces two joint products (A and B) and one by-product (C). Joint costs are apportioned on the basis of sales units.

The following information is relevant.

	Product A	Product B	Total
Sales units	2,000	8,000	10,000
Apportioned joint cost	$3,600	$14,400	$18,000

It is now possible to sell by-product C after further processing for $0.50 per unit. The futher processing costs are $0.20 per unit. 2,000 units of by-product C are produced.

Required:

How are the joint costs of $18,000 apportioned when by-product C is produced?

Expandable text

Test your understanding 8

A company operates a manufacturing process which produces joint products A and B, and by-product C.

Manufacturing costs for a period total $272,926, incurred in the manufacture of:

Product A 16,000 kgs (selling price $6.10 per kg)

Product B 53,200 kgs (selling price $7.50 per kg)

Product C 2,770 kgs (selling price $1.20 per kg)

Product B requires further processing after separation from the other two products. This costs a total of $201,930.

Product C also requires further processing to make it saleable, and this costs $0.40 per kg.

Calculate the total profit earned by Products A and B in the period, using the net realisable values (net income) to apportion joint costs.

11 Process accounts for joint and by-products

You may be required to deal with joint and by-products when preparing process accounts. Joint products should be treated as 'normal' output from a process. The treatment of by-products in process accounts is slightly more complicated.

- There is no information value in calculating a cost for a by-product or measuring its profitability. The accounting treatment of a by-product in process costing is similar to the treatment of normal loss.
- The by-product income could be treated as additional income in the income statement. However, it is more usual to deduct the by-product income from the cost of the process that produces it.
- The by-product income is therefore credited to the process account and debited to a by-product account.
- To calculate the EUs in a period, by-products (like normal loss) are zero EUs.
- When by-products are produced, the cost per EU of direct materials is calculated as follows:

$$\frac{\textbf{Material costs incurred – Scrap value of normal loss – Sales value of by-product}}{\textbf{EUs for materials}}$$

We can now consolidate what we have learned in this chapter by looking at how joint and by-products are recorded in a process account.

e.g

Illustration 13 - Process accounts for joint and by-products

The output from a process consists of two joint products, JP1 and JP2, and a by-product BP. The following data relates to November:

Materials input (18,500 units): cost	$134,000

Conversion costs	$89,500
Normal loss (1,000 units): scrap value	$500
Abnormal loss: (500 units)	$7,000
Output:	
By-product BP (2,000 units): sales value	$6,000
Joint-product JP1: 10,000 units	
Joint-product JP2: 5,000 units	

Required:

Prepare the process account for November, assuming that the common costs are apportioned to joint products on the basis of output units.

Expandable text

Test your understanding 9

The following information relates to a company that produces two joint products, X and Y and one by-product, Z, in a process.

Data for December

Materials input (43,750 units): cost	$266,500
Conversion costs	$105,000
Abnormal loss: (3,750 units)	$37,500
Joint products output:	
Joint-product X: 18,000 units	$180,000
Joint-product Y: 15,000 units	$150,000

Normal loss is 1,000 units. Units lost have a scrap value of $1 per unit.

6,000 units of by-product Z are produced. Each unit is sold for $0.50 per unit.
Prepare the process account for December.

12 Chapter summary

Joint and by-products

- Joint products are two or more products separated in processing, each having sufficiently high saleable value
- By-products are outputs of some value produced at the same time as joint products

Different types of production

Job costing

Form of specific order costing used when customer orders a specific job to be done. Each job is priced separately and is unique.

Batch costing

Form of specific order costing which is very similar to job costing. Each batch is a separately identifiable cost unit which is given a batch number.

Simple process costings

Costing method used when mass production of many identical products takes place, e.g. manufacture of bars of chocolate or cans of soup. All products manufactured are indistinguishable from each other and so an average cost per unit is calculated for each process.

Process costing with gains and losses

- Normal loss = expected loss. Value is $0 unless it has a scrap value.
- Abnormal loss is extra unexpected loss.
- Abnormal gain occurs when actual loss is less than expected.
- Abnormal loss and gain are valued at same value as good output.

Process costing with opening WIP

Weighted average method does not distinguish between opening WIP units and units added in process. FIFO method distinguishes between opening WIP, units started and finished in process ('fully- worked' units) and closing WIP.

Process costing with WIP and EUs

Not fair to allocate full unit cost to a part-processed unit. Idea behind the concept of EUs is that a part-processed unit can be expressed as a proportion of a fully-completed unit.

Process costs are allocated to units in a process on the basis of EUs.

Test your understanding answers

Test your understanding 1

Job costing is customer-driven with customers ordering a specific job to be done. It is also possible for production to be completed within a single accounting period and therefore the correct answer is C.

Test your understanding 2

	$
Direct materials 50 kg x $4	200
Direct labour 30 hours x $9	270
Variable production overhead 30 x $6	180
Fixed overheads $80,000/20,000 x 30	120
Other overheads	40
	———
Total cost (75%)	810
Profit $810.00 x 25/75	270
	———
Job cost (100%)	1,080
	———

Note: a net profit of 25% on sales means that total cost represents 75% of the job cost and the profit represents 25% of the job cost. The profit can therefore be calculated as 25/75ths of the total cost. Similarly, if a net profit of 30% on sales (or a 30% margin) were applicable, the total cost would represent 70% of the job cost and the profit would represent 30% of the job cost and could be calculated as 30/70ths of the total cost.

Test your understanding 3

$$\text{Average cost per unit} = \frac{\text{Costs of production}}{\text{Expected or normal output}}$$

Costs of production = $(24,000 + 5,000 + 4,000 + 3,000) = $36,000

Expected or normal output = 1,000 units

$$\text{Average cost per unit in Process 1} = \frac{\$36,000}{1,000} = \$36 \text{ per unit}$$

Process 2 Account

	Units	$		Units	$
Transfer from Process 1	1,000	24,000			
Additional raw materials		5,000			
Direct labour		4,000			
Departmental overheads		3,000	Transfer to Process 3	1,000	36,000
	1,000	36,000		1,000	36,000

Note that the units completed in Process 1 form the opening WIP in Process 2 and that the units completed in Process 2 form the opening WIP in Process 3.

Test your understanding 4

Process 2 Account

	Kg	$		Units	$
Transfer from Process 1	500	750	Normal loss	25	25
Additional raw materials	300	300	Finished goods	755	2,265
Direct labour		800	Abnormal loss	20	60
Departmental overheads		500			
	800	2,350		800	2,350

Normal loss = 5% of transfer from Process 1 = 500 kg x 0.05 = 25 kg

Scrap value of normal loss = 25 kg x $1 = $25

Cost per unit = ($2,350 – $25) / (800 kg – 25 kg) = $3

Abnormal gains and losses account

	$		$
Process 1	60.00	Scrap (20 $1)	20.00
		Income statement	40.00
	60.00		60.00

Scrap account

	$		$
Process 1 (normal loss)	25.00	Cash (45 $1)	45.00
Abnormal gain and loss	20.00		
	45.00		45.00

Test your understanding 5

Statement of EUs

	Output	Materials %	EUs	Conversion %	EUs
Fully-worked	6,650	100	6,650	100	6,650
Closing WIP	1,600	100	1,600	60	960
Total units			8,250		
Costs:			$453,750		$350,060
			$24,750		
Total cost			$478,500		$350,060
Cost per EU			$58		$46

The value of finished goods is (W1) $\boxed{\$691,600}$

The value of WIP is (W2) $\boxed{\$136,960}$

Workings

(W1)

Value of finished goods:

Materials: 6,650 x $58 = $385,700

Conversion: 6,650 x $46 = $305,900

Total = $691,600

(W2)

Value of closing WIP:

Materials: 1,600 x $58 = $92,800

Conversion: 960 x $46 = $44,160

Total = $136,960

Test your understanding 6

(a)

	%	Materials	%	Labour	%	Overheads
		EU		**EU**		**EU**
Output	100	900	100	900	100	900
Closing WIP	100	300	60	180	50	150
Total EUs		1,200		1,080		1,050
		$		**$**		**$**
Costs – period		98,000		60,000		39,000
Opening WIP		22,000		6,960		3,000
Total costs		120,000		66,960		42,000
Cost per unit		$100		$62		$40
(Total costs/total EUs)						

Value of completed output = 900 x $(100 + 62 + 40) = $181,800

(b)

		$
Materials	300 x $100	30,000
Labour	180 x $62	11,160
Overheads	150 x $40	6,000
Value of closing WIP		47,160

Test your understanding 7

Statement of EUs

	Output	**Materials**		**Conversion**	
		%	**EUs**	**%**	**EUs**
Opening WIP completed	200	0	0	25	50
Fully-worked in process	1,800	100	1,800	100	1,800
Closing WIP	100	100	100	50	50
Total	2,100		1,900		1,900
Costs			$19,000		$19,000 + $19,000*= $38,000
Cost per EU			$10		$20

* Overheads are absorbed at 100% of labour cost.

Labour cost = $19,000, [overheads = $19,000.

Value of units passed to Process 2:

Opening WIP value from last period = $1,800 + $4,000 = $5,800

Opening WIP completed this period:

Conversion only = 50 x $20 = $1,000

Fully-worked current period

Materials = 1,800 x $10 = $18,000

Conversion = 1,800 x $20 = $36,000

Total = $54,000

Total value of units transferred to Process 2 = $60,800

Test your understanding 8

Net revenue from product C = $(1.2 – 0.4) = $0.80

Costs to apportion = Joint process costs – net revenue from product C

= $272,926 – (2,770 x $0.80)

= $270,710

	A	B	Total
	$	$	$
Revenue	97,600	399,000	496,600
Further processing costs –	-	(201,930)	(201,930)
Net realisable values	97,600	197,070	294,670
Joint costs	(89,664)	(181,046)	(270,710)
Total profits	7,936	16,024	23,960

Total net realisable values = $(97,600 + 197,070) =$294,670

Joint costs apportioned to Product A = $\frac{97,600}{294,670}$ × $270,710 =$89,664

Joint costs apportioned to Product B = $\frac{197,070}{294,670}$ × $270,710 =$181,046

Test your understanding 9

Process Amount

	Units	$		Units	$
Direct materials	43,750	266,500	Normal loss (W1)	1,000	1,000
Conversion costs		105,500	Abnormal loss	3,750	37,500
			By-product Z (W2)	6,000	3,000
			Joint product X	18,000	180,000
			Joint product Y	15,000	150,000
	43,750	371,500		43,750	371,500

Workings:

(W1)

Scrap value of normal loss units = 1,000 x $1 = $1,000

(W2)

By-product Z value = 6,000 x $0.50 = $3,000

chapter

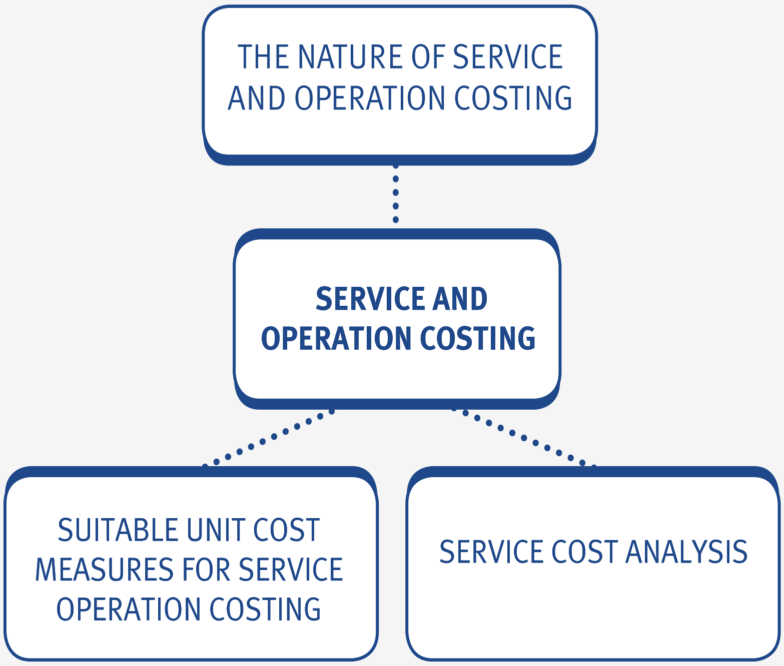
12

Service and operation costing

Chapter learning objectives

Upon completion of this chapter you will be able to:

- give examples of situations where the use of service/operation costing is appropriate
- illustrate, using simple numerical examples, suitable unit cost measures that may be used in different service/operation situations
- carry out service cost analysis in simple service industry situations using data supplied.



1 The nature of service and operation costing

Service costing

Service costing is used when an organisation or department provides a service. There are four main differences between the 'output' of service industries and the products of manufacturing industries.

- **Intangibility** – output is in the form of 'performance' rather than tangible ('touchable') goods.
- **Heterogeneity** – the nature and standard of the service will be variable due to the high human input.
- **Simultaneous production and consumption** – the service that you require cannot be inspected in advance of receiving it.
- **Perishability** – the services that you require cannot be stored.

e.g

Illustration 1 - The nature of service and operation costing

Examples of service industries include the following:

- hotel
- college
- hairdressers
- restaurant.

We can ask the following questions about, e.g. the hotel industry.

(1) Is output in the form of performance?Yes – the hotel provides a bed and possibly breakfast. You will judge the service of the hotel on how comfortable the bed was and how tasty the breakfast was. You cannot really 'touch' the performance of the hotel.

(2) Is the standard of the service variable?Yes – your stay at the hotel may vary each time you stay there. You may not have such a comfortable bed and your breakfast may not be very tasty each time you visit. The standard of service is therefore variable because lots of different staff work at the hotel – the standard of the service you receive may depend on which staff are on duty.

(3) Can you inspect the services in advance of receiving them?In general, you cannot sleep in a hotel bed or eat breakfast at the hotel until you have booked in and made a contract to buy the services of the hotel.

(4) Can the hotel services be stored?No! You cannot take your bed away with you, nor can you keep your breakfast – it must be eaten during the morning of your stay.

Test your understanding 1

List as many service industries that you can think of.

2 Suitable unit cost measures for service/operation costing

Unit cost measures for service costing

One of the main difficulties in service costing is the establishment of a suitable cost unit. In some situations it may be necessary to calculate a composite cost unit.

- A composite cost unit is more appropriate if a service is a function of two variables.
- Examples of composite cost units are as follows:
  - tonne-miles for haulage companies
  - patient-days for hospitals
  - passenger-miles for public transport companies
  - guest-days for hotel services.
- Alternatively, service organisations may use several different cost units to measure the different kinds of service that they are providing.
- The following cost units might be used by a hotel:

Service	Cost unit
Restaurant	Meals served
Function facilities	Hours

Cost per service unit

The total cost of providing a service will include labour, materials, expenses and overheads (the same as the costs associated with the products produced in manufacturing industry).

	$
Direct materials	X
Direct labour	X
Direct expenses	X
Overheads absorbed	X
	–––
TOTAL COST	XX

- In service costing, it is not uncommon for labour to be the only direct cost involved in providing a service and for overheads to make up most of the remaining total costs.
- In service costing it is sometimes necessary to classify costs as being fixed, variable or semi-variable. If costs are semi-variable, it is necessary to separate them into their fixed and variable constituents using the high/low method.
- The cost per service unit is calculated by establishing the total costs involved in providing the service and dividing this by the number of service units used in providing the service.
- The calculation of a cost per service unit is as follows.

$$\textbf{Cost per service unit} = \frac{\textbf{Total costs for providing the service}}{\textbf{Number of service units used to provide the service}}$$

Illustration 2 - Suitable unit cost measures for service/operation

The canteen of a company records the following income and expenditure for a month.

	$	$
Income		59,010
Food	17,000	
Drink	6,000	
Bottled water	750	
Fuel costs (gas for cooking)	800	
Maintenance of machinery	850	
Repairs	250	
Wages	15,500	
Depreciation	1,000	

During the month the canteen served 56,200 meals.

Required:

Calculate the average cost per meal served and the average income per meal served.

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Test your understanding 2

Which of the following are characteristics of service costing?

(i) High levels of direct labour costs as a proportion of total cost.

(ii) Use of composite cost units.

(iii) Use of equivalent units.

A (i) only

B (i) and (ii) only

C (ii) only

D (ii) and (iii) only

3 Service cost analysis

If organisations in the same service industry use the same service cost units then comparisons between the companies can be made easily, as the following example of a service cost statement shows.

e.g

Illustration 3 Service cost analysis

The following figures were taken from the annual accounts of two electricity supply boards working on uniform costing methods.
Meter reading, billing and collection costs:

	Board A	Board B
	$000	$000
Salaries and wages of:		
Meter readers	150	240
Billing and collection staff	300	480
Transport and travelling	30	40
Collection agency charges	–	20
Bad debts	10	10
General charges	100	200
Miscellaneous	10	10
	600	1,000
Units sold (millions)	2,880	9,600
Number of consumers (thousands)	800	1,600
Sales of electricity (millions)	$18	$50
Size of area (square miles)	4,000	4,000

Comparative costs for Boards A and B may be collected as follows and are useful in showing how well (or otherwise) individual services are performing.

Electricity Boards A and B
Comparative costs – year ending 31.12X5

	Board A $000	% of total	Board B $000	% of total
Salaries and wages:				
Meter reading	150	25.0	240	24.0
Billing and collection	300	50.0	480	48.0
Transport/travelling	30	5.0	40	4.0
Collection agency	-	-	20	2.0
Bad debts	10	1.7	10	1.0
General charges	100	16.6	200	20.0
Miscellaneous	10	1.7	10	1.0
	600	100.0	1,000	100.0

The information contained in the following table is much more useful for comparison purposes than the meter reading and billing costs information given above.

	$	$
Cost per:		
Million units sold	208	104
Thousand consumers	750	625
$m of sales	33,333	20,000
Square mile area	150	250

Test your understanding 3

Happy Returns Ltd operates a haulage business with three vehicles. The following estimated cost and performance data is available:

Petrol	$0.50 per kilometre on average
Repairs	$0.30 per kilometre
Depreciation	$1.00 per kilometre, plus $50 per week per vehicle
Drivers' wages	$300.00 per week per vehicle
Supervision and general expenses	$550 per week
Loading costs	$6.00 per tonne

During week number 26 it is expected that all three vehicles will be used, 280 tonnes will be loaded and a total of 3,950 kilometres travelled (including return journeys when empty) as shown in the following table:

Journey	**Tonnes carried**	**Kilometres**
	(one way)	(one way)
1	34	180
2	28	265
3	40	390
4	32	115
5	26	220
6	40	480
7	29	90
8	26	100
9	——	———
	280	1,975
	——	———

Calculate the average cost per tonne-kilometre in week 26.

4 Chapter summary

The nature of service and operation costing

- Intangibility
- Heterogeneity
- Simultaneous production and consumption
- Perishability

Examples of service industries

- Hotels
- Airlines
- Public transport
- College/university
- Accountancy/audit firms
- Utility companies
- Distribution/haulage companies

Service and operation costing

Used when an organisation or department provides a service.

Suitable unit cost measures for service and operation coasting

Composite cost unit used if service is function of two variables, for example:

- Patient-day
- Passenger-mile
- Guest-day
- Tonne-mile (or km)

Service cost analysis

If organisations in the same industry use the same service cost units then comparisons between companies can be made easily

Test your understanding answers

Test your understanding 1

Here are a few. You may have thought of others. Don't forget to ask questions 1-4 above if you are unsure whether it is appropriate to classify an industry as being a service industry:

- hotel
- airline, train and bus companies
- hairdressers
- taxi company
- college/university
- firm of accountants/auditors
- distribution company
- utility company (gas/electricity/telephone)
- banks
- insurance companies
- hospitals.

Test your understanding 2

B

Direct labour costs may be a high proportion on the total cost of providing a service and composite cost units are characteristic features of service costing. (i) and (ii) are therefore applicable and the correct answer is B.

Test your understanding 3

Total costs in Week 26

		$
	$/km	
Petrol	0.50	
Repairs	0.30	
Depreciation	1.00	
	——	
	1.80 x 3,950	7,110
	——	
	$/week	
Depreciation ($50 x 3)	150	
Wages ($300 x 3)	900	
Supervision and general expenses	550	
	——	
		1,600
Loading costs ($6 x 280)		1,680
		——
		10,390
		——
Tonne-km in week 26 (see working)		66,325

(Costs averaged over the outward journeys, not the return, as these are necessary, but carry no tonnes.)

$$\text{Average cost per tonne-kilometre} = \frac{\text{Total cost}}{\text{Total tonne-kilometres}} = \frac{\$10,390}{66,325} = \$0.157$$

Working

Journey	Tonnes carried one way)	Kilometres (one way)	Tonne- kilometres
1	34	180	6,120
2	28	265	7,420
3	40	390	15,600
4	32	115	3,680
5	26	220	5,720
6	40	480	19,200
7	29	90	2,610
8	26	100	2,600
9	25	135	3,375
	280	1,975	66,325

chapter

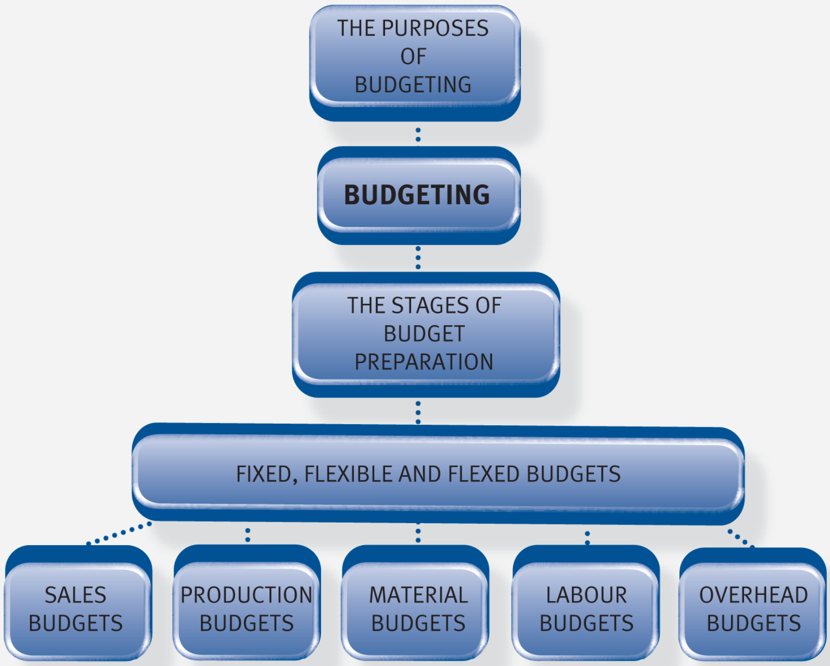
13

Budgeting

Chapter learning objectives

Upon completion of this chapter you will be able to:

- explain why organisations use budgeting, planning, control, communication, co-ordination, authorisation, motivation, evaluation
- explain the stages in the budget process, including the administrative procedures
- explain, giving examples, the term ‘principal budget factor’ (or ‘limiting factor’)
- from data supplied, prepare budgets for sales
- from data supplied, or derived, about the sales budget, prepare budgets for production
- from data supplied, or derived, about the production budget, prepare budgets for material usage
- from data supplied, or derived, about the materials usage budget, prepare budgets for material purchases
- from data supplied, or derived, about the production budget, prepare budgets for labour
- from data supplied, or derived, about the production budget, prepare budgets for overheads
- explain, and prepare from information provided: fixed, flexible, flexed budgets.



1 The purposes of budgeting

Budget theory

A budget is a quantitative expression of a plan of action prepared in advance of the period to which it relates.

Budgets set out the costs and revenues that are expected to be incurred or earned in future periods.

- For example, if you are planning to take a holiday, you will probably have a budgeted amount that you can spend. This budget will determine where you go and for how long.
- Most organisations prepare budgets for the business as a whole. The following budgets may also be prepared by organisations:
  - Departmental budgets.
  - Functional budgets (for sales, production, expenditure and so on).
  - Income statements (in order to determine the expected future profits).
  - Cash budgets (in order to determine future cash flows).

Purposes of budgeting

The main aims of budgeting are as follows:

- **Planning for the future** – in line with the objectives of the organisation.
- **Controlling costs** – by comparing the plan of the budget with the actual results and investigating significant differences between the two.
- **Co-ordination** of the different activities of the business by ensuring that managers are working towards the same common goal (as stated in the budget).
- **Communication** – budgets communicate the targets of the organisation to individual managers.
- **Motivation** – budgets can motivate managers by encouraging them to beat targets or budgets set at the beginning of the budget period. Bonuses are often based on 'beating budgets'. Budgets, if badly set, can also demotivate employees.
- **Evaluation** – the performance of managers is often judged by looking at how well the manager has performed 'against budget'.
- **Authorisation** – budgets act as a form of authorisation of expenditure.

2 The stages in budget preparation

How are budgets prepared?

Before any budgets can be prepared, the long-term objectives of an organisation must be defined so that the budgets prepared are working towards the goals of the business.

Once this has been done, the budget committee can be formed, the budget manual can be produced and the limiting factor can be identified.

- **Budget committee is formed** – a typical budget committee is made up of the chief executive, budget officer (management accountant) and departmental or functional heads (sales manager, purchasing manager, production manager and so on). The budget committee is responsible for communicating policy guidelines to the people who prepare the budgets and for setting and approving budgets.
- **Budget manual is produced** – an organisation's budget manual sets out instructions relating to the preparation and use of budgets. It also gives details of the responsibilities of those involved in the budgeting process, including an organisation chart and a list of budget holders.
- **Limiting factor is identified** – in budgeting, the limiting factor is known as the principal budget factor. Generally there will be one factor that will limit the activity of an organisation in a given period. It is usually sales that limit an organisation's performance, but it could be anything else, for example, the availability of special labour skills.

If sales is the principal budget factor, then the sales budget must be produced first.

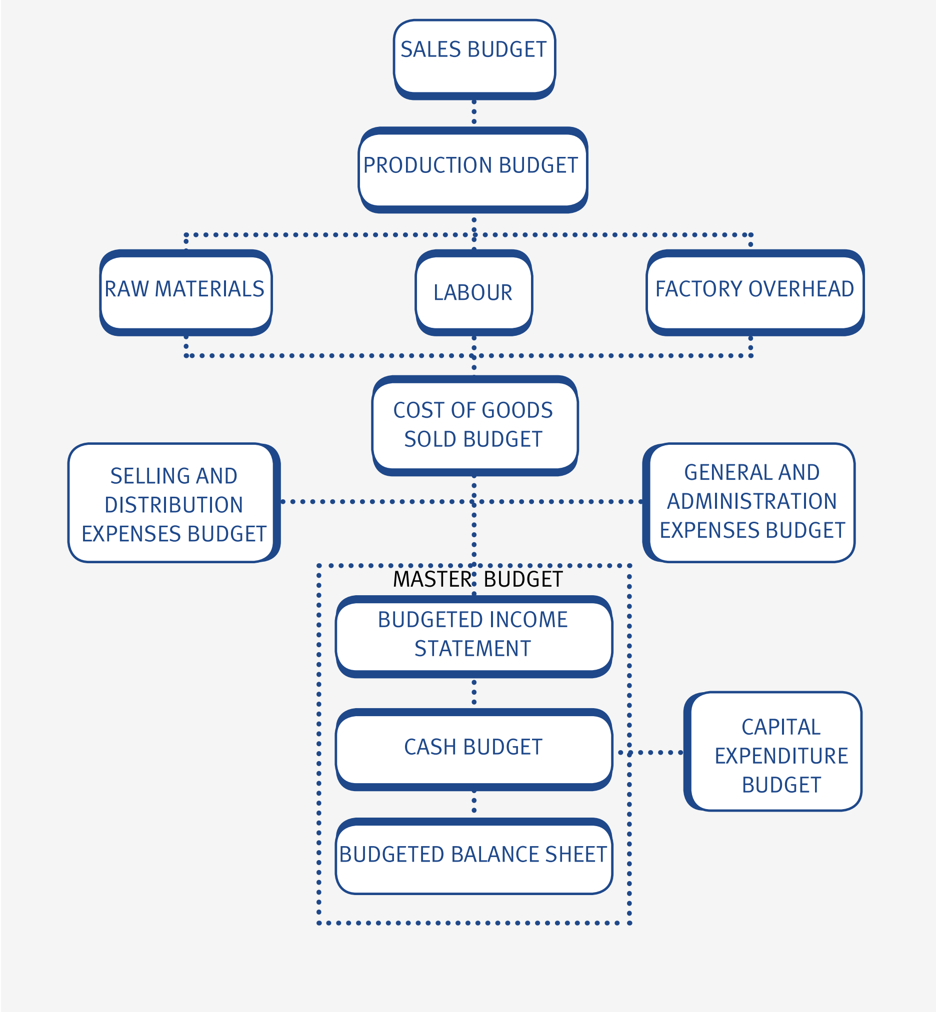
- **Final steps in the budget process** – once the budget relating to the limiting factor has been produced then the managers responsible for the other budgets can produce them. The entire budget preparation process may take several weeks or months to complete. The final stages are as follows.
  - 1 Initial budgets are prepared. Budget managers may sometimes try to build in an element of budget slack – this is a deliberate over-estimation of costs or under-estimation of revenues which can make it easier for managers to achieve their targets.
  - 2 Initial budgets are reviewed and integrated into the complete budget system.
  - 3 After any necessary adjustments are made to initial budgets, they are accepted and the master budget is prepared (budgeted income statement, balance sheet and cash flow). This master budget is then shown to top management for final approval.
  - 4 Budgets are reviewed regularly. Comparisons between budgets and actual results are carried out and any differences arising are known as variances.

Budget preparation

The preparation of budgets is illustrated as follows.

e.g

Illustration 1 – The stages in budget preparation



- The diagram shown above is based on sales being the principal budget factor.This is why the sales budget is shown in Step 1.
- Remember that if labour were the principal budget factor, then the labour budget would be produced first and this would determine the production budget.
- Once the production budget has been determined then the remaining functional budgets can be prepared.

Continuous budgets

- **Continuous budget** – this type of budget is prepared a year (or budget period) ahead and is updated regularly by adding a further accounting period (month, quarter) when the first accounting period has expired. If the budget period is a year, then it will always reflect the budget for a year in advance. Continuous budgets are also known as rolling budgets.

3 Sales budgets

Budget preparation – functional budgets

A functional budget is a budget of income and/or expenditure which applies to a particular function. The main functional budgets that you need to be able to prepare are as follows:

- sales budget
- production budget
- raw material usage budget
- raw material purchases budget
- labour budget
- overheads budget.

Sales budgets

We shall begin our preparation of functional budgets by looking at sales budgets. Sales budgets are fairly straightforward to prepare as the following illustration will demonstrate.

Illustration 2 - Sales budgets

A company makes two products – PS and TG. Sales for next year are budgeted to be 5,000 units of PS and 1,000 units of TG. Planned selling prices are $95 and $130 per unit respectively.

Required:

Prepare the sales budget for the next year.

Expandable text

Test your understanding 1

A company makes two products – A and B. The products are sold in the ratio 1:1. Planned selling prices are $100 and $200 per unit respectively. The company needs to earn $900,000 revenue in the coming year.Required:

Prepare the sales budget for the coming year.

4 Production budgets

Budgeted production levels can be calculated as follows:

Budgeted production = Forecast sales + Closing inventory of finished goods – Opening inventory of finished goods

Illustration 3 - Production budgets

A company makes two products, PS and TG. Forecast sales for the coming year are 5,000 and 1,000 units repectively.

The company has the following opening and required closing inventory levels.

	PS units	TG units
Opening inventory	100	50
Required closing inventory	1,100	50

Required:

Prepare the production budget for the coming year.

Expandable text

Test your understanding 2

Newton Ltd manufactures three products. The expected sales for each product are shown below.

	Product 1	Product 2	Product 3
Sales in units	3,000	4,500	3,000

Opening inventory is expected to be:

Product 1 500 units

Product 2 700 units

Product 3 500 units

Management have stated their desire to reduce inventory levels, and closing inventory is budgeted as:

Product 1	200 units
Product 2	300 units
Product 3	300 units

Required:

Complete the following:

(a) The number of units of product 1 to be produced is

(b) The number of units of product 2 to be produced is

(c) The number of units of product 3 to be produced is

5 Material budgets

There are two types of material budget that you need to be able to calculate, the usage budget and the purchases budget.

- The **material usage budget** is simply the budgeted production for each product multiplied by the quantity (e.g. kg) required to produce one unit of the product.
- The **material purchases budget** is made up of the following elements.

Material purchases budget = Material usage budget + Closing inventory – Opening inventory

e.g

Illustration 4 - Material budgets

A company produces Products PS and TG and has budgeted to produce 6,000 units of Product PS and 1,000 units of Product TG in the coming year.

The data about the materials required to produce Products PS and TG is given as follows.

	PS	TG
Finished products:	per unit	per unit
Kg of raw material X	12	12
Kg of raw material Y	6	8
Direct materials:		

	Raw material	
	X	Y
	kg	kg
Desired closing inventory	6,000	1,000
Opening inventory	5,000	5,000
Standard rates and prices:		
Raw material X		$0.72 per kg
Raw material Y		$1.56 per kg

Required:

Prepare the following:

(a) The material usage budget.

(b) The material purchase budget.

Expandable text

Test your understanding 3

Newton Ltd manufactures three products. The expected production levels for each product are shown below.

	Product 1	**Product 2**	**Product 3**
Budgeted production in units	2,700	4,100	2,800

Three types of material are used in varying amounts in the manufacture of the three products. Material requirements are shown below:

	Product 1	**Product 2**	**Product 3**
Material M1	2 kg	3 kg	4 kg
Material M2	3 kg	3 kg	4 kg
Material M3	6 kg	2 kg	4 kg

The opening inventory of material is expected to be:

Material M1	4,300 kg
Material M2	3,700 kg
Material M3	4,400 kg

Management are keen to reduce inventory levels for materials, and closing inventory levels are to be much lower. Expected levels are shown below:

Material M1 2,200 kg

Material M2 1,300 kg

Material M3 2,000 kg

Material prices are expected to be 10% higher than this year and current prices are $1.10/kg for material M1, $3.00/kg for material M2 and $2.50/kg for material M3.

Required:

Complete the following.

(a)
(i)The quantity of material M1 to be used is kg

(ii)The quantity of material M2 to be used is kg

(iii)The quantity of material M3 to be used is kg

(b)
(i)The quantity of material M1 to be purchased is kg

These purchases have a value of $ kg

(ii) The quantity of material M2 to be purchased is kg

These purchases have a value of $ kg

(iii) The quantity of material M3 to be purchased is kg

These purchases have a value of $ kg

6 Labour budgets

Labour budgets are simply the number of hours multiplied by the labour rate per hour as the following illustration shows.

Illustration 5 Labour budgets

A company produces Products PS and TG and has budgeted to produce 6,000 units of Product PS and 1,000 units of Product TG in the coming year.

The data about the labour hours required to produce Products PS and TG is given as follows.
Finished products:

	PS per unit	TG per unit
Direct labour hours	8	12

Standard rate for direct labour = $5.20 per hour
Required:

Prepare the labour budget for the coming year.

Expandable text

Test your understanding 4

Newton Ltd manufactures three products. The expected production levels for each product are shown below.

	Product 1	Product 2	Product 3
Budgeted production in units	2,700	4,100	2,800

Two types of labour are used in producing the three products. Standard times per unit and expected wage rates for the forthcoming year are shown below:

	Product 1	Product 2	Product 3
Hours per unit			
Skilled labour	3	1	3
Semi-skilled labour	4	4	2

Skilled labour is to be paid at the rate of $9/hour and semi-skilled labour at the rate of $6/hour.

Required:

Complete the following.

(a)The number of hours of skilled labour required is

The cost of this labour is $

(b)The number of hours of semi-skilled labour required is

The cost of this labour is $

Test your understanding 5

A contract cleaning firm estimates that it will take 2,520 actual cleaning hours to clean an office block. Unavoidable interruptions and lost time are estimated to take 10% of the workers' time. If the wage rate is $8.50 per hour, the budgeted labour cost will be:

A $19,278

B $21,420

C $23,562

D $23,800

7 Overhead budgets

The following illustration demonstrates the calculation of overhead budgets.

Illustration 6 - Overhead budgets

A company produces Products PS and TG and has budgeted to produce 6,000 units of Product PS and 1,000 units of Product TG in the coming year.

The following data about the machine hours required to produce Products PS and TG and the standard production overheads per machine hour is relevant to the coming year.

	PS per unit	TG per unit
Machine hours	8	12

Production overheads per machine hour

Variable	$1.54 per machine hour
Fixed	$0.54 per machine hour

Required:

Calculate the overhead budget for the coming year.

Expandable text

Test your understanding 6

Newton Ltd manufactures three products. The expected production levels for each product are shown below.

Product 1 Product 2 Product 3

Budgeted production in units 2,700 4,100 2,800

Two types of labour are used in producing the three products. Standard times per unit and expected wage rates for the forthcoming year are shown below:

Product 1 Product 2 Product 3

Hours per unit

Skilled labour 3 1 3

Semi-skilled labour 4 4 2

Production overheads per labour hour are as follows:

Variable $3.50 per labour hour

Fixed $5.50 per labour hour

Required:

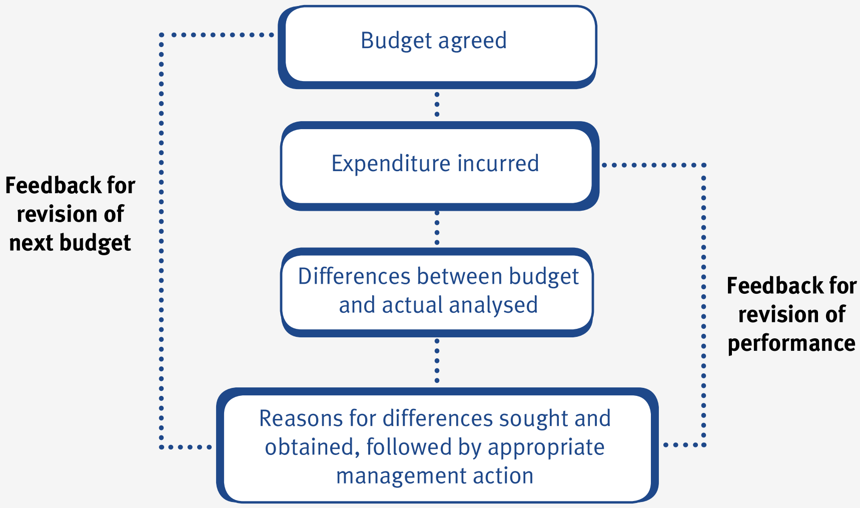
Calculate the overhead budget.

8 Fixed, flexible and flexed budgets

Budgetary control cycle

At the beginning of this chapter we listed the main purposes of budgeting, one of which was ‘controlling costs – by comparing the plan of the budget with the actual results and investigating any significant differences between the two’ – this is known as **budgetary control**.

The budgetary control cycle can be illustrated as follows.



Fixed budgets

The simplest form of budget report compares the original budget against actual results and is known as a fixed budget.

- Any differences arising between the original budget and actual results are known as variances. Variances may be either adverse or favourable.
- Adverse variances (Adv) or (A) decrease profits.
- Favourable variances (Fav) or (F) increase profits.
- An example of a budget report is shown below.

Budget	Actual	
Sales units	1,000	1,200
Production units		
	1,300	1,250

	$	$	$
			Variance
Sales revenue	10,000	11,500	1,500 Fav
- labour costs	2,600	125	475 Fav
- materials costs	1,300	1,040	260 Fav
- overheads	1,950	2,200	250 Adv
	5,850	5,365	485 Fav
Cost of sales			
Profit	4,150	6,135	1,985 Fav

- The fixed budget shown above is not particularly useful because we are not really comparing like with like. For example, the budgeted sales were 1,000 units but the actual sales volume was 1,200 units.
- The overall sales variance is favourable, but from the report shown we don't know how much of this variance is due to the fact that actual sales were 200 units higher than budgeted sales (or whether there was an increase in the sales price).
- Similarly, actual production volume was 50 units less than the budgeted production volume, so we are not really making a very useful comparison. It is more useful to compare actual results with a budget that reflects the actual activity level. Such a budget is known as a **flexed budget**.

Flexed budgets

A **flexed** budget is a budget which recognises different cost behaviour patterns and is designed to change as the volume of activity changes.

- When preparing flexed budgets it will be necessary to identify the cost behaviour of the different items in the original budget.
- In some cases you may have to use the high/low method in order to determine the fixed and variable elements of semi-variable costs.

e.g

Illusration 7 - Fixed, flexible and flexed budgets

Wye Ltd manufactures one product and when operating at 100% capacity can produce 5,000 units per period, but for the last few periods has been operating below capacity.

Below is the flexible budget prepared at the start of last period, for three levels of activity at below capacity:

Level of activity

	70%	80%	90%
	$	$	$
Direct materials	7,000	8,000	9,000
Direct labour	28,000	32,000	36,000
Production overheads	34,000	36,000	38,000
Administration, selling and distribution overheads	15,000	15,000	15,000
Total cost	84,000	91,000	98,000

In the event, the last period turned out to be even worse than expected, with production of only 2,500 units. The following costs were incurred:

	$
Direct materials	4,500
Direct labour	22,000
Production overheads	28,000
Administration, selling and distribution overheads	16,500
Total cost	71,000

Required:

Use the information given above to prepare the following.

(a) A flexed budget for 2,500 units.

(b) A budgetary control statement.

Expandable text

9 Chapter summary

The purposes of budgeting

Planning
Control
Co-ordination
Authorisation
Communication
Motivation
Evaluation

BUDGETING

The stages of budget preparation

1. Budget committee formed
2. Budget manual produced
3. Limiting factor identified and its corresponding budget prepared
4. Initial functional budgets prepared
5. Initial budgets reviewed and integrated into complete budget system
6. Top management give final approval
7. Budgets reviewed regulary and variances arising are investigated

Fixed, flexible and flexed budgets

- Original budget = fixed
- Flexed budget recognises different cost behaviour patterns and is designed to change volume of activity changes

SALES BUDGETS

PRODUCTION BUDGETS

MATERIAL BUDGETS

LABOUR BUDGETS

OVERHEAD BUDGETS

Note that if sales is the limiting factor (principal budget factor) then sales demand is what limits the activities of an organisation and so the sales budget must be prepared first.
Once the sales budget has been prepared, the production budget can be prepared, followed by the material, labour and overhead budgets.

Test your understanding answers

Test your understanding 1

Sales budget

	Total	A	B
Sales units (see working)	6,000	3,000	3,000
Selling price per unit		$100	$200
Sales value	$900,000	$300,000	$600,000

Working

Total sales revenue = $900,000

$300 revenue is earned every time a mix of one unit of Product A and one unit of Product B is sold ($100 + $200).

Number of 'mixes' to be sold to earn $900,000 = $\frac{\$900,000}{\$300}$ = 3,000 'mixes'

3,000 'mixes' = 3,000 units of Product A and 3,000 units of Product B.

Test your understanding 2

(a) The number of units of product 1 to be produced is

(b) The number of units of product 2 to be produced is

(c) The number of units of product 3 to be produced is

Workings:

	Product 1	**Product 2**	**Product 3**
Sales forecast	3,000	4,500	3,000
+ Closing inventory	200	300	300
– Opening inventory	500	700	500
Production budget	2,700	4,100	2,800

Test your understanding 3

(a)
(i)The quantity of material M1 to be used is 28,900 kg

(ii)The quantity of material M2 to be used is 31,600 kg

(iii)The quantity of material M3 to be used is 35,600 kg

(b)
(i)The quantity of material M1 to be purchased is 26,800 kg

These purchases have a value of $ 32,428

(ii) The quantity of material M2 to be purchased is 29,200 kg

These purchases have a value of $ 96,360

(iii) The quantity of material M3 to be purchased is 33,200 kg

These purchases have a value of $ 91,300

Working

	Material M1	**Material M2**	**Material M3**
Product 1 usage	5,400	8,100	16,200
Product 2 usage	12,300	12,300	8,200
Product 3 usage	11,200	11,200	11,200
Materials usage budget	28,900	31,600	35,600

	Material M1		**Material M3**
Material usage	28,900	31,600	35,600
+ Closing inventory	2,200	1,300	2,000
- Opening inventory	4,300	3,700	4,400
Material purchases budget(units)	26,800	29,200	33,200
Material price per kg	$1.21	$3.30	$2.75
Material purchases budget (value)	$32,428	$96,360	$91,300

Note: Material prices are as follows:

M1 : $1.10 x 1.1 = $1.21

M2 : $3.00 x 1.1 = $3.30

M3 : $2.50 x 1.1 = $2.75

Material usages are as follows:

Product 1 – Material M1 usage = 2 x 2,700 = 5,400

Product 2 – Material M1 usage = 3 x 4,100 = 12,300

Product 3 – Material M1 usage = 4 x 2,800 = 11,200

Product 1 – Material M2 usage = 3 x 2,700 = 8,100

Product 2 – Material M2 usage = 3 x 4,100 = 12,300

Product 3 – Material M2 usage = 4 x 2,800 = 11,200

Product 1 – Material M3 usage = 6 x 2,700 = 16,200

Product 2 – Material M3 usage = 2 x 4,100 = 8,200

Product 3 – Material M3 usage = 4 x 2,800 = 11,200

Test your understanding 4

The number of hours of skilled labour required is	20,600
The cost of this labour is $	185,400
The number of hours of semi-skilled labour required is	32,800
The cost of this labour is $	$ 96,800

Workings:

	Skilled	**Semi-skilled**
Product 1 hours	8,100	10,800
Product 2 hours	4,100	16,400
Product 3 hours	8,400	5,600
Labour budget (hours)	20,600	32,800
Labour rate per hour	$9	$6
Labour budget ($)	185,400	196,800

Product 1 – skilled hours = 3 x 2,700 = 8,100

Product 2 – skilled hours = 1 x 4,100 = 4,100

Product 3 – skilled hours = 3 x 2,800 = 8,400

Product 1 – semi-skilled hours = 4 x 2,700 = 10,800

Product 2 – semi-skilled hours = 4 x 4,100 = 16,400

Product 3 – semi-skilled hours = 2 x 2,800 = 5,600

Test your understanding 5

The correct answer is D.

The budgeted labour cost is $23,800

$$\text{Actual expected total time} = \frac{2{,}520}{0.9} = 2{,}800 \text{ hours}$$

Budgeted labour cost = 2,800 × $8.50 = $23,800.

Test your understanding 6

Number of hours of skilled labour = 20,600 (see working)

Number of hours of semi-skilled labour = 32,800 (see working)

Total hours worked = 20,600 + 32,800 = 53,400

		$
Variable costs	53,400 hours x $3.50	186,900
Fixed costs	53,400 hours x $5.50	293,700
		480,600

Working

Product 1 – skilled hours = 3 x 2,700 = 8,100

Product 2 – skilled hours = 1 x 4,100 = 4,100

Product 3 – skilled hours = 3 x 2,800 = 8,400

Total skilled hours = 8,100 + 4,100 + 8,400 = 20,600

Product 1 – semi-skilled hours = 4 x 2,700 = 10,800

Product 2 – semi-skilled hours = 4 x 4,100 = 16,400

Product 3 – semi-skilled hours = 2 x 2,800 = 5,600

Total semi-skilled hours = 10,800 + 16,400 + 5,600 = 32,800

chapter

14

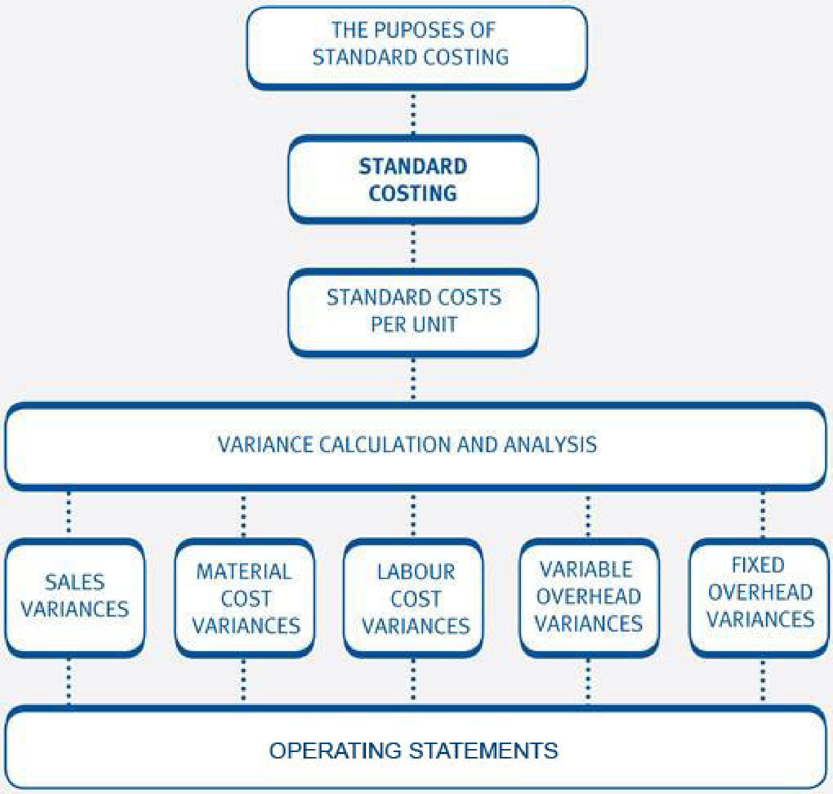
Standard costing

Chapter learning objectives

Upon completion of this chapter you will be able to:

- explain, for a manufacturing business, the purpose and principles of standard costing: targets, feedback/variances, ease of accounting
- establish, from information supplied, the standard cost per unit under marginal costing
- establish, from information supplied, the standard cost per unit under absorption costing
- calculate, from information supplied, the sales price and volume variances
- for given or calculated sales variances, interpret and explain possible causes, including possible interrelationships between them
- calculate, from information supplied, the materials total, price and usage variances
- for given or calculated materials variances, interpret and explain possible causes, including possible interrelationships between them
- calculate, from information supplied, the labour total, price and efficiency variances
- for given or calculated labour variances, interpret and explain possible causes, including possible interrelationships between them
- calculate, from information supplied, the variable overhead total, expenditure and efficiency variances
- for given or calculated variable overhead variances, interpret and explain possibe causes, including possible interrelationships between them
- calculate, from information supplied, fixed overhead total and expenditure variances

- calculate, from information supplied, where appropriate, fixed overhead volume, capacity and efficiency variances
- for given or calculated fixed overhead variances, interpret and explain possible causes, including possible interrelationships between them
- using given variances and standard figures, calculate actual figures
- using given variances and actual figures, calculate standard figures
- using given or derived variances, produce an operating statement to reconcile budgeted profit with actual profit under standard absorption costing
- using given or derived variances, produce an operating statement to reconcile budgeted profit or contribution with actual profit or contribution under standard marginal costing.



1 The purposes of standard costing

Introduction to standard costing

A **standard cost** is the planned unit cost of a product or service.

In order to prepare budgets we need to know what an individual unit of a product or service is expected to cost.

- Standard costs represent ‘target’ costs and they are therefore useful for planning, control and motivation.
- As well as being the basis for preparing budgets, standard costs are also essential for calculating and analysing variances.
- Variances provide ‘feedback’ to management indicating how well, or otherwise, the company is doing.
- Standard costs also provide an easier method of accounting since it enables simplified records to be kept.
- A standard cost may be computed on either a marginal cost basis, including only variable costs, or under absorption costing.

Types of cost standards

There are four main types of cost standards.

- **Basic standards.**
- **Ideal standards.**
- **Attainable standards.**
- **Current standards.**

Basic standards – these are long-term standards which remain unchanged over a period of years. Their sole use is to show trends over time for items such as material prices, labour rates, and labour efficiency. They are also used to show the effect of using different methods over time. Basic standards are the least used and the least useful type of standard.

Ideal standards – these standards are based upon perfect operating conditions. Perfect operating conditions include: no wastage; no scrap; no breakdowns; no stoppages; no idle time. In their search for perfect quality, Japanese companies use ideal standards for pinpointing areas where close examination may result in large cost savings. Ideal standards may have an adverse motivational impact because they are unlikely to be achieved.

Attainable standards – these standards are the most frequently encountered type of standard. They are based upon efficient (but not perfect) operating conditions. These standards include allowances for the following: normalp or expected material losses; fatigue; machine breakdowns. Attainable standards must be based on a high performance level so that with a certain amount of hard work they are achievable (unlike ideal standards).

Current standards – these standards are based on current levels of efficiency in terms of allowances for breakdowns, wastage, losses and so on. The main disadvantage of using current standards is that they do not provide any incentive to improve on the current level of performance.

2 Standard costs per unit

Standard cost cards

Once estimated, standard costs are usually collected on a standard cost card.

e.g

Illustration 1 - Standard costs per unit

K Ltd manufactures Product 20K. Information relating to this product is given below.

Budgeted output for the year: 900 units

Standard details for one unit:

Direct materials: 40 square metres at $5.30 per square metre

Direct wages: Bonding department 24 hours at $5.00 per hour

Finishing department 15 hours at $4.80 per hour

Budgeted costs and hours per annum are as follows:

	$	**Hours**
Variable overhead		
Bonding department	45,000	30,000
Finishing department	25,000	25,000
Fixed overhead apportioned to this product:		
Production	$36,000	
Selling, distribution and administration	$27,000	

Note: Variable overheads are recovered (absorbed) using hours, fixed overheads are recovered on a unit basis.

Required:

Prepare a standard cost card in order to establish the standard cost of one unit of Product 20K and enter the following subtotals on the card:

1 prime cost

2 marginal cost

3 total absorption cost

4 total standard cost.

Expandable text

Test your understanding 1

The following statement shows budgeted and actual costs for the month of October for Department X.

Month ended 31 October	**Original budget**	**Actual result**
	$	**$**
Sales	600,000	550,000
	———	———
Direct materials	150,000	130,000
Direct labour	200,000	189,000
Production overhead		
Variable with direct labour	50,000	46,000
Fixed	25,000	29,000
	———	———
Total costs	425,000	394,000
Profit	175,000	156,000
	———	———
Direct labour hours	50,000	47,500
Sales and production units	5,000	4,500

Note: There is no opening and closing inventory.

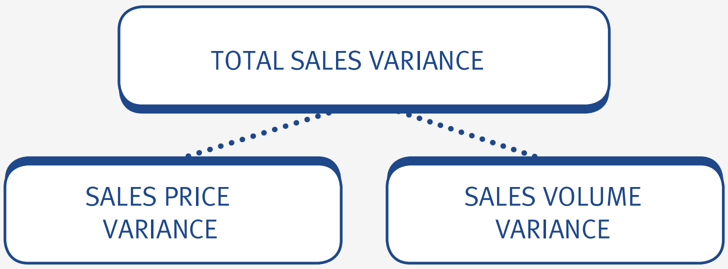
(a) Calculate the standard cost per unit under absorption costing.

(b) Calculate standard cost per unit under marginal costing.

3 Sales variances

Introduction

There are two causes of sales variances: a difference in the selling price, and a difference in the sales volume, giving:



Sales volume variance

The sales volume variance calculates the effect on profit of the actual sales volume being different from that budgeted. The effect on profit will differ depending upon whether a marginal or absorption costing system is being used.

- Under absorption costing any difference in units is valued at the standard profit per unit.
- Under marginal costing any difference in units is valued at the standard contribution per unit.

Sales volume variance

(Actual Sales Units – Budget Sales Units) × Standard Margin **

** Standard margin = Contribution per unit (marginal costing)

or

Profit per unit (absorption costing)

Sales price variance

The sales price variance shows the effect on profit of selling at a different price from that expected.

Sales price variance
(Actual Sales Price – Standard Sales Price) × Quantity sold

Illustration 2 - Sales variances

The following data relates to 20X8.

Actual sales:	1,000 units @ $650 each
Budgeted output and sales for the year:	900 units
Standard selling price:	$700 per unit
Budgeted contribution per unit:	$245
Budgeted profit per unit:	$205

Required:

Calculate the sales volume variance (under absorption and marginal costing) and the sales price variance.

Expandable text

Test your understanding 2

Radek Ltd has budgeted sales of 400 units at $2.50 each. The variable costs are expected to be $1.80 per unit, and there are no fixed costs.

The actual sales were 500 units at $2 each and costs were as expected.

Calculate the sales price and sales volume variances (using marginal costing).

Test your understanding 3

W Ltd budgeted sales of 6,500 units but actually sold only 6,000 units. Its standard cost card is as follows:

	$
Direct material	25
Direct wages	8
Variable overhead	4
Fixed overhead	18
Total standard cost	55
Standard gross profit	5
Standard selling price	60

The actual selling price for the period was $61.

Calculate the sales price and sales volume variances for the period (using absorption costing).

Possible causes of sales variances

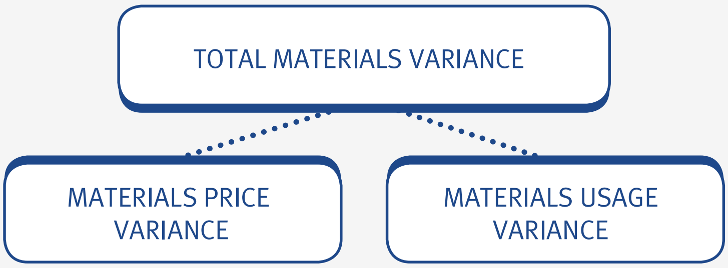
Causes of sales variances include the following:

- unplanned price increases (sales price variance)
- unplanned price reduction, for example, when trying to attract additional business (sales price variance)
- unexpected fall in demand due to recession (sales volume variance)
- additional demand attracted by reduced price (sales volume variance)
- failure to satisfy demand due to production difficulties (sales volume variance).

4 Materials cost variances

Introduction

There are three different materials variances that you need to know about:



Materials total variance

The materials total variance is the difference between:

(a) the standard material cost of the actual production (flexed budget) and

(b) the actual cost of direct material.

- It can be analysed into two sub-variances: a materials price variance and a materials usage variance.
- A materials price variance analyses whether the company paid more or less than expected for materials.
- The purpose of the materials usage variance is to quantify the effect on profit of using a different quantity of raw material from that expected for the actual production achieved.

Material variances

Actual Quantity	X	Actual Price	
			Price Variance
Actual Quantity	X	Standard Price	
			Usage Variance
Standard Quantity	X	Standard Price	

e.g

Illustration 3 - Material cost variances

The following information relates to the production of Product X.

Extract from the standard cost card of Product X

	$
Direct materials (40 square metres x $5.30 per square metre)	212

Actual results for direct materials in the period: 1,000 units were produced and 39,000 square metres of material costing $210,600 in total were used.

Required:

Calculate the materials total, price and usage variances for Product X in the period.

Expandable text

Test your understanding 4

James Marshall Ltd makes a single product with the following budgeted material costs per unit:

2 kg of material A at $10/kg

Actual details:

Output 1,000 units

Material purchased and used 2,200 kg

Material cost $20,900

Calculate materials price and usage variances.

Possible causes of materials variances

Materials price variances may be caused by:

- supplies from different sources
- unexpected general price increases
- changes in quantity discounts
- substitution of one grade of material for another
- materials price standards are usually set at a mid-year price so one would expect a favourable price variance early in a period and an adverse variance later on in a budget period.

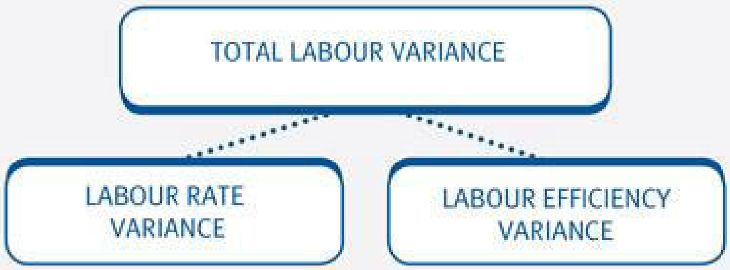
Materials usage variances may be caused by:

- a higher or lower incidence of scrap
- an alteration to product design
- substitution of one grade of material for another. A lower grade of material may be more difficult to work with, so there may be a higher wastage rate and, in turn, an adverse usage variance may arise.

5 Labour cost variances

Introduction

There are three different labour variances that you need to know about:



Labour total variance

The labour total variance is the difference between:

(a) the standard direct labour cost of the actual production (flexed budget); and

(b) the actual cost of direct labour.

- A labour price variance analyses whether the company paid more or less than expected for labour.
- A labour efficiency variance analyses whether the company used more or less labour than expected.

Actual Hours	X	Actual Rate	}	Rate Variance
Actual Hours	X	Standard Rate	}	Efficiency Variance
Standard Hours	X	Standard Rate		

Illustration 4 - Labour cost variances

The following information relates to the production of Product X.

Extract from the standard cost card of Product X

	$
Direct labour:	
Bonding (24 hrs @ $5 per hour)	120
Finishing (15 hrs @ $4.80 per hour)	72

Actual results for wages:

Production	1,000 units produced
Bonding	23,900 hours costing $131,450 in total
Finishing	15,500 hours costing $69,750 in total

Required:

Calculate the labour total, price and efficiency variances in each department for Product X in the period.

Expandable text

Test your understanding 5

Roseberry Ltd makes a single product and has the following budgeted information:

Budgeted production	1,000 units
Budgeted labour hours	3,000 hours
Budgeted labour cost	$15,000

Actual results:

Output 1,100 units

Hours paid for	3,400 hours
Labour cost	$17,680

Calculate labour total, price and efficiency variances.

Possible causes of labour variances

Labour price variances may be caused by:

- an unexpected national wage award
- overtime or bonus payments which are different from planned/budgeted
- substitution of one grade of labour for another higher or lower grade.

Labour efficiency variances may be caused by:

- changes in working conditions or working methods, for example, better supervision
- consequences of the learning effect
- introduction of incentive schemes or staff training
- substitution of one grade of labour for another higher or lower grade.

6 Variable overhead variances

Introduction

Variable overhead variances are very similar to those for materials and labour because, like these direct costs, the variable overhead cost also changes when activity levels change.

VARIABLE OVERHEAD TOTAL VARIANCE

VARIABLE OVERHEAD EXPENDITURE VARIANCE

VARIABLE OVERHEAD EFFICIENCY VARIANCE

Variable overhead total variance

It is normally assumed that variable overheads vary with direct labour hours of input and the variable overhead total variance will therefore be due to one of the following:

- the variable overhead cost per hour was different to that expected (an expenditure variance)
- working more or fewer hours than expected for the actual production (an efficiency variance)

Actual Hours	X	Actual Rate	
			Expenditure Variance
Actual Hours	X	Standard Rate	
			Efficiency Variance
Standard Hours	X	Standard Rate	

Expandable text

Illustration 5 - Variable overhead variances

The following information relates to the production of Product X.

Extract from the standard cost card of Product X

	$
Direct labour:	
Bonding (24 hrs @ $5 per hour)	120
Finishing (15 hrs @ $4.80 per hour)	72

Variable overhead:

Bonding (24 hrs @ $1.50 per hour)	36
Finishing (15 hrs @ $1 per hour)	15

Actual results for production and labour hours worked:

Production	1,000 units produced
Bonding	23,900 hours
Finishing	15,500 hours

Actual results for variable overheads:

Bonding Total cost	$38,240
Finishing Total cost	$14,900

Required:

Calculate the variable overhead total, expenditure and efficiency variances in each department for Product X for the period.

Expandable text

Test your understanding 6

The budgeted output for Carr Ltd for May was 1,000 units of product A. Each unit requires two direct labour hours. Variable overheads are budgeted at $3/labour hour.

Actual results:

Output	900 units
Labour hours worked	1,980 hours
Variable overheads	$5,544

Calculate variable overhead total, expenditure and efficiency variances.

Possible causes of variable overhead variances

Variable overhead expenditure variances may be caused by:

- incorrect budgets being set at the beginning of a period.
- overheads consisting of a number of items, such as: indirect materials, indirect labour, maintenance costs, power, etc. Consequently, any meaningful interpretation of the expenditure variance must focus on individual cost items.

Variable overhead efficiency variances may be caused by:

- changes in working methods and condition, for example, better supervision
- consequences of the learning effect
- introduction of incentive schemes or staff training
- substitution of one grade of labour for another higher or lower grade.

Note that the possible causes of variable overhead efficiency variances are the same as those for the labour efficiency variance.

7 Fixed overhead variances

Introduction

Fixed overhead variances show the effect on profit of differences between actual and expected fixed overheads.

- By definition, actual and expected fixed overheads do not change when there is a change in the level of activity, consequently many of the variances calculated are based upon budgets.
- However, the effect on profit depends upon whether a marginal or absorption costing system is being used.

Fixed overhead variances in a marginal costing system

As you know, marginal costing does not relate fixed overheads to cost units. There is no under- or over-absorption and the fixed overhead incurred is the amount shown in the income statement (as a period cost).

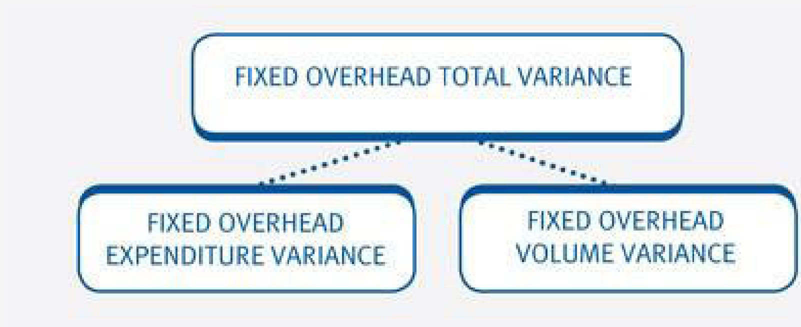
- Since fixed overhead costs are fixed, they are not expected to change when there is a change in the level of activity.
- Fixed overhead variances in a marginal costing system are as follows:

Fixed overhead variances (marginal costing principles)

	$
Actual Expenditure	x
Less: Budget Expenditure	x
	—
Fixed overhead expenditure variance	x
	—

Fixed overhead variances in an absorption costing system

- In absorption costing, fixed overheads are related to cost units by using absorption rates. This means that the calculation of fixed overhead variances in an absorption costing system is slightly more complex than in a marginal costing system.
- The fixed overhead variances in an absorption costing system are as follows:



- The fixed overhead total variance is equivalent to the under- or over-absorption of overhead in a period.

Fixed overhead total variance

- In an absorption costing system, the fixed overhead total variance measures the effect on profit of there being a difference between the actual cost incurred and the amount absorbed by the use of the absorption rate based on budgeted costs and activity. Under absorption costing, this variance can be further subdivided into an expenditure and volume variance.

Fixed overhead expenditure variance

The fixed overhead expenditure variance shows the effect on profit of the actual fixed overhead expenditure differing from the budgeted value. It is calculated in exactly the same way for both marginal and absorption costing.

	$
Actual expenditure	X
Less: Budgeted expenditure	X
Fixed overhead expenditure variance	X

Fixed overhead volume variance

The fixed overhead volume variance is a measure of the amount of under- or over-absorption of overheads that arises because the actual production volume is different from the budgeted production volume.

	$
Budgeted expenditure	X
Less: Actual units x Fixed overhead absorption rate per unit (FOAR)	X
Fixed overhead volume variance	

Alternatively:

	$
Budgeted expenditure	X
Less: Standard hours for actual production x FOAR per standard hour	X

Illustration 6 - Fixed overhead variances

The following information is available for a company for Period 4.

Budget

Fixed production overheads	$22,960
Unit	6,560

Actual

Fixed production overheads	$24,200
Unit	6,460

Required:

Calculate the following:

(a) FOAR per unit

(b) fixed overhead expenditure variance for marginal costing

(c) fixed overhead expenditure variance for absorption costing

(d) fixed overhead volume variance for marginal costing

(e) fixed overhead volume variance for absorption costing

(f) fixed overhead total variance for marginal costing

(g) fixed overhead total variance for absorption costing.

Expandable text

Fixed overhead capacity and efficiency variances

In absorption costing systems, the fixed overhead volume variance can be subdivided into capacity and efficiency variances.

- The capacity variance measures whether the workforce worked more or fewer hours than budgeted for the period:

	$
Budgeted expenditure	X
Less Actual hours x FOAR per hour	X
	––
Fixed overhead volume capacity variance	X
	––

- The efficiency variance measures whether the workforce took more or less time than expected in producing their output for the period:

	$
Standard hours for actual production x FOAR per hour	X
Less Actual hours x FOAR per hour	X
	––
Fixed overhead volume efficiency variance	X
	––

- Together, these two sub-variances explain why the level of activity was different from that budgeted, i.e. they combine to give the fixed overhead volume variance.

Illustration 7 - Fixed overhead variances

The following information is available for a company for Period 4.

Fixed production overheads	$22,960
Units	6,560

The standard time to produce each unit is 2 hours

Actual

Fixed production overheads	$24,200
Units	6,460
Labour hours	12,600 hrs

Required:

Calculate the following:

(a) FOAR per hour

(b) fixed overhead capacity variance

(c) fixed overhead efficiency variance

(d) fixed overhead volume variance.

Expandable Text

Test your understanding 7

Gatting Ltd produces a single product. Fixed overheads are budgeted at $12,000 and budgeted output is 1,000 units.

Actual results:

Output	1,100 units
Overheads incurred	$13,000

Calculate the following:

(a) under/over-absorption

(b) fixed overhead expenditure variance

(c) fixed overhead volume variance

(d) fixed overhead total variance.

Possible causes of fixed overhead variances

Fixed overhead expenditure variances may be caused by:

- changes in prices relating to fixed overhead expenditure, for example, increase in factory rent
- seasonal differences, e.g. heat and light costs in winter. When the annual budget is divided into four equal quarters, no allowances are given for seasonal factors and the fact that heat and light costs in winter are generally much higher than in the summer months. (Of course, over the year, the seasonal effect is cancelled out.)

Fixed overhead volume variances may be caused by:

- changes in the production volume due to changes in demand or alterations to stockholding policies
- changes in the productivitiy of labour or machinery
- lost production through strikes.

Possible interrelationships between variances

The cause of a particular variance may affect another variance in a corresponding or opposite way. This is known as interrelationships between variances. Here are some examples:

- if supplies of a specified material are not available, this may lead to a favourable price variance (use of cheaper material), an adverse usage variance (more wastage caused by cheaper material), an adverse fixed overhead volume variance (production delayed while material was unavailable) and an adverse sales volume variance (inability to meet demand due to production difficulties)
- a new improved machine becomes available which causes an adverse fixed overhead expenditure variance (because this machine is more expensive and depreciation is higher) offset by favourable wages efficiency and fixed overhead volume variances (higher productivity)
- workers trying to improve productivity (favourable labour efficiency variance) might become careless and waste more material (adverse materials usage variance)

- in each of these cases, if one variance has given rise to the other, there is an argument in favour of combining the two variances and ascribing them to the common cause. In view of these possible interrelationships, care has to be taken when implementing a bonus scheme. If the chief buyer is rewarded for producing a favourable price variance, this may cause trouble later as shoddy materials give rise to adverse usage variances.

8 Operating statements – total absorption costing

Variances are often summarised in an operating statement (or reconciliation statement).

e.g

Illustration 8 - Operating statements – total absorption costing

Absorption costing operating statement

			$
Budgeted profit			
Sales volume variance **(using profit per unit)**			——
Standard profit on actual sales			
Sales price variance			——
Cost variances:	**F** $	**A** $	
Material price			
Material usage			
Labour rate			
Labour efficiency			
Variable overhead rate			
Variable overhead efficiency			
Fixed overhead volume – Production			
Fixed overhead expenditure – Production			
Fixed overhead expenditure –Non-production	____	____	
Total			____
Actual profit			____

(Note the underlines in bold highlight the differences from the marginal costing version.)

Test your understanding 8

Chapel Ltd manufactures a chemical protective called Rustnot. The following standard costs apply for the production of 100 cylinders:

		$
Materials	500 kgs @ $0.80 per kg	400
Labour	20 hours @ $1.50 per hour	30
Fixed overheads	20 hours @ $1.00 per hour	20
		450

Chapel Ltd uses absorption costing.

The monthly production/sales budget is 10,000 cylinders.Selling price = $6 per cylinder.

For the month of November the following actual production and sales information is available:

Produced/sold		10,600 cylinders
Sales value		$63,000
Material purchased and used	53,200 kgs	$42,500
Labour	2,040 hours	$3,100
Fixed overheads		$2,200

The following variances were calculated for November:

Sales volume variance	$900 (F)
Sales price variance	$600 (A)
Materials price variance	$60 (F)
Materials usage variance	$160 (A)
Labour rate variance	$40 (A)
Labour efficiency variance	$120 (F)
Fixed overhead expenditure variance	$200 (A)

Fixed overhead volume variance	$120 (F)

Prepare an operating statement for November using the variances given above.

If you wish to gain further practice in calculating variances, use the information given in the question to calculate the variances. The Expandable Text after the solution shows how the individual entries in the operating statement were calculated.

Expandable text

9 Operating statements – marginal costing

The main differences between absorption and marginal costing operating statements are as follows:

- The marginal costing operating statement has a sales volume variance that is calculated using the standard contribution per unit (rather than a standard profit per unit as in absorption costing).
- There is no fixed overhead volume variance.

Illustration 9 - Operating statement – marginal costing

Marginal costing operating statement

			$
Budgeted contribution			
Sales volume variance (using contribution per unit)			——
Standard contribution on actual sales			
Sales price variance			——
Cost variances:			
	F	A	
	$	$	
Material price			
Material usage			
Labour rate			
Labour efficiency			

Variable overhead rate			
Variable overhead efficiency			
Actual contribution			
Budgeted fixed on			
Fixed overhead expenditure – Production			
Fixed overhead expenditure – Non-production	——	——	
Total			——
Actual profit			——

Note

The sales volume variance is calculated using the standard contribution per unit.

There is no fixed overhead volume variance (and therefore capacity and efficiency variances) in a marginal costing operating statement.

Test your understanding 9

Place an 'X' in the relevant column to show which variances are found in marginal costing (MC) and which are found in absorption costing (AC) operating statements.

Variance	**MC**	**AC**
Sales volume profit variance		
Sales volume contribution variance		
Material cost variances		
Labour cost variances		
Variable overhead variances		
Fixed overhead expenditure variance		
Fixed overhead volume variance		
Fixed overhead capacity variance		
Fixed overhead efficiency variance		
Fixed overhead total variance		

10 Working backwards

One way that the examiner can easily test your understanding of variances is to ask you to calculate the following instead of straightforward variance calculations:

- actual figures from variances and standards
- standards from variances and actual figures.

Illustration 10 - Working backwards

ABC Ltd uses standard costing. It purchases a small component for which the following data are available:

Actual purchase quantity	6,800 units
Standard allowance for actual production	5,440 units
Standard price	$0.85/unit
Material price variance (Adverse)	($544)

Required:

Calculate the actual price per unit of material.

Expandable text

Test your understanding 10

A business has budgeted to produce and sell 10,000 units of its single product. The standard cost per unit is as follows:

Direct materials	$15
Direct labour	$12
Variable overhead	$10
Fixed production overhead	$8

During the period the following variances occurred:

fixed overhead expenditure variance	$ 4,000 adverse
fixed overhead volume variance	$12,000 favourable

Calculate the following.

(a) Actual fixed overheads in the period.

(b) Actual production volume in the period.

Illustration 11 – Working backwards

In a period, 11,280 kilograms of material were used at a total standard cost of $46,248. The material usage variance was $492 adverse.

Required:

Calculate the standard allowed weight of material for the period.

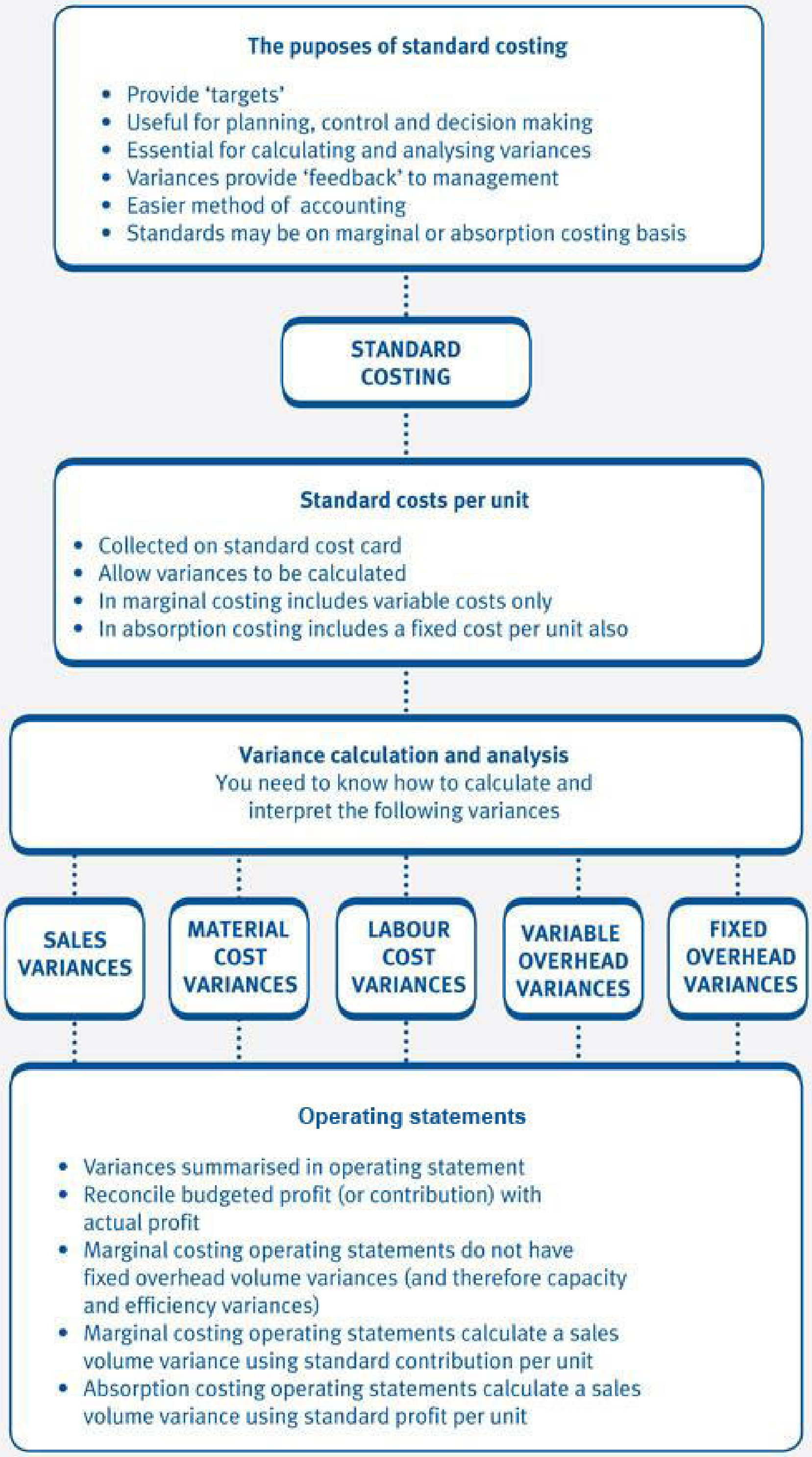
Expandable text

Test your understanding 11

In a period 6,500 units were made and there was an adverse labour efficiency variance of $26,000. Workers were paid $8 per hour, total wages were $182,000 and there was a nil rate variance.

Calculate how many standard labour hours there were per unit.

11 Chapter summary



Test your understanding answers

Test your understanding 1

(a) Standard cost per unit under absorption costing

	$
Direct materials	150,000
Direct labour	200,000
Production overhead	
Variable	50,000
Fixed	25,000
	________
Total production cost	425,000
	________

Budgeted production units = 5,000

Standard cost per unit – absorption costing = $425,000/5,000 units = $85

(b) Standard cost per unit under marginal costing

	$
Direct materials	150,000
Direct labour	200,000
Variable overhead costs	50,000
	________
Total variable cost	400,000
	________

Budgeted production units = 5,000

Standard cost per unit – marginal costing = $400,000/5,000 units = $80

Test your understanding 2

Price variance = ($2.50 – $2.00) x 500 = $250 (A)

Volume variance = (500 – 400) x $0.70* = $70 (F)

$180 (A)

*Standard contribution per unit = $(2.50 – 1.80) = $0.70

Test your understanding 3

Price variance = ($60 – $61) x 6,000 = $6,000 (F)

Volume variance = (6,500 – 6,000) x $5* = $2,500 (A)

$3,500 (F)

* Standard gross profit per unit

Test your understanding 4

Material variances	**$**	**$**
Actual quantity x Actual price =	20,900	
Price variance		1,100 (F)
Actual quantity x Standard price		
= 2,200 x $10 =	22,000	
Usage variance		2,000 (A)
Standard quantity x Standard price		
= 1,000 x 2 x $10 =	20,000	
Total variance		900 (A)

Test your understanding 5

Labour variances	$	$
Actual hours x Actual rate =	17,680	
Price variance		680 (A)
Actual hours x Standard rate		
= 3,400 hours x $5 ($15,000/3,000 = $5 per hour) =	17,000	
Efficiency variance		500 (A)
Standard hours x Standard rate		
= 1,100 x 3 hours x $5 =	16,500	
Total variance		1,180 (A)

Test your understanding 6

Variable overhead variances	$	$
Actual hours x Actual rate =	5,544	
Expenditure variance		396 (F)
Actual hours x Standard rate		
= 1,980 hours x $3 =	5,940	
Efficiency variance		540(A)
Standard hours x Standard rate		
= 900 x 2 hours x $3 =	5,400	
Total variance		144 (A)

Variable overhead variances: $ $

Actual hours x Actual rate = 5,544

Expenditure variance 396 (F)

Actual hours x Standard rate

= 1,980 hours x $3 = 5,940

Efficiency variance 540(A)

Standard hours x Standard rate

= 900 x 2 hours x $3 = 5,400

Total variance 144 (A)

The variable overhead total variance is the sum of the expenditure and efficiency variances, i.e.

$396 (F) + $540 (A) = $144 (A).

Proof: The variable overhead total variance in the Finishing department is the difference between:

(a) the standard variable overhead of the actual production (flexed budget) $5,400 and

(b) the actual cost of the variable overhead $5,544.

Difference = $(5,400 – 5,544) = $144 (A) which is what was calculated above.

Test your understanding 7

(a) FOAR per unit = $12,000/1,000 = $12 per unit	
Under/over-absorption	
Overheads absorbed = 1,100 x $12	$13,200
Overheads incurred	$13,000
Over-absorption	$200
(b) Fixed overhead expenditure variance	
Actual expenditure	$13,000
Budgeted expenditure	$12,000
	———
Expenditure variance	$1,000 (A)
	———
(c) Fixed overhead volume variance	
Budgeted expenditure	12,000
Actual units x FOAR (1,100 x $12)	13,200
	———
Volume variance	$1,200 (F)
(d) Fixed overhead total variance	
The fixed overhead total variance is the sum of the	
expenditure and volume variances.	
Fixed overhead expenditure variance	$1,000 (A)
Fixed overhead volume variance	$1,200 (F)
	———
Fixed overhead total variance	$200 (F)

Proof: the fixed overhead total variance is the same as the over-absorbed overhead ($200) as calculated in part (a) above.

Test your understanding 8

Chapel Ltd – Operating statement for November

Budgeted profit (W1)			15,000
Sales volume variance (W3)			900
Sales price variance (W3)			(600)
			15,300
Cost variances:			
	F	A	
Materials price (W4)	60		
Materials usage (W4)		160	
Labour rate (W5)		40	
Labour efficiency (W5)	120		
Fixed overhead expenditure (W6)		200	
Fixed overhead volume (W6)	120		
Total	300	400	(100)
Actual profit (W2)			15,200

Test your understanding 9

Variance	MC	AC
Sales volume profit variance		x
Sales volume contribution variance	x	
Material cost variances	x	x
Labour cost variances	x	x
Variable overhead variances	x	x
Fixed overhead expenditure variance	x	x
Fixed overhead volume variance		x
Fixed overhead capacity variance		x
Fixed overhead efficiency variance		x
Fixed overhead total variance	x	x

Test your understanding 10

(a) The actual fixed overheads are calculated as follows.

Actual fixed overheads	?
Budgeted fixed overheads	($8 x 10,000) $80,000
	————
Fixed overhead expenditure variance	$4,000 (A)
	————

Because the fixed overhead expenditure variance is adverse, this means that the actual fixed overheads were $4,000 more than budgeted.

Actual fixed overheads	=	Budgeted fixed overheads	+ Fixed overhead expenditure variance
	=	$80,000 + $4,000	= $84,000

(b) The actual production volume is calculated as follows.

The fixed overhead volume variance (in units) $= \dfrac{\text{Fixed overhead volume variance in \$}}{\text{Standard fixed overhead rate per unit}}$

$= \dfrac{\$12{,}000}{\$8}$ = 1,500 units (F)

Because the variance is favourable, this means that 1,500 more units were produced than budgeted, i.e. 10,000 + 1,500 = 11,500 units.

This can also be proved as follows.

Budgeted expenditure	$80,000
Actual units x standard fixed overhead rate per unit	?

? x $8 ————

Fixed overhead volume variance $12,000 (F)

Actual units x standard fixed overhead rate per unit = $80,000 + $12,000 = $92,000

$$\text{Actual units} = \frac{\$92{,}000}{\$8} = 11{,}500 \text{ units}$$

Test your understanding 11

	$	$
Actual hours × Actual rate	182,000	
Rate variance		Nil
Actual hours × Standard rate		
= ? × $8 = 4	182,000	
Efficiency variance		26,000 (A)
Standard hours × Standard rate		
= 6,500 × Standard hours × $8 =	156,000	

$$\text{Standard labour hours per unit} = \frac{\$156{,}000}{6{,}500 \times 8} = 3$$